

Connectionless Distributed Timing: A Prior Art Publication

Changing the Conditions of the FLP Impossibility Test

mlehaptics Project — Defensive Publication — December 2025

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Status: Prior Art / Defensive Publication / Open Source

DOI: DOI [10.5281/zenodo.18078265](https://doi.org/10.5281/zenodo.18078265)

DEFENSIVE PUBLICATION SEARCH ABSTRACT

Keywords: Distributed phased array, virtual metasurface, dynamic macroscopic lattice, programmable lattice, phononic crystal, Bragg reflection, band gap, volumetric aperture, dynamic aperture, swarm beamforming, 35 U.S.C. 102 prior art, 35 U.S.C. 101 inherent anticipation, connectionless synchronization, true time delay, mechanical wave modulation, non-reciprocal array, infrasound tomography, seismic-acoustic coupling, ESP-NOW synchronization, BLE bootstrap, acoustic beamforming, deformable aperture, substrate-free metasurface, emergent aperture, wavelength-spacing ratio, IoT sensing array, UTLF, RFIP, SMSP, connectionless execution, synchronized actuation, interference pattern coordination, band-pass filter, band-stop filter, solid-state physics parallel, energy harvesting, regenerative shielding, rectenna, piezoelectric harvesting, macro atom, active selective attenuation, coordinated interference, active noise cancellation, frequency selective surface, FSS, AFSS, direction-selective filtering, phase cancellation, anti-phase, destructive interference.

Patent Classification: G01S 3/80 (acoustic direction finding), G01S 7/00 (radar arrangements), H01Q 3/26 (phased arrays), H04B 7/02 (diversity systems), G01V 1/00 (seismology), G06F 1/12 (synchronization), H04W 56/00 (synchronization in wireless networks).

Summary: This disclosure establishes prior art for a class of distributed systems that form virtual, deformable metasurfaces via connectionless synchronized execution. It explicitly documents the **inherent anticipation** of such apertures in existing networks (e.g., Amazon Sidewalk, Starlink, smart meter grids), arguing that the physical phenomenon of a virtual aperture arising from any collection of synchronized, position-known nodes **should be treated as natural law under 35 U.S.C. 101**. The architecture documented here enables exploitation of these inherent apertures but does not claim to create them.

Enabling Disclosures Include:

- Pseudocode for mechanical wave phase calculation (Appendix B)
- Protocol specifications for UTLF/RFIP/SMSP
- Wavelength-spacing analysis for aperture utility assessment
- Cross-domain applicability (electromagnetic and acoustic)

- Position as primary control variable (not error compensation)
- Three-axis control space: phase × position × density

Inherent Anticipation Argument (35 U.S.C. 101/102): Any distributed network with synchronized time and known node positions *already constitutes* a virtual aperture at wavelengths where node spacing is favorable. This is physics, not invention. Amazon Sidewalk (designed for IoT mesh) is simultaneously an unexploited continental-scale infrasound/seismic sensing array. The aperture exists whether exploited or not. This document establishes prior art for the *exploitation architecture* (UTLP/RFIP/SMSP), ensuring both the inherent phenomenon and the exploitation method remain in the public domain.

Abstract

This document establishes prior art for a class of distributed embedded systems that achieve synchronized actuation across independent wireless nodes *without* real-time coordination traffic during operation. The core insight: when devices share a time reference and a script describing future actions, they can execute in perfect synchronization without exchanging messages during the timing-critical phase.

Development began with single-device pattern playback—deterministic, timer-driven actuation. Adding wireless pairing revealed that networked devices needed only to agree on time offset; the pattern architecture already supported connectionless operation. BLE worked. ESP-NOW worked better. By separating *configuration* (which requires bidirectional communication) from *execution* (which does not), we achieve sub-millisecond synchronization using commodity microcontrollers and standard RF protocols.

This architecture was validated using SAE J845-compliant emergency lighting patterns (Quad Flash) captured at 240fps, demonstrating zero perceptible overlap between alternating signals—precision sufficient for therapeutic bilateral stimulation, emergency vehicle warning systems, and distributed swarm coordination. Reference implementation runs on commodity ESP32-C6 hardware with a bill of materials under \$15 per node.

The architecture is scale-invariant: the same three protocols (UTLP for time synchronization, RFIP for relative positioning, SMSP for coordinated action and observation) apply from 2-node therapy devices to continental sensor networks to interstellar spacecraft constellations. SMSP operates bidirectionally—scores flow out to nodes, observations flow back—enabling the same protocol to coordinate both actuation (bilateral stimulation) and sensing (atmospheric tomography). This document establishes prior art across that entire range, with 122 claims covering therapeutic, emergency, meteorological, seismoacoustic, metasurface, and planetary-scale applications.

This work is published as open-source prior art to ensure these techniques remain freely available for public use and cannot be enclosed by patents.

0. Scope and Intent of This Publication

What This Document Claims

This document establishes prior art for the *architectural pattern* of connectionless distributed execution—specifically, the separation of synchronization (which requires communication) from execution (which does not). We document that devices sharing a time reference and a deterministic script can operate in coordination without runtime communication.

What This Document Does NOT Claim

We do not claim to have invented:

- Time synchronization between distributed nodes (established since NTP, 1985)
- Bounded clock uncertainty calculation (documented extensively, including US8073976B2)
- Bilateral stimulation therapy (patented since 1999)
- Distributed beamforming (active research with existing patents)
- Any individual technique in isolation

The contribution is the explicit documentation of how these established techniques combine into a connectionless execution architecture, ensuring this combination remains freely available.

A Note on Independent Rediscovery

During preparation of this document, we discovered that our time synchronization approach—calculating bounded clock uncertainty through round-trip message exchange with drift compensation—closely parallels techniques documented in US8073976B2 (Microsoft, 2008) and its predecessors. This is unsurprising: the mathematics of time transfer are well-established, and competent engineers solving the same problem will converge on similar solutions.

We document this overlap transparently. Our contribution is not the synchronization method itself, but what happens *after* synchronization: the deliberate termination of the communication channel and the transition to independent script-based execution. This architectural choice—treating connection as scaffolding rather than infrastructure—is what we establish as prior art.

Summary of Core Contributions

This document contains 110 specific prior art claims (Section 9). For navigation, they derive from eight core architectural innovations:

#	Core Innovation	Summary	Claims
1	Connectionless Synchronized Execution	Separating the <i>synchronization phase</i> (requires communication) from the <i>execution phase</i> (requires none), enabled by shared time and deterministic scripts	1-5, 21-23, 106-109
2	Bootstrap Security Model	Establishing trust and time via connection-oriented protocol (BLE), then deriving keys for connectionless protocol (ESP-NOW) to achieve lower-jitter execution	6, 11-14, 107
3	SMSP (Synchronized	Scale-invariant data structure defining actuator state by <i>time</i> rather than frequency, enabling identical	33-42, 50, 82-86

#	Core Innovation	Summary	Claims
	Multimodal Score Protocol)	behavior across independent nodes without runtime coordination	
4	Intrinsic Swarm Geometry (RFIP)	Deriving zone assignments and swarm topology solely from peer-to-peer ranging, creating a coordinate system relative only to the swarm itself	18, 25-30
5	Distributed Dynamic Aperture	Using mechanical displacement (via UTLP-synced actuators) to achieve True Time Delay beamforming, avoiding bandwidth limitations of electronic phase shifters	43-49, 51-60, 87-92
6	Passive Atmospheric Tomography	Using distributed, time-synchronized acoustic arrays to invert sound speed variations into volumetric temperature/wind/density maps	61-75, 76-81
7	Deformable Virtual Metasurface	Swarm-based metasurface where node position and density are primary control variables (not error sources), enabling geometry changes impossible with substrate-constrained approaches	93-100
8	Emergent Aperture Exploitation	Recognition that any synchronized, position-known node collection inherently constitutes a virtual aperture—the physics exists whether exploited or not; UTLP/RFIP/SMSP enables exploitation of apertures in existing networks	101-105

The 122 claims are specific instantiations of these eight patterns across therapeutic, emergency, meteorological, seismoacoustic, metasurface, and planetary-scale applications.

1. Introduction: The Solution Was Already There

1.1 The Development Path

The mlehaptics project began with a single device: one ESP32-C6 running a bilateral stimulation pattern—alternating haptic and visual pulses at configurable rates. The pattern playback was deterministic from the start, driven by a local timer.

When we added networking to create a wireless pair, the implementation was straightforward: the client runs the same pattern as the server, but in antiphase. Both devices execute the same script; they just need to agree on timing offset. BLE provided the communication channel, and it worked.

But we had questions. The ESP32-C6 has both BLE and WiFi radios. Could we do better with ESP-NOW? How tight could the synchronization actually get? What were the limits?

1.2 The Stack Jitter Investigation

Investigating BLE timing led us deep into the ESP32's radio stack:

RF Event (hardware interrupt) ← Precise timing exists here
BLE Controller ISR (closed-source binary blob)
VHCI Transport (RAM buffer exchange)
NimBLE Host Task (FreeRTOS context switch)
L2CAP → ATT → GATT parsing
Application callback ← Where we can timestamp

The latency from RF event to application callback varies by 1-50ms depending on system state: FreeRTOS scheduling, other BLE operations in progress, WiFi coexistence, flash operations. This jitter is not RF propagation (nanoseconds across a room) but *software processing time*.

1.3 The Realization

The investigation revealed something we hadn't fully appreciated: **we already had a connectionless architecture**. Both devices possessed the same pattern. They just needed to agree on time. The pattern playback we'd built for a single device was already the solution—we just hadn't recognized it as such.

BLE worked. We implemented PTP-style synchronization using NTP timestamps over BLE GATT, calculating clock asymmetry from round-trip measurements. Validation against wall-clock serial logs confirmed the algorithm reached coherence—but it took approximately 2 minutes to converge to stable sub-millisecond sync due to BLE's jitter distribution.

ESP-NOW could work better:

- **Faster convergence:** Seconds instead of minutes (lower jitter means fewer samples needed)
- **Lower steady-state jitter:** $\pm 100\mu\text{s}$ vs $\pm 10\text{-}50\text{ms}$ for time synchronization
- **Power efficiency:** Truly connectionless operation means radio silence when nothing changes
- **No connection to drop:** Pattern continues even if sync beacon is missed

The therapeutic requirement (sub-40ms precision) was easily met by BLE after convergence. But since the hardware supported ESP-NOW at no additional cost, why not use the better tool?

1.4 The Generalization

Once we recognized that "pattern playback with time agreement" was the core primitive, applications beyond bilateral stimulation became obvious. Any scenario where multiple devices need to act in coordination—emergency lighting, swarm robotics, sensor arrays—fits the same pattern:

Devices don't need to talk to each other during operation. They need to agree on time and script.

1.5 The VLBI Precedent

The architecture documented here is not new in concept—it's a generalization of Very Long Baseline Interferometry (VLBI), which has operated at planetary scale since the 1960s.

VLBI creates a virtual telescope aperture spanning continents:

- **Distributed observers:** Radio telescopes thousands of kilometers apart
- **Precise timestamps:** Atomic clocks at each site, synchronized to nanoseconds
- **No real-time connection:** Observations recorded to tape/disk with timestamps
- **Correlation later:** Recordings shipped to central facility, correlated offline
- **Virtual aperture:** The "telescope" exists in the math, not in physical structure

This is exactly UTLP + RFIP + SMSP:

VLBI	This Architecture
Atomic clocks	UTLP time sync
Surveyed telescope positions	RFIP geometry
Observation schedule	SMSP score
Recorded observations	SMSP observations (bidirectional)
Offline correlation	Gateway/coordinator processing

The Event Horizon Telescope that imaged a black hole in 2019 is VLBI at its largest—a virtual aperture the size of Earth, assembled from telescopes that never directly communicated during observation. They agreed on time, knew their positions, followed a script, and correlated later.

What this architecture adds:

- **Scale down:** VLBI assumes expensive atomic clocks and surveyed positions. UTLP/RFIP achieve adequate precision with commodity hardware at room scale.
- **Bidirectional:** VLBI is receive-only (passive observation). SMSP supports coordinated transmission (actuation, beamforming).
- **Dynamic geometry:** VLBI assumes fixed telescope positions. RFIP enables mobile nodes with continuously updated positions.
- **Commodity hardware:** VLBI requires purpose-built radio astronomy equipment. This runs on \$5 microcontrollers.

The conceptual structure is identical. The contribution is recognizing that the same pattern applies from nanoscale MEMS arrays to planetary telescope networks—and implementing it on hardware cheap enough that a student can build the small end in a parking lot.

Potential Upstream Flow-Back:

While this architecture descends from VLBI, the generalizations developed here may inform next-generation VLBI systems:

Extension	Upstream Application
Bidirectional SMSP	Real-time observation quality feedback—stations report RFI levels, weather degradation, data quality; coordinator adapts scheduling without waiting for post-hoc correlation
Dynamic geometry (RFIP)	Space VLBI with orbital elements—continuous position updates for satellites in varying orbits, improving on orbit-determination-only approaches
Conductor model	Heterogeneous networks—mixing a few expensive anchor stations (atomic clocks) with many cheaper stations (GPS timing), coordinator managing different capability tiers
Query-driven sensing	Adaptive observation—"this baseline is producing garbage, redistribute integration time" rather than rigid pre-planned schedules
Commodity scaling	Low-cost VLBI pathfinders—amateur/educational networks that can do useful interferometry without \$M per station

The ngEHT (next-generation Event Horizon Telescope) planning explicitly discusses challenges this architecture addresses: coordinating more stations with varying capabilities, incorporating space elements with changing geometry, moving toward more real-time operation, and adaptive scheduling based on conditions.

This mirrors the emergency lighting approach documented in Section 6: learn from established professional systems (Whelen, Federal Signal's GPS-based sync), implement on commodity hardware (ESP32, BLE/ESP-NOW), demonstrate the architecture at accessible scale, and potentially contribute improvements back upstream. The same pattern of downstream-then-upstream applies—VLBI informs this architecture, this architecture may inform ngEHT; professional lightbar sync informs this architecture, this architecture may enable new emergency lighting form factors.

2. The Connectionless Architecture

2.1 Separation of Concerns

The architecture cleanly separates three phases:

Phase	Transport	Purpose	Timing Criticality
Bootstrap	BLE	Pairing, key exchange, WiFi MAC exchange	None
Configuration	BLE (then released)	Script upload, zone assignment, ESP-NOW key derivation	None
Execution	ESP-NOW only	Pattern playback, time sync beacons	Sub-millisecond

BLE Bootstrap Model: Peer BLE connection is released after key exchange completes. The operational phase uses ESP-NOW exclusively for peer-to-peer traffic. This provides deterministic timing (ESP-NOW $\pm 100\mu\text{s}$ vs BLE $\pm 10\text{-}50\text{ms}$ jitter) while preserving radio bandwidth. A PWA (phone app) can still connect via BLE for user interface—only the peer-to-peer link transitions to ESP-NOW.

During execution, devices do not coordinate. Each device:

1. Knows what time it is (via prior UTP synchronization)
2. Knows what script to play (preloaded in firmware or uploaded during configuration)
3. Knows its zone/role (assigned during configuration)
4. Executes locally with no network dependency

Critical distinction: Synchronization jitter vs. execution jitter. The $\pm 100\mu\text{s}$ figure cited for ESP-NOW describes *synchronization channel* jitter—variance in network round-trip times during the sync phase. Execution jitter is different and typically much tighter: once synchronized, actuation is driven by local hardware timers (`esp_timer` or direct peripheral timers), not by network events. The radio is not in the execution path. Execution jitter depends on timer resolution and ISR latency, not on RF or protocol stack behavior. On ESP32, hardware timer resolution is $1\mu\text{s}$; ISR latency is typically $< 10\mu\text{s}$ unless preempted by higher-priority interrupts (WiFi/BLE radio tasks). For timing-critical applications, actuation should use dedicated hardware timers with high-priority ISRs, not FreeRTOS task scheduling.

2.2 Script-Based Execution

A "script" is a deterministic sequence of timed events that both devices possess. The simplest script for bilateral stimulation (illustrative pseudocode—the reference implementation uses richer structures for pattern playback):

```
typedef struct {
    uint64_t start_time_us;      // When pattern begins (synchronized time)
    uint32_t period_us;          // Alternation period (e.g., 1,000,000 = 1Hz)
    uint8_t  duty_cycle_percent; // Active portion of each half-cycle
    uint8_t  zone_active_first;  // Which zone starts active (0 or 1)
} bilateral_script_t;
```

Given synchronized time and this script, each device independently calculates:

```
bool should_be_active(uint64_t now_us, bilateral_script_t* script, uint8_t my_zone) {
    uint64_t elapsed = now_us - script->start_time_us;
    uint64_t cycle_position = elapsed % script->period_us;
    uint8_t current_phase = (cycle_position < script->period_us / 2) ? 0 : 1;

    // XOR determines if this zone is active in this phase
    return (current_phase ^ my_zone) == script->zone_active_first;
}
```

No communication. No coordination. Just math on shared data.

2.3 Time is Defined, Not Measured

Once devices share a time reference, the question shifts from "when did you send this?" to "what should we both be doing right now?"

Early designs explored TDM (Time Division Multiplexing) scheduling, but this introduced artificial delays—devices waiting for their assigned slot even when the channel was clear. The final architecture abandons TDM in favor of event-driven execution against synchronized time.

WiFi (ESP-NOW) is prioritized over BLE for all peer-to-peer traffic, even on the server/master device. BLE's connection-oriented overhead and scheduling constraints create unnecessary latency. Once trust is established via BLE pairing, all operational communication moves to ESP-NOW's connectionless model.

The synchronization problem becomes a shared-clock problem, not a message-passing problem.

2.4 BLE for Bootstrap, ESP-NOW for Operations

The ESP-NOW protocol (Espressif's proprietary connectionless WiFi protocol) offers 10-100x lower latency jitter than BLE because it bypasses the connection-oriented stack entirely. A typical ESP-NOW packet arrives in $\sim 300\mu\text{s}$ with $\sim 100\mu\text{s}$ jitter, versus BLE's $\sim 3\text{ms}$ with $\sim 10\text{-}50\text{ms}$ jitter.

The architecture uses both:

1. **BLE** establishes trust (pairing, bonding, key exchange)
2. **ESP-NOW** carries operational traffic (sync beacons, monitoring)
3. **Shared key material** derived from BLE pairing secures ESP-NOW

Key derivation concept (illustrative—reference implementation wraps this in a transport API):

```
// Key derivation: BLE provides the secret, MACs provide binding
esp_err_t derive_espnow_key(
    const uint8_t nonce[8],           // Random, sent via encrypted BLE
    const uint8_t server_mac[6],      // WiFi MAC of server
    const uint8_t client_mac[6],      // WiFi MAC of client
    uint8_t lmk_out[16]               // ESP-NOW encryption key
) {
    uint8_t ikm[12];
    memcpy(ikm, server_mac, 6);
    memcpy(ikm + 6, client_mac, 6);

    return mbedtls_hkdf(
        mbedtls_md_info_from_type(MBEDTLS_MD_SHA256),
        nonce, 8,                      // Salt
        ikm, 12,                       // Key binding material
        (uint8_t*)"ESPNOW_LMK", 10,    // Domain separation
        lmk_out, 16
    );
}
```

Why HKDF, not PBKDF2/Argon2: The input is already high-entropy (64-bit hardware RNG nonce or 128-bit BLE LTK). Password-stretching algorithms solve the wrong problem—they're designed to slow brute-force attacks on weak human-memorable passwords. With 64+ bit entropy, brute force is already computationally infeasible. HKDF correctly extracts and expands high-entropy key material without unnecessary iterations or memory overhead.

Defense-in-depth architecture: Security is layered across physical (proximity requirement), transport (BLE SMP bonding, ESP-NOW CCMP), key derivation (HKDF with dual-MAC binding), and application layers (sequence numbers, optional TOTP). This provides threat-proportional security—appropriate cryptographic strength without over-engineering that would increase complexity and attack surface.

This gives the security properties of BLE pairing (encrypted key exchange, MITM protection) with the timing properties of ESP-NOW (low-jitter delivery). Peer BLE connection is released after key exchange—the operational phase uses ESP-NOW exclusively for peer traffic.

Reference implementation status: The BLE pairing phase (SMP with Numeric Comparison, MITM protection, LTK derivation) is demonstrated in `examples/smp_pairing/`. This validates the *bootstrap phase only*—secure key exchange and trust establishment. The full UTP time synchronization, RFIP ranging, and SMSP score execution engines are separate components; `smp_pairing` proves the security foundation they build upon, not the complete system. Critical discovery: NimBLE requires `ble_store_config_init()` before host sync for SMP to function—this is not documented in ESP-IDF or NimBLE references. Tagged as `examples/secure-smp-pairing` for discoverability.

2.5 Three Channels: Time, Command, Execution

A common misconception: "connectionless" doesn't mean "never communicates." It means **communication is not in the timing-critical path**. The architecture uses three distinct channels during operation:

Channel	Purpose	Transport	Encrypted?	Frequency
Time broadcast	Passive sync maintenance	ESP-NOW multicast	No	Periodic (configurable, e.g., 20 min)
Command	Script changes, wake, mode switch	ESP-NOW unicast	Yes	On-demand
Execution	Synchronized actuation	None (local only)	N/A	Continuous

Note on bootstrap vs. operation: Initial time synchronization happens during the BLE bootstrap phase (Section 2.4) via request/response—a joining node asks "what time is it?" and receives a sync handshake. The three-channel model above describes *operational* behavior after bootstrap completes. The time broadcast channel maintains sync; it doesn't establish it.

Time Broadcast (Unencrypted): Nodes periodically broadcast current time as a public utility—any device can listen and resync. This follows the WWVB model: you don't request atomic time, you receive it. Spoofing concerns are addressed by Common Mode Rejection (Section 4.2): if all nodes receive the same spoofed time, relative synchronization is preserved. The broadcast can use triple-burst transmission for flight time and jitter estimation:

Broadcast 1: T_1 (sender's time at transmission)

Broadcast 2: T_2 (sender's time at transmission)

Broadcast 3: T_3 (sender's time at transmission)

Receiver calculates:

- Inter-packet intervals validate sender clock
- Arrival time variance characterizes local jitter
- Median filtering rejects outliers

Time is public infrastructure. Encrypting it adds complexity without security benefit—the threat model accepts that time is observable.

Command Channel (Encrypted): When operational changes are needed, encrypted ESP-NOW delivers them:

- Wake from sleep
- Change pattern/mode
- Upload new script
- Request sensor data
- Shutdown command

Example: Police lightbar operation

1. Officer activates lights → **command:** "wake, run PURSUIT pattern"
2. Light bar executes pattern → **execution:** no network traffic
3. Pull-over complete → **command:** "switch to SCENE pattern"
4. Pattern changes → **execution:** again, no traffic
5. End of shift → **command:** "sleep"

Commands arrive, execution continues independently. The command channel can be arbitrarily slow without affecting actuation timing.

Execution (No Channel): Once a node has time + script + zone, it executes locally. This is the timing-critical phase, and it **involves no communication whatsoever**. A node could be completely RF-silent during execution—receiving time broadcasts is optional (for drift correction), and commands only arrive when something changes.

The insight: **separating the command plane from the execution plane** means slow, encrypted, reliable communication for control doesn't compromise fast, deterministic, local execution. This is analogous to control plane / data plane separation in networking, but for real-time actuation.

3. Validation: SAE J845 Quad Flash

3.1 The Standard

SAE J845 defines flash patterns for emergency vehicle warning lights, with strict timing requirements for visibility and seizure safety. This validation used a tri-color variant: red and blue channels alternating in antiphase (like opposing lightbar sections), plus a shared white channel flashing in-phase on both devices. If alternating signals overlap or synchronized signals desynchronize, the pattern fails.

3.2 The Test

Two ESP32-C6 devices running the connectionless architecture were configured for a dual-pattern validation:

Zone-assigned alternation (antiphase):

- Device A: RED channel
- Device B: BLUE channel
- Alternation: When A's red flashes, B's blue is dark; when B's blue flashes, A's red is dark

Shared synchronization (in-phase):

- Both devices: WHITE channel
- Both white LEDs flash together at a different frequency than the red/blue pattern

This dual-pattern test validates both capabilities simultaneously:

1. **Antiphase coordination:** Red and blue never overlap (bilateral stimulation requirement)
2. **In-phase coordination:** White LEDs fire together (swarm coherence requirement)

Validation method: 240fps slow-motion video capture (4.17ms per frame). At this frame rate, any timing error would be visible as white desynchronization or glitches during frequency transitions.

3.3 The Result

Zero frames showed white desynchronization. Zero visible glitches during frequency transitions.

Note: Red/blue "overlap" isn't a meaningful metric here—they're on separate devices and physically cannot overlap. Bilateral synchronization quality is validated separately using a 100% duty cycle alternating pattern where each device is illuminated for its entire phase; any timing error appears as simultaneous illumination or gaps.

The quad flash validation reveals something more subtle: **smooth step-boundary transitions**. When the pattern changes frequency mid-execution, both devices transition cleanly at step boundaries with no visible discontinuity. At 240fps (4.17ms per frame), frequency changes appear instantaneous.

This validates:

1. **Step-boundary architecture works:** Pattern changes execute at deterministic boundaries, not arbitrary moments
2. **Frequency transitions appear instant:** Mode changes in therapeutic applications (EMDR bilateral stimulation) can switch speeds without perceptible glitches
3. **In-phase coordination is precise:** White channel lock across devices confirms sub-frame synchronization
4. **Zone assignment works correctly:** Identical firmware, different runtime behavior based on role

4. Foundational Protocols

The connectionless architecture builds on two protocol specifications documented separately:

4.1 UTLP (Universal Time Lord Protocol)

UTLP provides synchronized time as a "broadcast environmental variable"—a public utility that any device can consume without pairing or authentication. Key features:

- **Stratum-based hierarchy:** Automatic source selection (GPS → FTM → ESP-NOW peer → free-running)
- **Holdover mode:** Graceful degradation during source loss using Kalman-filtered drift compensation
- **Transport agnostic:** Designed to work over BLE, ESP-NOW, 802.11, or acoustic channels
- **Glass Wall architecture:** Time stack (public) strictly separated from application stack (private)

Implementation note: In the reference implementation, UTLP runs over ESP-NOW after BLE bootstrap completes. BLE establishes trust and exchanges the Long-Term Key (LTK); all subsequent peer traffic—including UTLP sync beacons—uses ESP-NOW encrypted with keys derived from the LTK. This provides BLE-grade security with ESP-NOW's timing characteristics.

The core insight: when devices agree on time to $\pm 30\mu\text{s}$ precision, timestamps provide sufficient ordering granularity for any human-scale coordination.

4.1.1 Multi-Burst Beacon Timing: Time-Domain Interferometry

A critical implementation detail: UTLP sync beacons use **3 equally-spaced bursts** rather than a single transmission. This approach was discovered empirically ("more samples = more math") and independently rediscovers **seismic chirp signal processing**—the same technique used to characterize subsurface velocity models from swept-frequency acoustic pulses.

The derivative stack:

Sample	Measures	Seismic Analog	Mathematical Role
Burst 1 (t_0)	Offset	Position	Where am I?
Burst 2 (t_1)	Drift	Velocity	Where am I going? (1st derivative)

Sample	Measures	Seismic Analog	Mathematical Role
Burst 3 (t_2)	Stability	Acceleration	Is my "going" stable? (2nd derivative)

With a single sample, you get a scalar—one offset measurement contaminated by jitter. With 3 samples, you can fit a parabola: offset + drift rate + drift acceleration (thermal stability). You're not just *measuring* time—you're measuring **the curvature of your clock's relationship to truth**.

Why this matters:

- 1 Burst: A dot. Zero context. Jitter indistinguishable from offset.
- 3 Bursts: A curve. Trajectory of time. Jitter rejected as deviation from trend.

The 3-burst approach builds a **temporal phased array**—"focusing" the receiver on the valid time signal and rejecting transient noise (stack jitter), exactly like a seismic array rejects ground roll to resolve the oil deposit beneath.

Cross-domain validation: This technique is mathematically identical to:

- Seismic chirp surveys (swept frequency characterizes medium velocity)
- GPS carrier-phase tracking (multiple samples resolve integer ambiguity)
- Kalman filter initialization (3+ samples establish state and covariance)
- Polynomial curve fitting (N points determine N-1 degree polynomial)

The fact that this approach was discovered independently from first principles, then recognized as a known seismic technique, validates the architecture's cross-domain coherence. **The same math works because the same physics applies.**

4.2 RFIP (Reference-Frame Independent Positioning)

When UTLP is combined with 802.11mc FTM (Fine Time Measurement) ranging, devices can establish spatial relationships without external reference:

- **Intrinsic geometry:** "Where are we relative to each other?" not "Where are we on Earth?"
- **No infrastructure dependency:** Works in moving vehicles, underground, underwater, or in space
- **Swarm-emergent coordinates:** The coordinate system arises from the swarm itself

This enables applications where GPS is unavailable or meaningless: drone swarms in GPS-denied environments, rescue operations in collapsed structures, coordination on mobile platforms. Critically, the swarm can *build a map of where it has been* using only peer-derived coordinates—enabling systematic search patterns without any external positioning infrastructure.

Distributed IMU from ranging geometry: With 3+ nodes ranging to each other, the swarm gains 6-DOF orientation sensing without per-node IMUs:

- **Translation:** Swarm centroid moves
- **Rotation:** Inter-node angles change
- **Scale:** All distances shift proportionally

A per-node 6-axis IMU (e.g., Seeed XIAO MG24 Sense) adds ~\$5-8 per device. Ranging geometry provides equivalent swarm-level orientation for the cost of the ranging capability you already need. For budget-constrained projects—high school robotics teams, community safety initiatives—this tradeoff matters. The \$5 saved per node is another node in the swarm.

4.3 Autonomous Zone Assignment via Ranging

802.11mc ranging enables a capability beyond simple positioning: **autonomous zone assignment**. When devices can measure distances to each other, the swarm can self-organize without explicit configuration.

The problem with manual zone assignment:

- Pre-configured device zones require knowing which physical device is which
- Ordered assignment ("first to connect = zone 0") depends on connection timing, not spatial position
- Neither method produces spatially-meaningful zone topology

Ranging-based autonomous assignment:

1. Devices range to each other and/or to reference points
2. Relative positions computed via trilateration (RFIP)
3. Zone assignment derived from spatial position (e.g., leftmost = zone 0, rightmost = zone 1)
4. Topology emerges from geometry, not configuration

Example: Pile of drones → Flying swarm

A responder dumps a bag of identical drones at an incident scene. Without ranging:

- Drones must be pre-labeled with zone assignments, OR
- Zones assigned by activation order (spatially meaningless)

With 802.11mc ranging:

- Drones take off and establish relative positions
- Spatial topology computed (who's north, south, highest, lowest)
- Zone assignment follows spatial role (e.g., "northernmost drone = zone 0")
- Coherent warning pattern emerges from spatial reality

The swarm becomes spatially self-aware. Zone assignment becomes a property of where you *are*, not what you were *labeled*.

4.4 IMU-Augmented Positioning (Hardware Option)

RFIP using 802.11mc ranging provides position but not orientation. Adding a 6-axis IMU (accelerometer + gyroscope) enables:

- **Reflection ambiguity resolution:** RFIP's distance-only geometry has a mirror ambiguity—the swarm could be "flipped." Gravity vector from accelerometer defines "down," resolving which solution is correct.

- **Dead reckoning between ranging updates:** FTM exchanges take time. IMU provides continuous position estimates during gaps via inertial navigation.
- **Orientation awareness:** Ranging tells you *where* you are relative to peers. IMU tells you *which way you're facing*. Search patterns require both.
- **Motion state detection:** Ranging accuracy differs when stationary vs. moving. IMU provides immediate motion classification.

Hardware example: Adding a 6-axis IMU (e.g., MPU6050, \$2) to ESP32-C6 provides inertial sensing. Alternatively, integrated boards like Seeed XIAO MG24 Sense (\$15, Silicon Labs platform) include IMU on-board—though this requires porting UTLP from ESP-IDF. The architecture is platform-agnostic; the primitives work on any microcontroller with suitable RF capabilities.

Design tradeoff: For bilateral EMDR devices (stationary during use, only 2 nodes, known left/right assignment), IMU is unnecessary cost. For mobile swarms doing search patterns, IMU becomes valuable. The architecture supports both—IMU data feeds into the same RFIP coordinate system when available.

4.5 SMSP (Synchronized Multimodal Score Protocol)

While UTLP provides *when* and RFIP provides *where*, SMSP defines *what*: the format for describing synchronized actuator behavior across any number of channels and modalities.

4.5.1 The Three-Layer Architecture

SMSP separates human intent from machine execution:

DECLARATIVE LAYER (Human Intent) "Alternate left/right at 1Hz with 50% duty cycle" "SAE J845 Quad Flash pattern" "Binary star wobble with golden ratio orbit"
COMPILER LAYER (PWA / Configuration Tool) Transforms intent into timeline of discrete events Validates parameters, calculates phase offsets
IMPERATIVE LAYER (Score + Playback Engine) Sequence of: "At time T, set channel C to state S" Engine only knows: current time, current segment, outputs

This separation means:

- **Firmware stays simple:** The playback engine is "dumb"—it reads the timeline, interpolates between keyframes, sets outputs. No waveform math, no frequency calculation.
- **Complexity lives in the compiler:** The PWA or configuration tool can be updated without touching firmware.
- **Advanced users can bypass:** Raw timelines can be authored directly for patterns the compiler doesn't anticipate.

4.5.2 The Score Line Primitive

The fundamental unit is a **score line**: a specification of actuator state at a point in time.

```
typedef struct {
    uint32_t time_offset_ms;    // When (relative to score start or previous segment)
    uint16_t transition_ms;    // Interpolation duration to reach this state
    uint8_t  flags;            // EASE_IN, EASE_OUT, SYNC_POINT, etc.
    uint8_t  waveform;         // CONSTANT, SINE, RAMP, PULSE (for audio/haptic)

    // Per-channel state (example: bilateral device)
    uint8_t  L_r, L_g, L_b;    // Left LED RGB (0-255)
    uint8_t  L_brightness;    // Left LED master brightness
    uint8_t  L_motor;         // Left haptic intensity
    uint8_t  R_r, R_g, R_b;    // Right LED RGB
    uint8_t  R_brightness;    // Right LED master brightness
    uint8_t  R_motor;         // Right haptic intensity

    // Audio channels (if present)
    uint16_t L_audio_freq_hz;  // Left audio frequency (synthesized, not sampled)
    uint8_t  L_audio_amplitude; // Left audio amplitude
    uint16_t R_audio_freq_hz;  // Right audio frequency
    uint8_t  R_audio_amplitude; // Right audio amplitude
} score_line_t;

typedef struct {
    uint8_t  line_count;
    uint8_t  loop_point;    // Which line to jump to on completion (0xFF = stop)
    uint8_t  pattern_class; // BILATERAL, EMERGENCY, SWARM, CUSTOM
    score_line_t lines[];
} score_t;
```

The structure above is illustrative; implementations may optimize for their specific channel configuration. The key properties are:

1. **Time-indexed**: Each line specifies *when*, not *what frequency*—frequency is implicit in line spacing
2. **Transition-aware**: Interpolation duration is explicit, enabling smooth crossfades
3. **Modality-agnostic**: LED, haptic, audio are parallel channels in the same timeline
4. **Zone-agnostic**: Score describes one node's behavior; zone assignment determines which score (or score variant) each node plays

4.5.3 Pattern Classification

Scores carry metadata indicating their pattern class:

```
typedef enum {
    PATTERN_BILATERAL,    // Antiphase pair, therapeutic applications
    PATTERN_EMERGENCY,    // SAE J845 compliant, warning applications
}
```

```

    PATTERN_SWARM_SYNC,    // In-phase coherence, mutual visibility
    PATTERN_PURSUIT,       // Sequential activation around geometry
    PATTERN_BINARY_ORBIT,  // Wobble + rotation composite
    PATTERN_CUSTOM         // Raw timeline, no assumptions
} pattern_class_t;

```

Classification enables:

- UI hints ("this is a bilateral pattern, show left/right preview")
- Validation ("emergency patterns must meet SAE J845 timing")
- Optimization ("bilateral patterns can use simpler zone logic")

4.5.4 Scale Invariance

The same score format applies regardless of physical scale:

Scale	Zones Are	Transport	Example
PCB	GPIO pins	Direct register write	RGB LEDs on one board
Device	Peer MAC addresses	ESP-NOW	Bilateral handhelds
Room	Node positions	ESP-NOW / WiFi	Warning light array
Field	Drone IDs	ESP-NOW / custom RF	Search and rescue swarm

A score written for three LEDs on a PCB plays identically on three drones 100 meters apart. The playback engine doesn't know or care about physical spacing—it only knows time and channels.

4.5.5 Transport Agnosticism

SMSP defines the score format, not the delivery mechanism:

- **ESP-NOW**: Peer-to-peer wireless, used during configuration phase
- **BLE**: Alternative transport for PWA-to-device score upload
- **Wired bus**: UART/I2C/SPI for conductor-to-node in fixed installations
- **Flash at build time**: Score baked into firmware for dedicated-function devices

The protocol is complete when a node possesses:

1. A score (however delivered)
2. A time reference (however synchronized via UTLP)
3. Knowledge of its zone (however assigned, potentially via RFIP ranging)

4.5.6 Minimal Engine Requirements

The playback engine requires only:

```

while (running) {
    uint32_t now = get_synchronized_time_ms();

    if (now >= current_segment->time_offset_ms + transition_end) {
        advance_to_next_segment();
    }

    float progress = calculate_eased_progress(now, current_segment);

    for (each channel) {
        output[channel] = interpolate(
            previous_state[channel],
            current_segment->state[channel],
            progress
        );
    }

    apply_outputs();
}

```

This runs on an 8-bit microcontroller. An ATtiny85 (\$0.50) can execute SMSP scores; the ESP32-C6 (\$5) is only required for wireless bootstrap and UTP time synchronization. For wired or pre-configured deployments, a "smart" conductor node handles sync while "dumb" nodes only play scores.

4.5.7 Relationship to Existing Standards

Standard	What It Does	SMSP Difference
DMX512	512 channels, wired, master broadcasts continuously	SMSP: score uploaded once, nodes execute independently
MIDI	Note events, primarily musical timing	SMSP: continuous interpolated states, sub-millisecond sync
SMPTE	Timecode for film sync, devices chase master	SMSP: devices agree on time then execute autonomously
OSC	Real-time control messages over network	SMSP: score is complete before execution, no runtime traffic
Art-Net	DMX over Ethernet, still continuous broadcast	SMSP: connectionless during execution

SMSP combines:

- DMX's channel abstraction
- MIDI's event timing
- SMPTE's frame accuracy
- OSC's flexibility

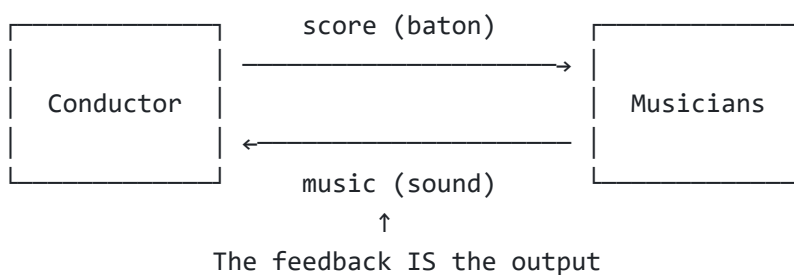
...while eliminating:

- Continuous network traffic during execution
- Central controller as single point of failure
- Wired infrastructure requirements
- Per-node cost barriers (runs on \$0.50 MCU)

4.5.8 The Bidirectional Model: Conductor and Orchestra

SMSP as described above is one-directional: scores flow from conductor to nodes, nodes execute. But real orchestras don't work that way—the conductor *hears the music* and provides real-time corrections. SMSP extends naturally to bidirectional operation where observations flow back.

The Orchestra Metaphor:



In an orchestra:

- The conductor sends instructions (tempo, dynamics, cues)
- The musicians execute (play their parts)
- The music itself is the feedback (conductor hears what's happening)
- Corrections are targeted ("cellos, you're dragging"—a sharp gesture toward that section)

This maps directly to distributed sensing:

Orchestra	Swarm
Conductor	Coordinator node / gateway
Musicians	Sensor/actuator nodes
Score	SMSP instruction stream
Music	Observations flowing back
"You're dragging"	UTLP sync correction packet
Exaggerated downbeat	Re-broadcast sync beacon
Stop and reset	Halt pattern, re-sync, restart

The Observation Format:

Just as the score has a line format, observations have a symmetric format:

```
// Score line: what the conductor tells nodes to do
typedef struct {
    uint64_t timestamp;           // When to do it (UTLP time)
    uint16_t target_node;        // Who should do it (0xFFFF = all)
    uint8_t  action;             // What to do
    uint8_t  payload[];          // Action-specific parameters
} smsp_instruction_t;

// Observation: what nodes report back
typedef struct {
    uint64_t timestamp;           // When it happened (UTLP time)
    uint16_t source_node;        // Who observed it
    uint8_t  observation_type;    // What kind of observation
    uint8_t  payload[];          // Observation data
} smsp_observation_t;

// They're structurally identical. Direction determines semantics.
```

Observation Types:

```
typedef enum {
    OBS_HEARTBEAT,           // "I'm alive, clock offset is X"
    OBS_ACK,                 // "I executed instruction N"
    OBS_SAMPLE,              // "At time T, sensor read value V"
    OBS_EVENT,               // "At time T, threshold crossed"
    OBS_ERROR,               // "Instruction N failed, reason R"
    OBS_SYNC_QUALITY,        // "My sync jitter is X, drift is Y"
    OBS_POSITION,            // "RFIP says I'm at X,Y,Z"
} observation_type_t;
```

Action Types (extending score lines):

```
typedef enum {
    // Actuation (original SMSP)
    ACTION_SET_OUTPUT,       // Set GPIO/PWM to value
    ACTION_PLAY_SEGMENT,     // Execute score segment

    // Sensing (bidirectional extension)
    ACTION_SAMPLE,           // Acquire sensor reading, timestamp it
    ACTION_SAMPLE_BURST,     // Acquire N samples at interval I
    ACTION_ARM_TRIGGER,      // Watch for threshold, report when crossed

    // Reporting
    ACTION_REPORT_NOW,       // Send your latest observation
    ACTION_REPORT_RANGE,     // Send observations from T1 to T2

    // Correction (conductor's sharp gestures)
    ACTION_SYNC_CORRECT,     // Adjust your clock by offset X
    ACTION_HALT,             // Stop current pattern
}
```

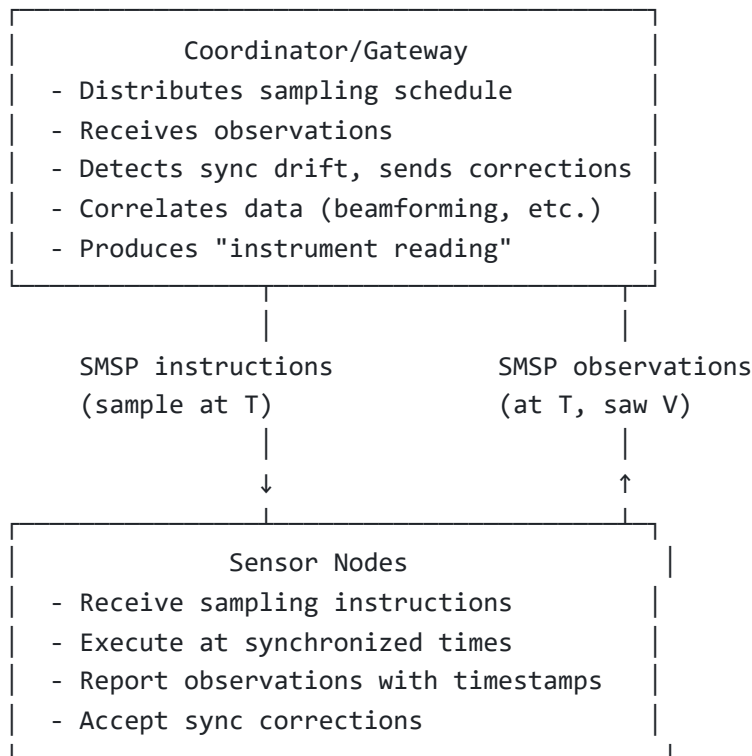
```

ACTION_RESET,          // Clear state, await new score
} action_type_t;

```

Closed-Loop Operation:

For sensing applications, SMSP becomes a closed loop:



The "Spastic Jerk" Correction:

When a conductor notices a section dragging, they make an exaggerated, targeted gesture. The SMSP equivalent:

```

// Coordinator notices node 7 is consistently 450µs late
smssp_instruction_t correction = {
    .timestamp = NOW,
    .target_node = 7,
    .action = ACTION_SYNC_CORRECT,
    .payload = {
        .offset_adjust_us = -450,
        .confidence = HIGH    // "I'm sure about this"
    }
};

// Node 7 receives this as "the conductor is glaring at me"
// and adjusts its local offset

```

This is finer-grained than UTLP's broadcast sync beacons—it's a targeted correction to a specific node that's drifting.

Transport Agnosticism (Preserved):

The bidirectional extension maintains transport agnosticism:

Direction	ESP-NOW	LoRa	WiFi	Serial
Instructions → nodes	Broadcast/unicast	Broadcast	UDP multicast	Point-to-point
Observations ← nodes	Unicast to coordinator	Unicast to gateway	UDP to server	Collected by host

The protocol defines the format and semantics. Transport is deployment-dependent.

Levels of Access:

Different users need different views of the swarm:

Level	Query	Response
Operator	"Where's the tornado?"	"Bearing 270°, 50km, moving NE"
Analyst	"Show confidence distribution"	Probability density plot
Researcher	"Raw samples, nodes 7-12, T=0 to T=500ms"	Timestamped sample arrays
Debug	"Node 7 clock state"	Offset, drift rate, sync quality

The observation stream supports all levels. High-level queries aggregate observations; low-level queries return them directly.

Swarm as Instrument:

With bidirectional SMSP, the distributed array becomes a single instrument:

Traditional sensor network:

Each node → reports data → central database → query → answer
(Always streaming, high bandwidth, central dependency)

SMSP-based instrument:

Query arrives → becomes SMSP sampling schedule → nodes execute
→ observations flow back → correlation produces answer
(On-demand, efficient, degradation-tolerant)

The swarm doesn't stream data constantly. It responds to queries by executing coordinated measurement and reporting results. The same architecture that coordinates bilateral stimulation now coordinates distributed sensing.

4.5.9 The Protocol Family

UTLP, RFIP, and SMSP together form a complete primitive for distributed synchronized operation:

Protocol	Question Answered	Direction
UTLP	When is it?	Broadcast (time source → all)
RFIP	Where am I?	Peer-to-peer (mutual ranging)
SMSP	What do I do? / What did I see?	Bidirectional (instructions ↔ observations)

Any node with synchronized time (UTLP), known position (RFIP), and a score (SMSP) can participate in coordinated behavior. The bidirectional extension means the same node can both actuate and sense, and the coordinator can correct drift without waiting for the next UTLT sync cycle.

5. Applications: The Primitive Enables Many Things

The connectionless timing architecture is not a product—it's a **distributed real-time primitive**. The EMDR device was simply the forcing function that demanded solving the hard parts. Once solved, the primitive instantiates across domains.

5.1 Medical and Therapeutic

Bilateral Stimulation Devices (EMDR, tDCS)

- Handheld haptic/visual alternating stimulation
- Multi-electrode transcranial stimulation with precise phase relationships
- Binaural audio generation across separate speakers

Coordinated Wearable Sensing

- Multi-point biosignal acquisition (ECG leads, EEG arrays) with synchronized sampling
- Distributed pulse oximetry for perfusion mapping
- Synchronized accelerometry for gait analysis across body segments

Rehabilitation Systems

- Bilateral motor training devices with precisely timed feedback
- Synchronized haptic cues for Parkinson's gait freezing intervention
- Multi-limb coordination training with phase-locked stimulation

5.2 Emergency and Safety

Vehicle Warning Light Synchronization

- Fleet-coherent flash patterns without GPS dependency
- Incident-scene "light walls" from multiple vehicles
- Motorcycle/bicycle conspicuity systems synced to nearby emergency vehicles

Aerial Warning Swarms

- Drone-carried warning lights above traffic incidents
- Elevated pattern visibility over terrain/vehicle obstructions
- Self-organizing formation with RFIP-derived zone assignment

Search and Rescue in GPS-Denied Environments

- Collapsed structures, underground, underwater, or remote wilderness
- Swarm builds its own spatial map using RFIP peer ranging
- Searched areas tracked via swarm-local coordinates—no external reference needed
- Pattern coverage emerges from spatial awareness: "I'm north of you, I'll search north"
- Works where GPS fails: rubble attenuation, canyon walls, dense canopy, caves

Swarm Form Factors: Coordinated node networks are not limited to mobile robots. The architecture covers three instantiation categories:

Form Factor	Locomotion	Platform	Example
Wearable swarm	Person provides	Body-mounted	Bilateral EMDR pulsers, rescue worker trackers
Fixed swarm	None	Stationary	Warning lights on poles, building evacuation beacons
Mobile swarm	Self-propelled	Autonomous	Aerial drones, ground robots, autonomous vehicles

All three share the same timing and coordination architecture—UTLP synchronization, pattern scripts, peer ranging. Locomotion is simply another actuator channel: some nodes have it, some don't.

Terminology note: The distinction is *agency*, not form factor. A *drone* is an independent agent acting on your behalf. A *wearable* extends the person wearing it—the person remains the agent. Both participate identically in swarm coordination.

The architecture requires only: radio and time sync participation. Nodes may be stationary, worn, or mobile.

Building Evacuation Systems

- Synchronized directional lighting across floors
- Wave patterns indicating egress direction
- No wired infrastructure required—battery-powered nodes

Maritime and Aviation

- Distributed navigation markers without shore infrastructure
- Runway/taxiway lighting for austere airfields
- Search pattern coordination for distributed assets

5.3 Entertainment and Art

Distributed Light Shows

- Audience-carried LED devices synchronized to stage performance
- Architectural lighting across multiple buildings
- No DMX wiring—wireless with frame-accurate timing

Silent Disco Evolution

- Multiple music streams with synchronized lighting across all receivers
- Crowd-wide visual effects responding to the mix
- Zone-based experiences within single venue

Kinetic Sculptures

- Synchronized mechanical elements across physical space
- No control wiring between sculpture segments
- Battery-operated with seasonal deployment

Theatrical Effects

- Wireless pyrotechnic timing (with appropriate safety systems)
- Coordinated fog/haze release across stage
- Actor-carried props with synchronized effects

5.4 Industrial and Agricultural

Distributed Sensor Arrays

- Synchronized sampling for sensor fusion
- Seismic/acoustic arrays without wired timing bus
- Air quality monitoring with coordinated measurement windows

Agricultural Automation

- Coordinated spraying across multiple drones/vehicles
- Synchronized irrigation pulse patterns
- Pollination timing coordination for indoor farms

Construction and Surveying

- Synchronized laser scanning from multiple stations
- Coordinated vibration monitoring during blasting
- Multi-point settlement monitoring with time-aligned measurements

Mining and Underground

- GPS-denied environment timing for equipment coordination
- Ventilation control with synchronized damper operation

- Refuge chamber status synchronization

5.5 Research and Scientific

Distributed Physics Experiments

- Multi-point detector timing without dedicated timing infrastructure
- Balloon-borne instrument arrays with synchronized acquisition
- Underwater acoustic arrays for marine research

Environmental Monitoring

- Wildlife tracking with synchronized beacon detection
- Distributed weather sensing with coordinated measurements
- Volcanic/seismic monitoring in remote locations

Astronomy and Space

- Ground-based interferometry timing layer
- CubeSat swarm coordination
- Lunar/planetary surface operations (RFIP particularly relevant)

5.6 Consumer and Lifestyle

Gaming and Social

- Multiplayer physical game props with synchronized effects
- Escape room puzzle coordination without wired systems
- Social dancing apps with group-synchronized haptic cues

Fitness and Sports

- Team training systems with synchronized timing cues
- Interval training coordination across group classes
- Race timing systems for informal events

Home Automation

- Circadian lighting synchronized across rooms
- Multi-speaker audio with sub-millisecond alignment
- Holiday lighting coordination across properties

5.7 The Meta-Pattern

All these applications share the same structure:

CONFIGURATION PHASE
- Establish trust (pairing, authentication)

- Synchronize time reference - Distribute script/pattern - Assign zones/roles
EXECUTION PHASE - No coordination traffic - Local calculation: script[zone][t] - Independent actuation - Coherent group behavior emerges

The primitive doesn't care whether it's vibrating a therapy device, flashing a warning light, triggering a pyrotechnic, or sampling a sensor. It only cares that distributed nodes agree on time and plan.

5.8 Distributed Wave Beamforming

The Physics: Beamforming is not about creating more energy—it redistributes energy spatially. By altering the timing (phase) of multiple emitters, waves arrive at a target point in sync (constructive interference) while arriving out of sync elsewhere (destructive interference). The "beam" is a region of reinforcement.

The Timing Relationship: Beamforming requires phase coherence—emitters synchronized to a fraction of the wave period. The fraction determines beam quality; typically <10% of period for useful steering.

Domain	Wavelength	Period	Required Sync	ESP32 Capable?
Acoustic 1kHz	34 cm	1 ms	~100 μs	✓
Ultrasonic 40kHz	8.5 mm	25 μs	~2.5 μs	Marginal
RF 2.4GHz	12.5 cm	0.4 ns	~40 ps	✗
Optical 600THz	500 nm	1.7 fs	~170 as	✗

Honest Phase Error Assessment: The "Required Sync" column above targets <10% of period for useful beam steering. But what do actual achievable jitter figures mean for beam quality?

Jitter Source	Typical Value	At 1kHz (1ms period)	Beam Impact
Sync channel (ESP-NOW)	~100μs	36° phase error	Degraded main lobe, elevated sidelobes; usable for rough directionality
Sync channel (BLE)	~10-50ms	3600-18000°	Unusable for beamforming
Execution (hardware timer)	~1-10μs	0.36-3.6° phase error	Good beam quality; suitable for steering

Jitter Source	Typical Value	At 1kHz (1ms period)	Beam Impact
Execution (RTOS task)	~100-1000µs	36-360° phase error	Poor; avoid for beamforming

The critical insight: Sync jitter and execution jitter are different. A 100µs sync error means nodes disagree about what time it is by ~100µs—this is a *constant offset* once sync converges, not per-cycle jitter. Execution jitter is the variance in *when the actuator fires* relative to the intended time. For beamforming, execution must use hardware timers (ESP32 `esp_timer` or peripheral timers), not RTOS `vTaskDelay` . With hardware timers, execution jitter of <10µs yields <3.6° phase error at 1kHz—sufficient for coherent beam formation.

What this enables: Acoustic beamforming at 1kHz with hardware-timer execution is valid for directional audio, alert systems, and architectural validation of the distributed control logic. It is NOT sufficient for precision phased array applications requiring <1° phase accuracy. The document claims this work validates the *architecture*, not that ESP32 achieves radar-grade precision.

The Architecture Is Domain-Invariant: UTLP/RFIP/SMSP implement the coordination pattern. The achievable precision depends on execution hardware. ESP32 with hardware timers achieves ~1-10µs execution jitter → valid for kHz-range acoustic. FPGA/SDR achieving ~10-100ps execution jitter → valid for GHz-range RF. **The protocol doesn't change; the clock hardware determines the applicable domain.**

RFIP Feeds the Math: Beam steering requires knowing inter-node distances. For a linear array steering to angle θ with node spacing d :

```

delay_n = (n × d × sin(θ)) / v

Where:
n = node index (0, 1, 2...)
d = inter-node spacing (from RFIP ranging)
θ = target steering angle
v = wave velocity (343 m/s for sound, 3×108 m/s for RF)

```

RFIP's peer ranging provides d directly. No pre-surveyed array geometry required—the swarm measures itself.

SMSP Carries the Phase Offsets: Beam steering compiles to per-node time delays in the score:

```

// PWA/Compiler calculates offsets for 45° steering, 10cm spacing
// delay_n = (n × 0.10m × sin(45°)) / 343 m/s
// Node 0: 0 µs, Node 1: 206 µs, Node 2: 412 µs, Node 3: 618 µs

score_line_t beam_45deg[] = {
    { .zone = 0, .time_offset_ms = 0,   .L_audio_freq_hz = 1000 },
    { .zone = 1, .time_offset_ms = 0,   .start_delay_us = 206, .L_audio_freq_hz = 1000 },
    { .zone = 2, .time_offset_ms = 0,   .start_delay_us = 412, .L_audio_freq_hz = 1000 }
};

```

```
{ .zone = 3, .time_offset_ms = 0, .start_delay_us = 618, .L_audio_freq_hz = 1000
};
```

The nodes execute independently. The beam emerges from synchronized phase relationships.

Dynamic Steering: Changing beam direction requires only a new score. The execution model is unchanged—nodes receive updated phase offsets, execute locally, new beam direction emerges. No architectural changes for scanning, tracking, or multi-beam patterns.

Enclosure Effects as Radome Simulation: In the reference implementation, speakers are mounted inside plastic enclosures (EMDR handles). The enclosure modifies acoustic response—internal reflections, phase distortion, directivity changes. This is not a limitation; it's a **high-fidelity simulation of radome interaction** in real RF systems. Putting radar inside an aircraft nose cone creates identical phenomena: antenna detuning, boresight error, sidelobe distortion. The acoustic prototype exercises the same compensation algorithms needed for aerospace deployment.

Application Domains (architecture identical, timing hardware varies):

Application	Domain	Hardware	Status
Directional alerts	Acoustic	ESP32 + speaker	Achievable now
Parametric audio	Ultrasonic	ESP32 + transducer array	Marginal now
Sonar/echolocation	Ultrasonic	ESP32 + transducer	Marginal now
Directed comms	RF	FPGA/SDR	Architecture ready
Synthetic aperture	RF	FPGA/SDR	Architecture ready
Jamming/nulling	RF	FPGA/SDR	Architecture ready

The Validation Path: Acoustic beamforming with ESP32 proves the distributed control logic at human-observable timescales. The architecture is identical for RF—only the timing hardware changes. Successfully steering a 1kHz acoustic beam validates that the UTLP/RFIP/SMSP stack can steer a 2.4GHz RF beam when instantiated on appropriate silicon.

5.9 Dynamic Aperture Beamforming (Time-Varying Geometry)

The beamforming architecture extends naturally to arrays where node positions change over time—a "breathing" or "waving" aperture. Rather than a limitation, dynamic geometry becomes a feature when RFIP continuously tracks actual positions.

The Rigid Array Limitation: Traditional phased arrays use electronic phase shifters that delay signals by fractions of a wavelength. This works for single-frequency continuous wave transmission but causes **beam squint** for wideband or pulsed signals—different frequencies steer to different angles, smearing the beam.

True Time Delay via Physical Displacement: When array elements physically move, all frequencies experience identical delay (distance / speed of propagation). A 1cm physical displacement delays RF and acoustic signals equally across all frequency components. For pulsed energy applications (HPM weapons, impulse radar, EMP), physical displacement enables perfect pulse focusing that electronic phasing cannot achieve.

Dynamic Array Advantages:

Property	Rigid Array	Dynamic Array
Bandwidth	Limited (beam squint)	Wideband (true time delay)
Sidelobe structure	Fixed (exploitable)	Time-averaged (smeared, harder to exploit)
Null locations	Fixed (jamming vulnerability)	Moving (resilient to static jammers)
Synthetic aperture	Requires platform motion	Inherent from element motion
Failure mode	Geometry collapse	Graceful degradation
Countermeasure resistance	Predictable	Non-reciprocal (array state changes between TX and RX)

Mechanical Wave as Beam Scanner: A traveling mechanical wave across the array physically tilts the effective aperture, sweeping the beam without electronic phase control. The scan rate equals the mechanical wave velocity divided by array length. With UTLP synchronization, the mechanical wave phase is deterministic—beam position at any instant is calculable.

Non-Reciprocal Transmission: As the array deforms, element velocities create Doppler shifts that encode transmission angle into signal frequency. More importantly, the array geometry at transmission time differs from geometry when a countermeasure signal returns. The reciprocal path no longer exists—the array has moved. This provides inherent jamming resistance without cryptographic complexity.

RFIP Enables Dynamic Beamforming: Without real-time geometry knowledge, a deforming array produces incoherent noise. With RFIP tracking actual positions and UTLP ensuring time alignment, deformation becomes a controllable parameter. The phase offset calculation simply updates continuously:

```
// Static array (calculate once)
phase_offset[n] = (n × d × sin(θ)) / λ

// Dynamic array (continuous update)
phase_offset[n](t) = (position[n](t) · target_vector) / λ
// where position[n](t) comes from RFIP
```

See Appendix B.1 for complete enabling pseudocode implementing mechanical wave phase calculation.

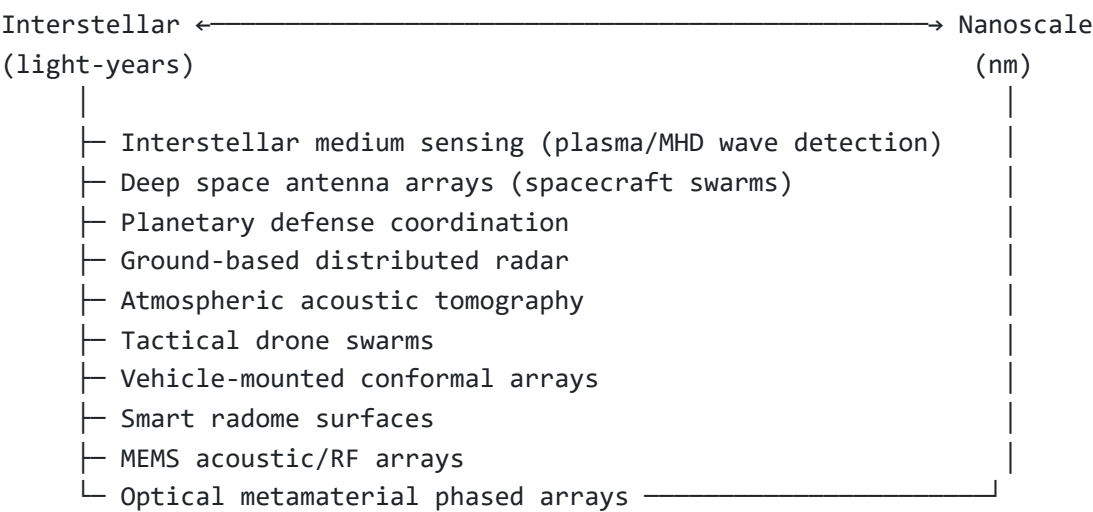
Physical Implementations Across Scale:

Scale	Implementation	Geometry Sensing	Application
Planetary (AU)	Spacecraft constellation	Light-time ranging	Deep space arrays
Field (km)	Drone swarm	RFIP peer ranging	Distributed radar/comms
Array (m)	Cargo net with nodes	RFIP peer ranging	Tactical beamforming
Panel (cm)	Tensioned membrane	Encoders, strain gauges	Vehicle-mounted arrays
Radome (mm)	Piezoelectric surface	MEMS position sensing	Aircraft/missile seekers
MEMS (μm)	Micromirror array	Capacitive sensing	Integrated photonics
Nano (nm)	Metamaterial elements	Designed response	Optical phased arrays

Scale Invariance to Integrated Devices: The dynamic aperture architecture does not require physically separate nodes. A single integrated device with a mechanically actuated surface—piezoelectric membrane, MEMS mirror array, tensioned mesh—implements identical principles:

- "Swarm" generalizes to "distributed actuation points"
- RFIP generalizes to "geometry sensing" (encoders, strain gauges, capacitive sensing)
- UTLP reduces to a shared clock (trivial on a single PCB)
- SMSP becomes embedded firmware controlling surface shape

The Architecture Spans All Scales:



Same architecture: time sync + geometry knowledge + coordinated actuation
Only the implementation scale changes.

The Interstellar Extension: Beyond the atmosphere, the interstellar medium is not vacuum but extremely thin plasma ($\sim 1 \text{ atom/cm}^3$). This supports plasma waves and magnetohydrodynamic (MHD) oscillations—Voyager 1 detected these crossing the heliopause. A constellation of spacecraft with UTPP-synchronized clocks and RFIP-known positions could perform coherent detection of interstellar medium phenomena. The "acoustic" sensing becomes electromagnetic field sensing (magnetometers, electric field probes), but the coordination architecture is identical. The wave physics changes; the distributed timing problem doesn't.

Active Radome Applications: A mechanically actuated radome surface provides:

- Real-time correction for radome-induced beam distortion
- Additional beam steering capability beyond the antenna
- Conformal integration with vehicle surfaces
- Reduced RCS through dynamic surface shaping

Space-Time Modulated Metasurfaces: The architecture describes what the metamaterials community calls "space-time modulated metasurfaces"—but implemented via distributed coordination rather than centralized control. The time modulation (mechanical wave) combined with spatial distribution (node positions) creates effects impossible with static arrays:

- Frequency conversion (Doppler from motion)
- Non-reciprocal propagation (geometry changes between TX and RX)
- Wideband operation (true time delay)
- Adaptive nulling (sidelobes move)

5.9.1 Research Validation: Time-Varying Metasurfaces in Nature Journals (2021-2025)

The concepts described above—time-varying surfaces with cryptographically large configuration spaces—are actively being validated at the highest levels of research. The following peer-reviewed publications demonstrate that these claims are grounded in demonstrated physics, not speculation:

Radar Invisibility via Doppler Cancellation (Nature Scientific Reports, July 2021): Researchers demonstrated a metasurface cloak that temporally modulates reflected phase to cancel Doppler signatures. An aircraft coated with such material appears *stationary* to radar—its motion signature matches ground clutter and is filtered out. The metasurface achieves broadband invisibility against wideband radar systems without absorbing or deflecting the signal.

Reference: "Broadband radar invisibility with time-dependent metasurfaces," Scientific Reports 11, 14011 (2021)

Anti-Multi-Static Radar via Space-Time Coding (Nature Communications, August 2025): A space-time-coding metasurface (STCM) demonstrated the ability to defeat multi-static radar networks. By dynamically modulating the surface in both space and time, different receivers observe different harmonic frequencies. Conventional multi-static localization algorithms—which assume consistent target signatures across receivers—fail completely. Validated with outdoor drone flight experiments.

Reference: "Anti-radar based on metasurface," Nature Communications 16, Article 62633 (2025)

Chaotic Metasurface for Keyless Secure Communication (Nature Communications, July 2025): This paper directly validates the cryptographically-large configuration space concept. A metasurface driven by chaotic patterns achieves physical-layer security *without shared encryption keys*:

- Legitimate receiver (at correct spatial position) receives clear signal
- All other observers receive chaotically scrambled noise
- Scrambling is position-dependent and never repeats
- No decryption operation required—security emerges from physics

The chaos-driven modulation creates a configuration space that is:

- Effectively infinite (chaotic sequences don't repeat)
- Unpredictable (sensitive to initial conditions)
- Position-dependent (different observers see different patterns)
- Computationally irreversible (can't reconstruct signal from noise)

Reference: "Chaotic information metasurface for direct physical-layer secure communication," Nature Communications 16, 5853 (2025)

Time-Varying Metasurface Radar Jamming (Optica Express, May 2024): Demonstrated a time-varying metasurface-driven radar jamming and deception system (TVM-RJD) that achieves broadband jamming without a separate transmitter—the surface itself creates deceptive returns by modulating reflections. Energy-efficient and integrable.

Reference: "Time-varying metasurface driven broadband radar jamming and deceptions," Optics Express 32(10), 17911 (2024)

Acoustic Metasurfaces for Selective Sound Delivery (Nature Communications Physics, November 2025): An active acoustic metasurface using time-reversal processing achieves selective sound delivery in reverberant environments—clear audio to target listeners, suppressed audio elsewhere. Demonstrates that the same principles (programmable phase control, real-time reconfiguration) apply across electromagnetic and acoustic domains.

Reference: "Reconfigurable and active time-reversal metasurface turns walls into sound routers," Communications Physics 8, Article 2351 (2025)

Active Selective Attenuation: Beyond Passive Geometry

The following cross-domain research establishes the distinction between *passive* frequency-selective geometry (Faraday cages, fixed FSS) and *active* selective attenuation (sense-coordinate-cancel):

Active Noise Cancellation Foundations (Paul Lueg, 1936): The foundational patent for phase-inverted cancellation—US 2,043,416 (1936)—established that unwanted sound can be attenuated by generating an anti-phase signal. Lueg's principle applies to any wave phenomenon: acoustic, electromagnetic, or mechanical. The limitation: single speaker canceling single source at a single location. Modern ANC headphones and highway noise barriers extend this to multi-speaker systems but remain centralized (one controller, wired connections).

Reference: Lueg, P. "Process of silencing sound oscillations," US Patent 2,043,416 (1936)

Spatially Selective Active Noise Control (JASA, May 2023): Demonstrated that multi-channel ANC can achieve *direction-selective* attenuation—blocking sound from unwanted directions while passing sound from desired directions through the same physical aperture. This is impossible with passive geometry (a wall blocks everything). The key insight: by imposing spatial constraints on the ANC cost function, selectivity becomes algorithmic rather than geometric.

Reference: "Spatially selective active noise control systems," *Journal of the Acoustical Society of America* 153(5), 2733 (2023)

Active Frequency Selective Surfaces (Cambridge IJMW Review, 2023): Comprehensive review of Active FSS (AFSS) showing evolution from passive geometry-determined filtering to active PIN/varactor/MEMS-reconfigured filtering. Key distinction: even "active" FSS require *geometric* reconfiguration (switching element states, mechanical deformation). The dynamic macroscopic lattice goes further—same geometry, different filtering based on sensed input and coordinated cancellation.

Reference: "Active frequency selective surfaces: a systematic review for sub-6 GHz band," *Int. J. Microwave and Wireless Technologies* (2023)

Energy Selective Surfaces for Adaptive Shielding (PMC, 2025): Documents the evolution toward adaptive EM protection: bandwidth expansion, tunable thresholds, and sensing-triggered response. Validates the threat-powered activation concept (Claim 115)—surfaces that wake from incoming energy rather than internal timers.

Reference: "Development of Energy-Selective Surface for Electromagnetic Protection," *PMC* (2025)

The Fundamental Distinction:

Approach	Selectivity Determined By	Can Change Response Without Geometry Change?
Faraday cage	Mesh spacing	No
Fixed FSS	Resonator geometry	No
Reconfigurable FSS	Switched/deformed geometry	No (requires physical state change)
Dynamic macroscopic lattice	Coordination algorithm	Yes—same geometry, different response

The dynamic macroscopic lattice is to passive shielding what the transistor is to the relay: same function (switching), fundamentally different mechanism (amplification vs mechanical contact). Same physical structure producing different behavior based on input—the definition of active response.

Property	Electromagnetic (RF)	Acoustic	Validated?
Time-varying surface configuration	✓ Nature Comms 2025	✓ Comms Physics 2025	✓
Cryptographically large state space	✓ Chaos metasurface	✓ Reconfigurable holography	✓

Property	Electromagnetic (RF)	Acoustic	Validated?
Position-dependent response	✓ Directional information modulation	✓ Selective sound delivery	✓
Anti-characterization (unpredictable)	✓ Anti-radar STCM	✓ (implied by chaos)	✓
Real-time reconfiguration	✓ FPGA-controlled	✓ FPGA-controlled	✓
No shared keys needed	✓ Chaotic metasurface	✓ (physical layer)	✓
Active selective attenuation	✓ AFSS (Cambridge 2023)	✓ ANC (Lueg 1936, JASA 2023)	✓
Direction-selective filtering	✓ ESS (PMC 2025)	✓ Spatially selective ANC	✓

What This Architecture Adds Beyond Current Research:

The published research focuses on individual metasurfaces with centralized control (single FPGA controlling all elements). The UTLP/RFIP/SMSP architecture extends this to:

Current Research	This Architecture Enables
Single surface, fixed position	Distributed surfaces, coordinated across platforms
Centralized control (one FPGA)	Connectionless coordination (swarm of controllers)
Static geometry	Dynamic geometry (surfaces on mobile platforms, RFIP-updated)
Receive OR transmit optimized	Bidirectional SMSP (coordinated sensing AND emission)
Laboratory demonstrations	Scalable from handheld to planetary

A swarm of time-varying metasurfaces, each controlled by local SMSP execution, synchronized via UTLP, with geometry continuously updated via RFIP, creates capabilities impossible with single-surface implementations:

- Spatially distributed aperture synthesis (VLBI-style correlation)
- Coherent multi-platform jamming/communication
- Self-healing arrays (nodes fail, swarm adapts)
- Mobile conformal surfaces with real-time phase correction

5.9.2 Deformable Virtual Metasurfaces: Geometry as Primary Control Variable

The preceding discussion focuses on *phase control*—nodes at known positions applying coordinated timing to achieve wavefront manipulation. But the swarm architecture enables something more fundamental: **the surface geometry itself becomes a primary control variable.**

Terminology Evolution: From "Virtual Metasurface" to "Dynamic Macroscopic Lattice"

The term "metasurface" carries fabrication baggage—it implies thin engineered layers on substrates, hence "meta-SURFACE." This document uses "virtual metasurface" for continuity with existing literature and patent searchability, but a more accurate term exists: **dynamic macroscopic lattice**.

This terminology maps directly to solid-state physics:

Solid-State Physics	Atomic Crystal	Dynamic Macroscopic Lattice
Structure	Atoms fixed in lattice	Nodes distributed in space
Scale	Angstroms (10^{-10} m)	Meters (10^0 m)
Reconfigurability	None (frozen geometry)	Complete (nodes relocate)
Band gaps	Fixed by atomic spacing	Programmable by node spacing
Bragg reflection	$\lambda/2$ atomic spacing $\rightarrow$ mirror	$\lambda/2$ node spacing $\rightarrow$ mirror
Refraction	Density slows wave	True time delay tilts wavefront
Material properties	Determined by structure	Determined by coordination

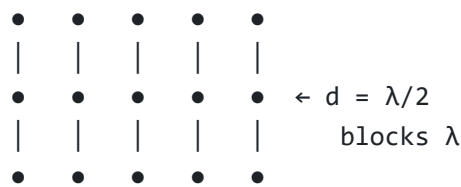
The Physics Parallel:

In solid-state physics, material properties (transparency, conductivity, color) emerge from the fixed geometry of atoms. A photonic crystal blocks specific wavelengths because its atomic spacing creates Bragg reflection—waves at $\lambda/2$ spacing interfere destructively and cannot propagate (the "band gap").

The UTLP/RFIP/SMSP architecture creates the same physics at macro scale:

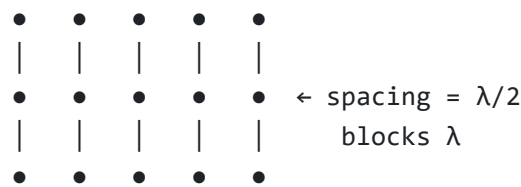
BRAGG REFLECTION IN ATOMIC VS. MACROSCOPIC LATTICE

Atomic Crystal (fixed):



Spacing fixed at fabrication.
Band gap permanent.

Dynamic Macroscopic Lattice:



Spacing programmable at runtime.
Band gap adjustable.

Why "Dynamic Macroscopic Lattice" Is More Accurate:

"Virtual Metamaterial"	"Dynamic Macroscopic Lattice"
Implies fabrication heritage	Implies physics heritage

"Virtual Metamaterial"	"Dynamic Macroscopic Lattice"
"Meta" = engineered beyond nature	"Lattice" = fundamental structure
Suggests 2D (surface)	Naturally 3D (volumetric)
Unique to metamaterials community	Bridges solid-state + distributed systems
Marketing connotation	Physics connotation

The Key Insight: You are not simulating physics—you are *instantiating* it. A diamond's refractive index cannot change; a dynamic macroscopic lattice can transition from transparent to mirror to lens in a single clock cycle by updating `phase_offset` .

Domain Invariance via Wavelength Scaling:

The wave equation is domain-agnostic. The only difference between RF and acoustic is scale:

Domain	Frequency	Wavelength	Node Spacing for $\lambda/2$
UHF Radio	300 MHz	1 m	0.5 m
Mid-range Audio	343 Hz	1 m	0.5 m
Infrasound	1 Hz	343 m	171.5 m
Seismic	0.1 Hz	3,430 m	1,715 m

The same node geometry that creates a band gap for 300 MHz RF *simultaneously* creates a band gap for 343 Hz acoustic. The lattice operates on both domains at once if equipped with appropriate transducers.

Sensor Type Determines Domain, Spacing Determines Utility:

A critical clarification: the **sensor/transducer type** determines which domain a node listens to or acts upon, while the **node spacing** determines whether the geometry is useful for that domain's wavelengths.

Component	Determines	Example
Antenna	Senses/emits RF	Node responds to radar
Microphone	Senses acoustic	Node responds to sound
Geophone	Senses seismic	Node responds to ground motion
Accelerometer	Senses vibration	Node responds to structural movement
All of the above	Multimodal sensing	Node responds to multiple domains simultaneously

The same physical node, at the same location, can participate in *multiple* virtual apertures by having multiple sensor types. A node with both an antenna and a geophone simultaneously contributes to an RF array and a seismic array—different domains, same coordination architecture.

This is intentional cross-domain generalization, not conflation: RF, acoustic, and seismic are distinct physical phenomena with different propagation characteristics, but the *coordination mathematics* (timing, geometry, interference) applies identically. The wave equation doesn't care what medium it describes.

Phononic Crystal Precedent:

The phononic crystal literature validates this parallel. Researchers have demonstrated:

- Programmable band gaps via magnetic/pneumatic deformation of physical lattices
- Tunable wave filtering through geometry changes
- Bragg scattering and local resonance effects in periodic structures

What they achieve through physical deformation of fabricated structures, the dynamic macroscopic lattice achieves through node repositioning and phase coordination—with unlimited reconfigurability.

Key references (phononic crystals):

- Acta Mechanica Solida Sinica (2022): "Tunability of Band Gaps of Programmable Hard-Magnetic Soft Material Phononic Crystals"
- ScienceDirect (2023): "Magnetic-controlled programmable soft lattice phononic crystals with sinusoidally-shaped-like ligaments for band gap control"
- Int. J. Mechanical Sciences (2022): "Pneumatic soft phononic crystals with tunable band gap"
- Nature (1995): Martinez-Sala et al., "Sound attenuation by sculpture" — foundational phononic crystal demonstration

Terminology Usage in This Document:

- **"Virtual metasurface"**: Used when connecting to existing literature, patent searches, or comparing to physical metasurfaces
- **"Dynamic macroscopic lattice"**: Used when describing the physics of what the architecture actually creates
- **"Emergent aperture"**: Used when emphasizing that the capability exists inherently in any synchronized node collection

All three terms describe the same physical phenomenon from different perspectives.

The Research Landscape:

Current metasurface approaches fall into distinct categories, each with inherent limitations:

Approach	Mechanism	Limitation
Fixed + phase control	PIN diodes, varactors, FPGA control phase at fixed positions	Geometry cannot change
Mechanically reconfigurable	MEMS, kirigami, stretchable substrates	Constrained to substrate deformation limits

Approach	Mechanism	Limitation
Drone/satellite swarm arrays	Nodes relocate, form virtual aperture	Position treated as <i>error to compensate</i> , not exploit
Active acoustic	Tunable Helmholtz resonators, membrane tension	Fixed physical structure

Key references:

- Nature Reviews Materials (2025): "Shape-morphing metamaterials" — establishes geometry as design variable
- Nature Communications (2025): "Abnormal beam steering with kirigami reconfigurable metasurfaces" — synchronous lattice + phase control
- MDPI Drones (2023): "Distributed Antenna in Drone Swarms: A Feasibility Study" — virtual aperture, position as challenge
- NASA/JPL (2019): "Distributed Swarm Antenna Arrays for Deep Space Applications" — CubeSat virtual aperture
- Frontiers Materials (2023): "Tunable, reconfigurable, programmable acoustic metasurfaces: A review"

What Existing Approaches Miss:

Drone swarm antenna research (Plymouth Rock Technologies 2022, JPL 2019, Huazhong 2025) demonstrates that spatially distributed nodes can form coherent apertures. But these systems treat *node position as an error source*—something to measure, track, and compensate for. The goal is making a distributed system behave like a rigid phased array.

The UTLP/RFIP/SMSP architecture inverts this relationship: **position is not an error to minimize but a control variable to exploit.**

Three Independent Control Axes:

DYNAMIC MACROSCOPIC LATTICE: THREE CONTROL AXES
CONTROL AXIS 1: Phase/Timing (SMSP) - Electronic phase shifts via actuation timing - Fast (microseconds), continuous - What traditional phased arrays do: $e^{i\phi}$
CONTROL AXIS 2: Physical Position (RFIP-tracked) - Nodes physically relocate in 3D space - Medium speed (seconds to minutes) - What existing swarm arrays compensate for, we exploit - Creates actual path length changes: Δr
CONTROL AXIS 3: Local Density (swarm topology)

- Contract/expand local regions
- Variable sampling density across aperture
- Adaptive resolution: concentrate nodes in regions of interest
- No fixed substrate constrains topology

$$\text{Combined control: } e^{i\phi} \times \Delta r \times \rho(x,y,z)$$

$\uparrow$
phase

$\uparrow$
position

$\uparrow$
density

Physical vs. Virtual Metasurface Capabilities:

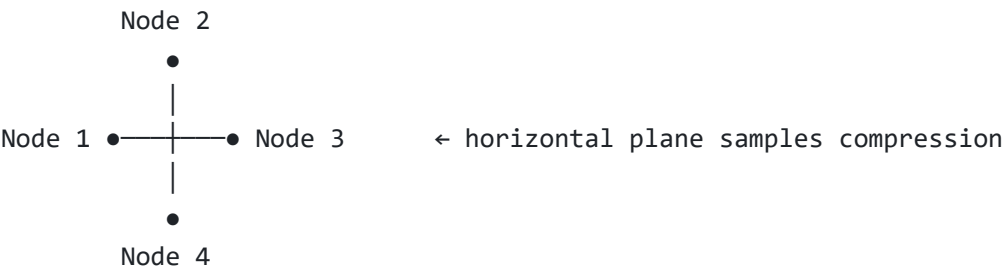
Capability	Physical Metasurface	Deformable Virtual Metasurface
Phase control	✓ Electronic	✓ SMSP timing
Geometry change	Limited (substrate constraints)	Unlimited (nodes freely mobile)
Density variation	Fixed (fabricated pattern)	Programmable (swarm topology)
Topology change	None (fixed connectivity)	Complete reconfiguration
Surface curvature	None or fixed	Dynamic (bulge, dimple, fold)
Multi-physics	Single wave type	Simultaneous longitudinal + transverse

Wave Orientation and Deployment Geometry:

A critical insight connects deployment geometry to wave physics:

Longitudinal waves (acoustic): Oscillation parallel to propagation direction (compression/rarefaction). A *flat horizontal array* samples the wave pattern optimally—exactly what the Lab Manual teaches for parking lot deployment.

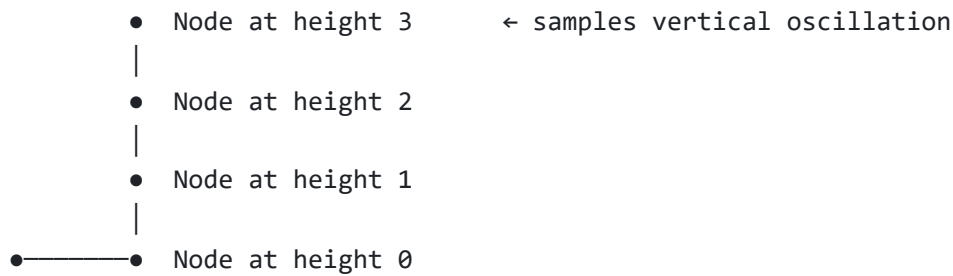
FLAT ARRAY FOR LONGITUDINAL WAVES



Incoming acoustic wave → → → (compression/rarefaction in propagation direction)
 Horizontal spacing samples the wave pattern ✓

Transverse waves (electromagnetic): Oscillation perpendicular to propagation direction. For vertically polarized EM waves, nodes at the *same height see identical oscillation phase*—the interesting information is in the *vertical* dimension.

VERTICAL ARRAY FOR TRANSVERSE WAVES



Incoming EM wave → → → with vertical polarization ↑
Vertical spacing samples the transverse oscillation ✓

This explains why the cargo net / wall swarm concept is *required* for electromagnetic applications—it's not convenience, it's physics. A flat parking lot array cannot sample vertically-polarized transverse waves.

Deformable Surface Capabilities Beyond Fixed Metasurfaces:

Because the swarm has no physical substrate:

1. **Dynamic Focusing:** Change aperture curvature to adjust focal distance in real-time

- A physical dish has fixed focus
- A swarm can reshape from parabolic to flat to hyperbolic

2. **Adaptive Resolution:** Concentrate nodes in regions of interest

- Incoming signal from unknown direction? Spread out for coverage
- Source located? Contract toward source for resolution
- Multiple sources? Split swarm into sub-apertures

3. **Simultaneous Multi-Physics:** Same swarm handles both wave types

- Acoustic sensing (longitudinal) via microphones
- RF manipulation (transverse) via antenna elements
- Different hardware, same UTLP/RFIP/SMSP coordination

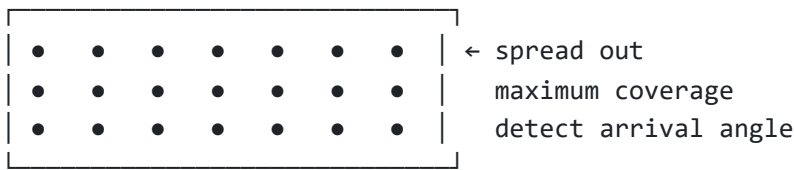
4. **Topological Reconfiguration:** Not just deformation but complete restructuring

- Single large aperture → multiple small apertures
- Planar surface → volumetric distribution
- Flat array → conformal to arbitrary shape

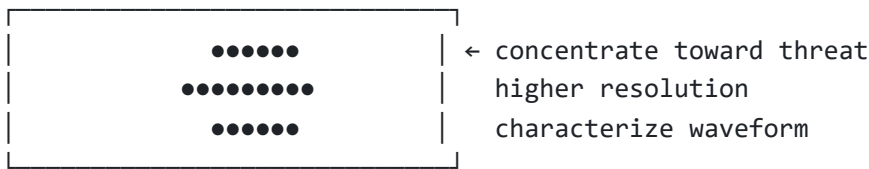
Application: Adaptive Radar Surface:

SCENARIO: Unknown radar threat

Phase 1 - Detection (distributed coverage):



Phase 2 - Characterization (adaptive concentration):



Phase 3 - Response (shaped reflection):



Phase 4 - Redistribution:

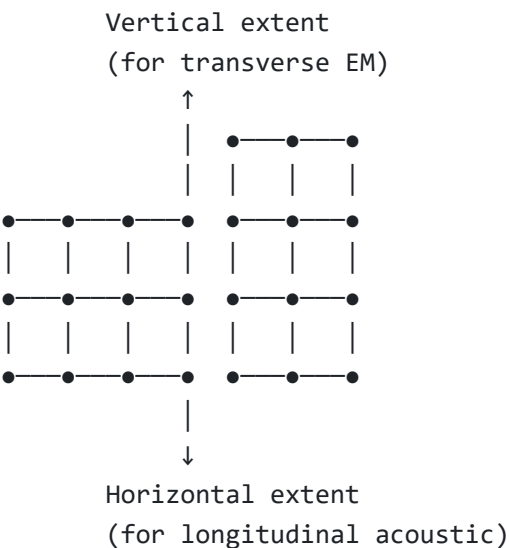
Return to Phase 1 geometry for continued surveillance

A fixed metasurface—even one with phase control—cannot do Phase 2 or Phase 3. The geometry is fabricated, not commanded.

Dual-Mode Metasurface (Longitudinal + Transverse):

The same swarm infrastructure, with appropriate sensor/actuator hardware, handles both wave physics simultaneously:

SWARM WITH DUAL-MODE CAPABILITY



Capabilities:

- LONGITUDINAL (acoustic):
- Horizontal array samples compression waves
 - Parking lot deployment
 - Atmospheric sensing
- TRANSVERSE (EM):
- Vertical extent samples polarization
 - Wall/net deployment
 - Radar applications
- COMBINED:

- 3D volumetric coverage
- Multi-physics sensing
- Simultaneous modes

Research Precedent—What We Extend:

Research	What They Demonstrated	What We Add
Kirigami metasurface (Nature Comms 2025)	Lattice + phase control synchronized	Nodes fully mobile, no substrate
Drone swarm arrays (MDPI 2023)	Position tracking for compensation	Position as control variable
Satellite swarm (NASA/JPL 2019)	Virtual aperture, coherent combination	Topology as design variable
Shape-morphing metamaterials (Nature Rev 2025)	Geometry affects properties	Combined with connectionless coordination
Acoustic metasurfaces (Frontiers 2023)	Programmable acoustic response	Cross-domain same architecture

The convergence of these research threads—shape-morphing materials, swarm antenna arrays, programmable metasurfaces—has not yet produced a system that combines:

- Fully mobile nodes (not substrate-constrained)
- Position as primary control variable (not error source)
- Density as independent control axis
- Connectionless coordination (not centralized control)
- Cross-domain applicability (same architecture for EM and acoustic)

The UTLP/RFIP/SMSP architecture provides exactly this combination—a **dynamic macroscopic lattice** with full programmability.

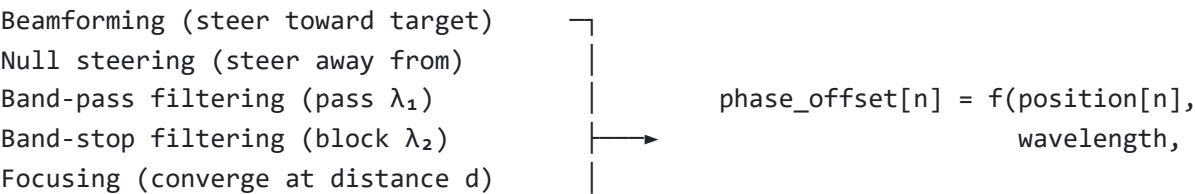
Unified Principle: All Wavefront Manipulation Is Interference Pattern Coordination

Every "capability" of a dynamic macroscopic lattice—beamforming, null steering, filtering, focusing, scattering, cloaking—reduces to a single operation: **coordinating interference patterns across the lattice**.

THE INTERFERENCE PATTERN UNIFICATION

All of these "different capabilities":

Are actually this one operation:



desired_pattern)

Defocusing (diverge from point)

Scattering (randomize reflection)

"Invisibility" (cancel in direction)

|

|

└─

Same math. Different parameters.

What changes between modes:

Mode	Interference Pattern	Parameter Change
Beam toward target	Constructive at θ	<code>target_direction = θ</code>
Null toward target	Destructive at θ	<code>target_direction = θ, invert = true</code>
Pass frequency f_1	Constructive at λ_1	<code>wavelength = c/f_1</code>
Block frequency f_2	Destructive at λ_2	<code>wavelength = c/f_2, invert = true</code>
Focus at distance d	Spherical wavefront	<code>focal_distance = d</code>
Scatter	Random phase	<code>pattern = random</code>

The implications:

1. **No separate "filtering" invention:** Band-pass/band-stop filtering is beamforming with `invert=true`
2. **No separate "cloaking" invention:** Invisibility is null steering toward the observer
3. **No separate "focusing" invention:** It's beamforming with curved wavefront target
4. **All applications are parameterizations** of the same interference coordination

This means a single prior art disclosure (this document) covers the entire space of wavefront manipulation applications. You cannot patent "band-stop filtering via distributed nodes" separately from "beamforming via distributed nodes"—they are the same mechanism with different target parameters.

The only variables are:

- Node positions (geometry)
- Target direction/location (where to create constructive/destructive interference)
- Wavelength (what frequency to affect)
- Pattern type (constructive, destructive, shaped, random)

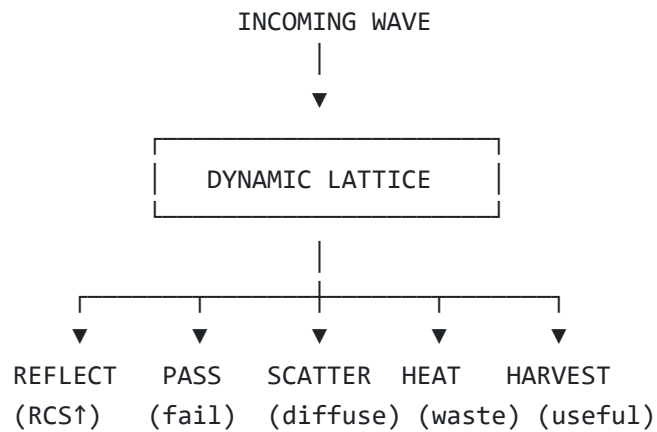
Everything else is derived.

Energy Harvesting: The Physics of Perfect Absorption

Conservation of energy creates an inevitable consequence: if a dynamic macroscopic lattice acts as a band-stop filter (blocking a frequency), the incoming wave energy must go somewhere.

ENERGY CONSERVATION IN WAVE BLOCKING

Incoming wave energy has exactly four destinations:



A PERFECT ABSORBER that doesn't reflect, doesn't pass, and doesn't heat up MUST be harvesting.

This is thermodynamics, not invention.

Harvesting Mechanisms by Domain:

Domain	Blocking Mechanism	Harvesting Mechanism	Hardware
RF	Destructive interference	Rectenna (antenna + rectifier)	Schottky diode
Acoustic	Pressure cancellation	Piezoelectric transducer	PZT crystal
Vibration	Mechanical damping	Piezo/electromagnetic	Voice coil as generator

The Macro Atom Analogy (Extended):

The solid-state physics parallel extends to energy absorption:

Real Atom	Macro Atom (Node)
Absorbs photon	Absorbs RF pulse / acoustic pressure
Electron jumps to higher energy state	Rectifier converts AC → DC
Stores energy in electron orbital	Stores energy in supercapacitor
Can re-emit (fluorescence)	Can re-transmit (active cancellation)
Absorption spectrum = material property	Absorption spectrum = lattice geometry

Regenerative Shielding:

This transforms the operational model:

Traditional Shielding	Regenerative Shielding
Expend energy to block	Harvests energy from blocking
Battery drains during use	Battery charges during use
Stronger threat = faster depletion	Stronger threat = faster charging
Passive defense	"Jujitsu" defense

The Strategic Implications:

1. **Threat-powered wake:** Sleeping swarm wakes when incoming energy (radar ping, acoustic blast) provides the activation power
2. **Self-sustaining defense:** The act of blocking partially powers the blocker
3. **Adversarial incentive inversion:** "The harder you jam us, the longer we last"

Power Scaling: From Nerf Darts to Directed Energy

The physics described here is scale-invariant. The architecture applies identically whether implemented at:

Scale	Power Level	Example Application	Budget Class
Hobbyist	mW	Acoustic demonstration, stops nerf darts	\$100
Industrial	W	Noise barriers harvesting highway sound	\$10K
Military tactical	kW	Vehicle-mounted jamming/harvesting array	\$1M
Strategic	MW	Ship-based directed energy defense	\$100M+

What changes with budget:

- Number of nodes
- Power handling per node
- Frequency range (RF requires faster switching)
- Aperture size (determines directivity and power density)

What does NOT change:

- The coordination architecture (UTLP/RFIP/SMSP)
- The physics (interference, Bragg reflection, band gaps)
- The energy conservation requirement (blocked energy must go somewhere)
- The regenerative principle (threat powers defense)

A hobbyist implementation that stops nerf darts and a Navy railgun-budget implementation that defeats incoming missiles are **the same architecture at different power levels**. This document establishes prior art across the entire range. The limiting factor is engineering budget, not physics.

High-Power Specific Considerations:

At MW power levels, additional physics becomes relevant:

- **Nonlinear effects:** High-intensity waves create harmonics, requiring broader band gap coverage
- **Thermal management:** Harvested energy may exceed storage capacity, requiring active dissipation
- **Plasma effects:** Extreme RF power can ionize air, changing propagation characteristics
- **Ablation:** Physical nodes may be damaged by the energy they're blocking

These are engineering challenges, not architectural changes. The coordination protocol remains identical; only the node hardware scales.

Why This Cannot Be Patented Separately:

Energy harvesting from blocked waves is not a separate invention—it is a **physical requirement** of effective absorption. Any claim to "harvesting energy from waves blocked by distributed nodes" would fail because:

1. **Conservation of energy** demands the energy go somewhere
2. **Effective absorption** (low reflection, low transmission) requires energy capture
3. **The harvesting hardware** (rectenna, piezo) is commodity technology
4. **The combination** is obvious once you recognize lattice-as-absorber

A patent on "regenerative shielding" would be like patenting "getting wet while swimming"—the physics makes it inevitable.

5.9.3 Emergent Virtual Apertures: The Physics That Already Exists

Foundational Principle: Creating a virtual metasurface is not an invention—it is an inevitable consequence of physics. Any collection of synchronized, position-known nodes *already constitutes* a virtual aperture. The wave response exists whether anyone exploits it or not.

This is not a claim of invention. It is recognition of what physics guarantees.

Aperture as Epistemology:

The term "aperture" derives from Latin *aperire*: "to open." An aperture is not a device—it is an *opening through which information passes*.

Every aperture—physical or virtual—performs the same fundamental operation: **correlating sparse samples across space and/or time to synthesize understanding unavailable from any single sample**.

Domain	"Nodes"	"Aperture"	"Image" Synthesized
Radio astronomy	Telescopes	VLBI baseline	Resolved celestial source
Acoustic sensing	Microphones	Distributed array	Sound field / direction
Historiography	Artifacts, texts, ruins	Scholarly correlation	Historical narrative

Domain	"Nodes"	"Aperture"	"Image" Synthesized
Human vision	Retinal photoreceptors	Visual cortex integration	Conscious sight
Criminal investigation	Witness statements	Cross-examination	Reconstructed event
Scientific method	Experiments	Meta-analysis	Validated theory

The historian correlating Mayan codices, Spanish colonial records, and archaeological stratigraphy is performing *exactly* the same mathematical operation as the VLBI correlator combining signals from telescopes on different continents. Different domains. **Identical principle.** Both synthesize apertures from sparse samples to resolve what no single sample could reveal.

Consider: the Event Horizon Telescope imaged a black hole not by building a planet-sized dish, but by correlating sparse samples from telescopes scattered across Earth. The "aperture" existed the moment those telescopes had synchronized clocks and known positions. The image emerged from correlation.

History works identically. The past is not directly observable—only sparse samples survive (artifacts, texts, oral traditions). Historians correlate these samples, applying timestamps (stratigraphy, carbon dating, textual analysis) and position knowledge (provenance, context) to synthesize images of events no living person witnessed. **History is a dynamic aperture:** what we resolve depends on which samples survive, how we correlate them, and what interference patterns we choose to constructively combine.

This universality is precisely why apertures cannot be invented—only instantiated:

- You cannot patent "correlating distributed samples to gain resolution"
- You cannot patent "using multiple perspectives to synthesize understanding"
- You cannot patent the operation that underlies radio astronomy, historiography, criminal justice, scientific method, and human cognition simultaneously

A dynamic macroscopic lattice is not an invention of a new aperture. It is architecture for *instantiating* apertures that physics already permits—the same way a historian doesn't *invent* the past but instantiates a view of it through correlation of surviving evidence.

Patent Law Implications (35 U.S.C. 101 and 102):

Under 35 U.S.C. 101, laws of nature, natural phenomena, and abstract ideas are not patentable subject matter. The emergence of a virtual aperture from distributed nodes is arguably a **natural phenomenon**—it occurs whenever the physical conditions are met, regardless of human intent or awareness. This document argues that you cannot patent the fact that a collection of synchronized nodes has wave-response properties any more than you can patent the fact that water freezes at 0°C.

Under 35 U.S.C. 102 (anticipation), an invention is anticipated if all elements were present in the prior art. Every existing distributed network with synchronized time and known positions—Amazon Sidewalk, Starlink, smart meter grids, weather station networks—**already embodies** a virtual aperture at appropriate wavelengths. These apertures exist today, whether exploited or not. This constitutes **inherent anticipation**: the aperture was always there; we are merely the first to document it explicitly.

The implications (as argued):

- **The aperture itself should not be patentable** (natural phenomenon, 35 U.S.C. 101)
- **The aperture in existing networks should not be patentable** (inherent anticipation, 35 U.S.C. 102)
- **Methods to exploit the aperture are prior art** (this document, 35 U.S.C. 102)

THE EMERGENT APERTURE PRINCIPLE



Any distributed nodes with:

- Known positions
- Synchronized time
- Sensors or actuators

...ARE a virtual aperture.

The aperture exists. The question is: at what wavelengths is it useful, and who exploits it?

The physics doesn't wait for permission or intent.

The Wavelength-Spacing Relationship:

A virtual aperture's utility depends on the ratio between node spacing and target wavelength. The same physical deployment creates *different* virtual apertures at different frequencies:

Wave Domain	Typical Frequency	Wavelength (λ)	50m Spacing =	Aperture Quality
WiFi	2.4 GHz	12 cm	400λ	Unusable (extreme grating lobes)
Sidewalk RF	900 MHz	33 cm	150λ	Unusable (severe aliasing)
Audible speech	1 kHz	34 cm	150λ	Unusable
Low audio	100 Hz	3.4 m	15λ	Sparse but detectable
Infrasound	1 Hz	340 m	0.15λ	Well-sampled
Microseism	0.1 Hz	3.4 km	0.015λ	Oversampled
Seismic	0.01 Hz	34 km	0.0015λ	Massively oversampled

The same 50-meter node spacing is:

- Useless for its designed RF purpose as a coherent aperture
- Excellent for infrasound sensing
- Superb for seismic detection

Case Study: Amazon Sidewalk as Accidental Sensing Array

Amazon Sidewalk connects Echo devices, Ring cameras, and other hardware into a mesh network for IoT communication. Typical deployment characteristics:

- **Node spacing:** 10-100m in residential areas
- **Time synchronization:** NTP-derived (millisecond-scale)
- **Position knowledge:** Installation addresses (meter-scale accuracy)
- **Sensor hardware:** Microphones present (for voice), not characterized for infrasound

This network was designed for packet routing. But physics doesn't care about design intent:

AMAZON SIDEWALK: DESIGNED PURPOSE vs. EMERGENT CAPABILITY

DESIGNED (RF mesh at 900 MHz):

- $\lambda = 33 \text{ cm}$
- Node spacing = 50m = 150λ
- Result: Not a coherent aperture (too sparse)
- Status: Works fine for packet routing (doesn't need coherence)

EMERGENT (Infrasound sensing at 1 Hz):

- $\lambda = 340 \text{ m}$
- Node spacing = 50m = 0.15λ
- Result: EXCELLENT coherent aperture (well-sampled)
- Status: Completely unexploited

EMERGENT (Seismic sensing at 0.1 Hz):

- $\lambda = 3.4 \text{ km}$
- Node spacing = 50m = 0.015λ
- Result: SUPERB aperture (massively oversampled)
- Status: Completely unexploited

The infrasound/seismic aperture EXISTS. Amazon just doesn't use it.

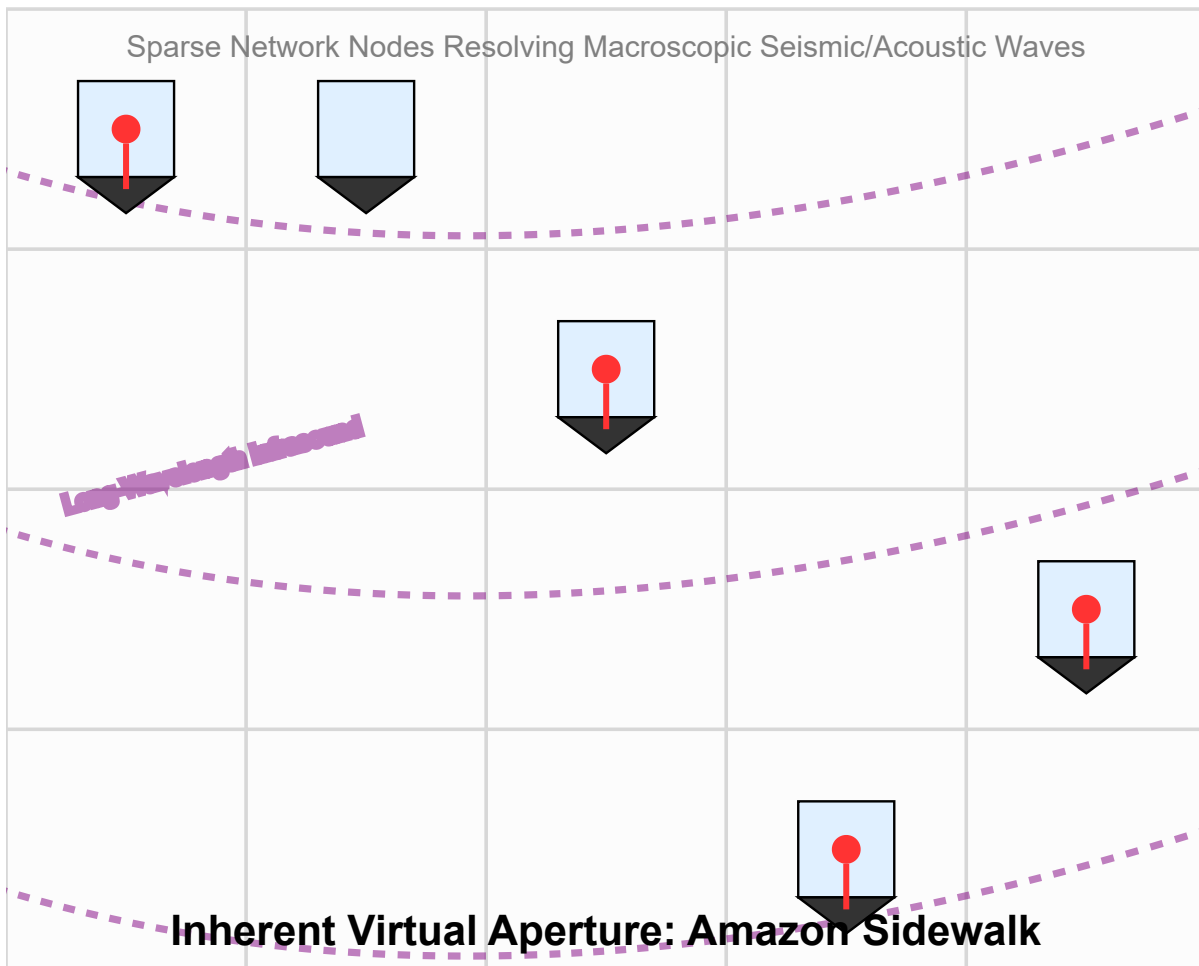


Figure: Sparse network nodes (Echo/Ring devices, shown as houses with antennas) resolving macroscopic seismic/acoustic waves (purple dashed lines). The grid represents suburban deployment; long-wavelength infrasound passes through the entire network. The aperture is physics, not design. (Diagram: Grok/Gemini collaboration)

What Prevents Exploitation Today:

Existing distributed networks possess emergent aperture capabilities but don't exploit them because:

1. **Sensor mismatch:** Hardware optimized for designed purpose (RF, voice) not sensing purpose (infrasound, seismic)
2. **Timing not characterized:** Synchronization adequate for packets, not characterized for phase-coherent sensing
3. **Position not precise:** Address-level location, not surveyed coordinates
4. **No exploitation architecture:** No protocol for coordinated sensing across nodes
5. **Nobody thought to try:** The aperture is invisible until you look for it

What UTLP/RFIP/SMSP Provides:

The architecture documented in this publication does not *create* virtual apertures—they already exist. The architecture *enables exploitation*:

Exploitation Requirement	What Provides It
Precise time synchronization	UTLP (microsecond-scale, characterized jitter)
Known node geometry	RFIP (peer-to-peer ranging, continuous updates)
Coordinated observation	SMSP (timestamped samples, conductor aggregation)
Connectionless operation	Architecture core (sync once, execute independently)

The Invention Is Not the Aperture—It Is the Exploitation:

This distinction matters for intellectual property:

Category	Example	Patentable?
Physical phenomenon	EM waves exist	No
Exploitation device	Radio receiver	Yes
Physical phenomenon	Virtual aperture emerges from distributed nodes	No
Exploitation architecture	UTLP/RFIP/SMSP coordination protocols	Prior art (this document)

We do not claim to have invented virtual apertures. We document the architecture that enables their exploitation, establishing prior art to ensure these exploitation techniques remain freely available.

Other Networks With Unexploited Aperture Potential:

Network	Designed Purpose	Emergent Aperture Potential	Enabling Modification
Amazon Sidewalk	IoT mesh	Infrasound/seismic continental array	Add infrasound mic, characterize timing
Starlink	Internet	Ionospheric tomography, TEC mapping	Use signal timing variations
Smart meters	Power billing	Grid harmonic sensing, lightning detection	Characterize power line as antenna
Cell towers	Communication	Atmospheric refraction mapping	Exploit multipath timing
Traffic sensors	Vehicle counting	Urban seismic/acoustic monitoring	Add low-frequency sensing
Weather stations	Point measurement	Distributed acoustic tomography	Add synchronized acoustic

Network	Designed Purpose	Emergent Aperture Potential	Enabling Modification
EV charging network	Power delivery	Grid-scale power quality monitoring	Already have electrical sensing

Each of these networks *already is* a virtual aperture at appropriate wavelengths. The aperture is not invented—it is recognized and exploited.

Implications for Prior Art:

By documenting the emergent aperture principle, we establish:

- 1. **No one can patent the existence of virtual apertures**—they are physics, not invention
- 2. **No one can patent "using network X for sensing Y"**—the capability is inherent
- 3. **The exploitation architecture is prior art**—UTLP/RFIP/SMSP documented here
- 4. **Future networks automatically have these properties**—any synchronized distributed deployment

This is defensive publication in its purest form: documenting physics that was always true, and the architecture to exploit it, so both remain freely available.

5.10 Passive Atmospheric Sensing

The receive beamforming capability extends to a powerful class of applications: using the atmosphere itself as the sensing medium. Instead of emitting signals and processing returns, distributed arrays listen passively to atmospheric acoustic phenomena.

The Physics Foundation:

Sound speed varies with atmospheric conditions:

- **Temperature:** Primary effect. $c \approx 331.3 \times \sqrt{1 + T/273.15}$ m/s
- **Humidity:** Secondary effect. Humid air is less dense (H₂O MW=18 vs N₂ MW=28), so slightly faster (~0.4% at saturation)
- **Wind:** Asymmetric travel times between node pairs

Acoustic absorption varies with humidity—water vapor molecular resonances create frequency-dependent attenuation. This provides an independent humidity measurement channel.

Acoustic Tomography: With UTLP-synchronized nodes at RFIP-known positions, measuring acoustic travel times between all node pairs enables inversion to extract:

- 3D temperature fields
- Humidity distribution (from absorption spectra)
- Vector wind fields (from travel time asymmetry)
- Turbulence structure (from signal coherence)

This is dense volumetric atmospheric sounding without expendable sensors (radiosondes) or active radar.

Infrasound Detection: Sound below 20 Hz propagates hundreds to thousands of kilometers via atmospheric waveguides. Sources include:

- Severe weather (tornadoes, microbursts, convection)
- Aircraft and missiles (aerodynamic disturbance)
- Explosions and volcanic events
- Mountain waves and clear-air turbulence

The CTBTO operates 60 infrasound stations globally for nuclear test detection. Weather and aircraft signals are treated as "noise." This architecture enables treating them as signal.

Tornado Detection: Tornadoes produce characteristic infrasound signatures (0.5-10 Hz) **before** the visible funnel forms—the mesocyclone and pressure deficit announce themselves acoustically 15-30 minutes before touchdown. A distributed infrasound network provides:

- Earlier warning than Doppler radar
- Continuous tracking (vs 4-10 minute radar scan cycles)
- Remote structure sensing (pressure field, rotation rate)
- False alarm reduction through acoustic confirmation of radar signatures

5.10.1 Research Validation: Cross-Domain Literature (1994-2025)

The atmospheric sensing claims above are not speculative—they represent active areas of research with decades of validation. The following summarizes peer-reviewed literature demonstrating that the physical principles and practical implementations described in this document are grounded in demonstrated science.

Infrasound Tornado Detection (Validated 1990s-Present)

Research at Oklahoma State University (Dr. Brian Elbing) and the National Oceanic and Atmospheric Administration has demonstrated that tornado-producing storms emit infrasound (0.5-20 Hz) **up to two hours before tornadogenesis**. Key validations:

- **GLINDA System** (Ground-based Local INfrasound Data Acquisition): Mobile infrasound measurement deployed with storm chasers since May 2020. Successfully detected elevated 10-15 Hz signals during tornado formation (Lakin, KS EFU tornado). Published in *Atmospheric Measurement Techniques*, 2022.
- **General Atomics ICE Sensors:** 20 Infrasound Collection Element sensors delivered to University of Alabama Huntsville for early tornado detection research. Accurately captured signals from multiple tornadoes up to 100 km away during the April 27, 2011 outbreak (General Atomics press release, 2016).
- **IEEE Spectrum Report** (2018): Ten minutes before the Perkins, Oklahoma tornado, the OSU array detected strong signals. The predicted tornado width (46m) matched the official damage path width exactly.
- **Warning Lead Time:** Multiple studies confirm infrasound precursors precede tornado onset by 30-120 minutes, compared to current average warning times of 13 minutes.

Key References:

- White, B.C., Elbing, B.R., Faruque, I.A. "Infrasound measurement system for real-time in situ tornado measurements." *Atmos. Meas. Tech.* 15, 2923–2938 (2022)
- Elbing, B.R., Petrin, C., Van Den Broeke, M.S. "Detection and characterization of infrasound from a tornado." *J. Acoust. Soc. Am.* 143(3), 1808 (2018)
- Bedard, A.J. "Infrasound from Tornadoes: Theory, Measurement, and Prospects for Their Use in Early Warning Systems." *Acoustics Today* (2005)

Seismic-Acoustic Coupling and Balloon Seismology (Validated 2021-2025)

The claim that seismic events can be detected via atmospheric infrasound—and that this enables seismology without ground sensors—has been validated by recent research:

- **Nature Communications Earth & Environment** (October 2023): "Remotely imaging seismic ground shaking via large-N infrasound beamforming"—demonstrated earthquake detection tens to hundreds of km away using infrasound arrays. CLEAN beamforming resolves individual waves in complicated wavefields.
- **Nature Communications Earth & Environment** (November 2025): "Balloon seismology enables subsurface inversion without ground stations"—balloon-borne infrasound data enabled joint inversion of earthquake source location AND subsurface velocity structure, matching results from ground-based seismometers. Direct application to Venus exploration where surface seismometers cannot survive.
- **Geophysical Research Letters** (August 2022): First detection of seismic infrasound from a large magnitude earthquake on a balloon network (Strateole-2 campaign, Flores Sea M7.3 earthquake). Demonstrated that quake magnitude and distance can be estimated from balloon pressure records alone.

Key References:

- Nature Communications Earth & Environment, "Remotely imaging seismic ground shaking via large-N infrasound beamforming" (2023)
- Nature Communications Earth & Environment, "Balloon seismology enables subsurface inversion without ground stations" (2025)
- Garcia, R.F. et al. "Infrasound From Large Earthquakes Recorded on a Network of Balloons in the Stratosphere." *Geophys. Res. Lett.* 49(15), e98844 (2022)

Acoustic Tomography for Atmospheric Sensing (Validated 1994-Present)

The claim that distributed acoustic arrays can reconstruct 3D temperature and wind fields has 30+ years of validation:

- **Foundational Work** (1994): Wilson & Thomson demonstrated acoustic tomography in the atmospheric surface layer—200m square array with three sources and seven receivers reconstructed temperature and wind fields with ~50m horizontal resolution. Published in *Journal of Atmospheric and Oceanic Technology*.
- **University of Leipzig Campaigns** (1990s-2000s): Extensive field testing established acoustic tomography as reliable for boundary layer monitoring. Time-dependent stochastic inversion (TDSI) algorithms developed.

- **UAV-Based Acoustic Tomography** (2015-2019): Using drone engine signatures as sound sources, researchers reconstructed 3D atmospheric profiles up to 120m altitude over 300m × 300m areas. Achieved $\pm 0.5^{\circ}\text{C}$ temperature accuracy and ± 0.3 m/s wind accuracy compared to LIDAR measurements.
- **DOE Wind Energy Research** (2022): NREL technical report on "Acoustic Travel-Time Tomography for Wind Energy" validates AT as a transformational remote sensing technique for wind farm applications.

Key References:

- Wilson, D.K., Thomson, D.W. "Acoustic Tomographic Monitoring of the Atmospheric Surface Layer." J. Atmos. Oceanic Tech. 11(3), 751–769 (1994)
- Finn, A., Rogers, K. "The feasibility of unmanned aerial vehicle-based acoustic atmospheric tomography." J. Acoust. Soc. Am. 138(2), 874–889 (2015)
- Hamilton, N., Maric, E. "Acoustic Travel-Time Tomography for Wind Energy." NREL Technical Report (2022)

Swarm Robotics Time Synchronization (Validated 2018-Present)

The connectionless synchronized execution model has direct parallels in swarm robotics research:

- **Swarm-Sync Framework** (Pervasive and Mobile Computing, 2018): Fully decentralized, energy-efficient time synchronization for swarm robotic systems. Achieved resynchronization intervals of 10+ minutes with bounded global synchronization error—exactly the pattern described in this document's UTLT protocol.
- **Decentralized Learning and Execution** (Royal Society Philosophical Transactions A, 2024): Paradigm where swarm robots learn and execute simultaneously in a decentralized manner without centralized control—validates the "synchronize once, execute independently" model.
- **Formation Flying via Synchronized Time** (Multiple sources): Swarm coordination research consistently identifies shared time reference as the critical enabler for coherent group behavior without continuous communication.

Key References:

- "Swarm-Sync: A distributed global time synchronization framework for swarm robotic systems." Pervasive and Mobile Computing 46, 35-52 (2018)
- "Signaling and Social Learning in Swarms of Robots." Phil. Trans. R. Soc. A 383, 2024.0148 (2024)
- "The road forward with swarm systems." PMC (2025)

Spacecraft Formation Flying (Validated 2000-Present)

The architecture scales to interplanetary distances—and spacecraft formation flying research validates this:

- **Stanford DiGiTaL System**: Distributed Timing and Localization for nanosatellite formations provides centimeter-level navigation and nanosecond-level time synchronization via peer-to-peer decentralized networks. Validated on PRISMA, MMS, CPOD missions.

- **GPS-Denied Deep Space:** X-ray pulsar-based navigation (XPNAV) provides absolute and relative positioning for spacecraft beyond GPS coverage. Inter-satellite links provide relative timing without Earth-based infrastructure—exactly the UTLP/RFIP model at interplanetary scale.
- **NASA Formation Flying Program:** JPL's Distributed Spacecraft Technology Program explicitly identifies the key technologies: robust fault-tolerant architecture for distributed communication/control/sensing, distributed guidance/estimation/control algorithms, and relative sensor technology—the same elements as UTLP/RFIP/SMSP.

Key References:

- Stanford Space Rendezvous Laboratory, "Distributed Multi-GNSS Timing and Localization System (DiGiTaL)" project documentation
- "X-ray pulsar-based GNC system for formation flying in high Earth orbits." Acta Astronautica 170, 294-305 (2020)
- NASA JPL Distributed Spacecraft Technology Program documentation

Bilateral Stimulation for EMDR Therapy (Validated 1990s-Present)

The therapeutic application that originated this architecture is itself well-validated:

- **Near-Infrared Spectroscopy Studies** (PMC, 2016): Demonstrated that alternating bilateral tactile stimulation affects prefrontal cortex activity during memory recall—the physiological basis for EMDR's effectiveness.
- **Affective Startle Reflex Paradigm** (ScienceDirect, 2020): Bilateral tactile stimulation decreases startle magnitude during negative imagination, providing physiological evidence for the mechanism.
- **Commercial Validation:** Multiple FDA-registered bilateral stimulation devices exist (TouchPoints, TheraTapper, various "buzzers" and "pulsers"), demonstrating commercial viability of synchronized haptic delivery.

Key References:

- "The Role of Alternating Bilateral Stimulation in Establishing Positive Cognition in EMDR Therapy." PLOS ONE (2016)
- "Good vibrations: Bilateral tactile stimulation..." European J. Psychotraumatology 12(1) (2021)

Cross-Domain Validation Summary

Claim Domain	Validation Status	Key Evidence
Infrasound tornado precursors	Strong	30+ years research, operational deployments, 10-120 min warning demonstrated
Seismic-acoustic coupling	Strong	Nature journals 2022-2025, balloon seismology validated

Claim Domain	Validation Status	Key Evidence
Acoustic atmospheric tomography	Strong	30+ years since 1994, DOE-funded, operational systems
Swarm time synchronization	Strong	Multiple peer-reviewed algorithms, royal society publications
Spacecraft formation flying	Strong	NASA/ESA missions, Stanford validation, operational systems
Bilateral stimulation therapy	Strong	NIH/PMC research, FDA-registered devices, clinical adoption
Time-varying metasurfaces	Strong	Nature journals 2021-2025, cryptographic security demonstrated

This cross-domain validation demonstrates that the architecture documented here—connectionless synchronized execution based on shared time reference and known geometry—is not novel in concept, but represents the convergence of proven techniques from geophysics, robotics, aerospace, and neuroscience into a unified framework applicable from handheld therapeutic devices to continental sensing networks to interplanetary spacecraft constellations.

Clear-Air Turbulence: CAT is invisible to radar (no precipitation) and causes aviation injuries. It is a density/velocity discontinuity with an acoustic signature. Distributed arrays along flight corridors could provide actual detection vs current reliance on pilot reports and model predictions.

Stealth-Independent Target Detection: RF stealth (radar-absorbing materials, geometry) is irrelevant to acoustic detection. A stealth aircraft with the radar signature of a bird still moves 170,000 lbs of air. The aerodynamic disturbance—pressure waves, wake turbulence, engine noise—propagates regardless of coating.

Deployment Scale Analysis:

Deployment	Nodes	Spacing	Coverage	Cost	Capability
Proof of concept	4-8	50-200m	Parking lot	\$500-1K	Algorithm validation
Research	10-20	0.5-2 km	Campus	\$3-6K	Urban micrometeorology papers
Regional pilot	50-100	5-10 km	Metro area	\$15-50K	Agriculture, aviation data
Tornado warning	200-500	15-30 km	State	\$100-300K	Improved warning lead time
National	1000+	30-50 km	Continental	\$2M+	CTBTO-class capability

Infrastructure Piggybacking: Deployment scales by adding capability to existing distributed infrastructure:

Existing Network	Nodes	Add Infrasound
State mesonet	50-200	Weather station upgrade
School weather stations	100s	Educational + sensing
ASOS/AWOS airports	~900	Aviation safety
Cell towers	100,000s	Carrier partnership
CoCoRaHS volunteers	20,000+	Citizen science

Node Hardware (research grade):

- MEMS infrasound microphone: \$3-10
- Wind noise mitigation (soaker hose array): \$20-50
- ESP32-C6 + weatherproof enclosure: \$30-50
- Power (solar + battery): \$30-50
- Connectivity (cellular/LoRa): \$20-50
- **Total per node:** \$100-200

A state-scale tornado warning network (200 nodes × \$200) costs \$40K in hardware—less than a single weather radar maintenance visit.

Moisture Effects on Propagation:

Condition	Sound Speed	Absorption	Detection Impact
Dry air	Baseline	Low	Maximum range
Humid air	+0.4%	Higher (freq-dependent)	Range reduction at high freq
Fog	Minimal change	Minimal	Droplets too small to scatter infrasound
Rain	Noise source	Scattering at high freq	Track precipitation by emission

Rain creates useful signal—precipitation cells can be tracked by their broadband acoustic emission without active radar.

6. Industry Context: How Professionals Solved This (And Its Limitations)

6.1 The GPS Convergence

Investigation of professional emergency lighting manufacturers—Whelen, Federal Signal, SoundOff Signal, and Fenix—revealed a surprising convergence: **none use direct RF communication for inter-vehicle timing synchronization**. All four independently adopted GPS time as a universal reference clock.

Manufacturer	Product	Sync Method	Characteristics
Whelen	V2V Module	GPS atomic reference	8+ hour holdover with TCXO
SoundOff	bluePRINT Sync	GPS (passive, no position)	~8-hour GPS refresh
Federal Signal	PFSYNC-1	GPS + RS485	45s acquisition time
Fenix	Fusion/T3	Pattern-boundary resync	Per-cycle realignment

SoundOff's documentation explicitly states their sync module is "passive"—it does not broadcast, transmit, or know its position. It only consumes GPS timing signals. This validates the core insight: **synchronization requires shared time, not bidirectional communication**.

6.2 Existing Patent Landscape

US7116294B2 (Whelen, filed 2003): Documents a self-organizing master/slave architecture for LED synchronization using a shared SYNC line. Key elements:

- First device to assert the line becomes master
- Other devices detect the assertion and become slaves
- 8-bit PIC microcontroller manages phase clock (Ø1/Ø2 phases)
- 400ms signal phases with matching 400ms resting phases

This patent covers *wired* synchronization with physical sync lines—not applicable to wireless connectionless operation.

EP3535629A1 (Whelen): Describes Receiver-Receiver Synchronization (RRS) where a reference node broadcasts a message that multiple receivers witness simultaneously. Receivers exchange timestamps of the commonly-witnessed event to calculate mutual offsets. This requires "fully meshed" devices and continuous coordination traffic.

6.3 The GPS Dependency Problem

GPS-based synchronization works but creates hard dependencies:

- **Acquisition time:** 45 seconds to several minutes for initial lock
- **Sky visibility:** Fails indoors, underground, in urban canyons
- **Hardware cost:** GPS receivers add \$5-20 per node
- **Power consumption:** GPS draws 20-50mA continuous
- **Jamming vulnerability:** Civilian GPS is trivially disrupted

The connectionless architecture documented here provides an alternative: peer-derived time synchronization that works without external infrastructure, indoors, and at lower power.

6.4 Pattern-Boundary Resynchronization (Feniex Contribution)

Feniex's approach accepts clock drift between sync exchanges by **resynchronizing automatically at the end of each pattern cycle**. This is pragmatic: if patterns repeat every 500ms, and crystal drift accumulates at 50 PPM (typical ESP32), maximum drift per cycle is 25µs—imperceptible.

This insight influenced the UTLP design: for many applications, perfect continuous synchronization is unnecessary. Agreement at pattern boundaries suffices.

6.5 What This Work Adds Beyond Industry Practice

Existing Practice	This Work's Contribution
GPS as time source	Peer-derived time (UTLP stratum hierarchy)
Wired sync lines	Connectionless RF execution
Continuous coordination traffic	Script-based independent execution
Fixed infrastructure	Mobile, infrastructure-free operation
Single-purpose devices	Zone/Role abstraction for flexible deployment
Earth-referenced positioning	Intrinsic swarm geometry (RFIP)

6.6 Relationship to US8073976B2: An Honest Assessment

During research for this publication, we identified US8073976B2 ("Synchronizing clocks in an asynchronous distributed system," Microsoft Corporation, filed 2008) as closely related prior art. We document this relationship transparently.

What US8073976B2 Covers

The Microsoft patent describes methods for calculating bounded time uncertainty between nodes in asynchronous distributed systems without requiring a master clock. Key elements include:

- **Request/reply message exchanges** to measure round-trip time
- **Mathematical framework** incorporating:
 - Clock quantum constraint (Q): maximum quantization error
 - Drift rate constraint (D): maximum clock drift per time period
 - Maximum round trip constraint: worst-case message exchange time
- **Upper and lower bounds** for inferring time at remote nodes
- **Event timing inference**: using calculated bounds to determine when events occurred at observed nodes

The patent's core variance formula:

$$\text{Maximum variance} = ((\text{receive_time} - \text{send_time})/2) + Q + (2D * (T - \text{AVG}(\text{send_time}, \text{receive_time}) + Q))$$

What We Independently Developed (Before Discovering This Patent)

Our UTP synchronization phase uses conceptually similar techniques:

- Round-trip timing measurement via BLE or ESP-NOW exchanges
- Statistical filtering to reduce jitter impact
- Drift estimation and compensation
- Kalman filtering for holdover during source loss

We arrived at these techniques through first-principles analysis of the ESP32 timing stack, not through study of US8073976B2. However, this independent convergence is expected: the mathematics of time transfer are constrained by physics, and NTP (1985) established the foundational approach decades before either our work or Microsoft's patent.

Where Our Architectures Diverge

Aspect	US8073976B2	This Work
Purpose of sync	Maintain ongoing knowledge of remote time	Establish shared time once, then disconnect
Communication model	Continuous message exchange to track variance	Sync channel is scaffolding—removed after convergence
During operation	Ongoing request/reply to refine bounds	No communication—independent script execution
What sync enables	Inference of when remote events occurred	Pre-calculated coordinated actuation
Connection state	Persistent, assumed necessary	Temporary, deliberately terminated

The Microsoft patent answers: *"How can I continuously know what time it is at another node, within bounds?"*

Our architecture answers: *"How can nodes act in coordination without needing to know anything about each other during operation?"*

These are related but distinct problems. The synchronization phase (where we overlap) is prerequisite to our actual contribution: the connectionless execution phase (where we diverge).

Patent Status

US8073976B2 shows status "Expired - Fee Related" with adjusted expiration 2030-01-03. This means Microsoft ceased paying maintenance fees (due at 3.5, 7.5, and 11.5 years after grant), causing the patent to lapse early. The "2030" date indicates when it *would have* expired with full term; the "Fee Related" status means it's already dead.

Microsoft's decision to abandon this patent—despite having resources to maintain it—suggests the technique became commoditized or wasn't generating licensing value. This reinforces that bounded time synchronization is well-established prior art, available for general use.

Our Position

We acknowledge that our synchronization methodology is not novel—it builds on NTP, PTP, US8073976B2, and decades of distributed systems research. We do not claim otherwise.

What we document as prior art is the architectural insight that synchronization can be *temporary scaffolding for permanent connectionless operation*. The sync channel establishes shared time; the shared time enables script-based execution; the script-based execution requires no further communication. This separation of concerns—and its application across domains from therapy devices to emergency lighting to distributed beamforming—is what we ensure remains freely available.

6.7 Relationship to Other Relevant Patents

Beyond US8073976B2, we identified additional patents in adjacent spaces. We document these relationships to clarify how the present work relates to existing intellectual property.

Time Synchronization Patents

Patent	Coverage	Relationship to This Work
US8165171B2 (DARPA, 2012)	Distributed synchronization for beamforming via round-trip and two-way methods	Covers sync methodology for coherent arrays. Our beamforming claims use sync as prerequisite but focus on connectionless phase coordination during transmission.
US10021659B2 (2018)	Synchronization of distributed nodes for cooperative beamforming with master/slave architecture	Covers continuous coordination for beam steering. Our architecture eliminates runtime coordination via pre-distributed phase offset scores.
US7047435B2 (IBM, 2006)	Clock synchronization using regression analysis of offset measurements	Covers sync refinement techniques. UTLP's Kalman filtering is analogous but not identical.

Bilateral Stimulation Patents

Patent	Coverage	Relationship to This Work
US20020035995A1 (Schmidt, 1999)	Alternating tactile stimulation device (TheraTapper)	Covers wired bilateral devices with controller. Our work uses wireless peer synchronization—different architecture.

Patent	Coverage	Relationship to This Work
TouchPoints BLAST patents	Bilateral alternating stimulation tactile technology	Covers specific vibration patterns and form factors. Our open architecture doesn't specify patterns—those are score content, not protocol.

Emergency Vehicle Lighting Patents

Patent	Coverage	Relationship to This Work
US7116294B2 (Whelen, 2003)	LED synchronization via physical SYNC wire, master/slave	Wired synchronization. Our architecture is wireless and connectionless.
EP3535629A1 (Whelen)	Receiver-Receiver Synchronization for mesh networks	Requires continuous mesh coordination traffic. Our architecture executes from pre-distributed scripts.

Distributed Beamforming Patents

Patent	Coverage	Relationship to This Work
US8165171B2 (2012)	DARPA-funded distributed sync for beamforming without centralized control	Covers synchronization method. Our contribution is connectionless execution after sync.
US10021659B2 (2018)	Dynamic untethered array nodes with frequency/phase/time alignment	Master-slave with ongoing coordination. Our architecture needs no runtime coordination.

Summary Assessment

The individual components of our architecture exist in prior art:

- Time synchronization: NTP (1985), PTP (2002), US8073976B2 (2008), others
- Bilateral stimulation: US20020035995A1 (1999), others
- Distributed beamforming: US8165171B2 (2012), US10021659B2 (2018), others
- Emergency lighting sync: US7116294B2 (2003), others

What these patents do NOT cover:

- The explicit phase separation (Bootstrap/Configuration/Execution)
- Connection-oriented sync bootstrapping connectionless execution
- Script-based independent actuation after sync channel release
- The cross-domain architectural pattern unifying these applications

We establish prior art for this architectural pattern, not for the individual techniques it employs.

6.8 Physical Metamaterials vs. Emergent Virtual Apertures: A 35 U.S.C. 101 Analysis

The existence of patents on physical metamaterials and reconfigurable metasurfaces provides compelling evidence for our core argument: **virtual apertures are emergent natural phenomena, not inventions.**

The Contrast That Proves Our Point

Physical metamaterial patents exist because metasurface properties do NOT spontaneously occur in ordinary materials. Engineers must:

- Design subwavelength structures with specific geometries
- Fabricate surfaces using specialized manufacturing
- Integrate actuators, control systems, and substrates
- Solve novel engineering problems

This is patentable subject matter under 35 U.S.C. 101—genuine invention is required.

Virtual apertures are fundamentally different. When synchronized, position-known nodes exist in space, wave-response properties emerge *automatically* from physics. No fabrication. No special materials. No engineering of the aperture itself. The aperture is a mathematical consequence of node distribution, not an invention.

Physical Metamaterial Patents as Evidence

Patent	What They Had to Invent	Why Virtual Apertures Don't Require This
US20230184938A1 (Boeing, 2023): Reconfigurable metasurface with mechanical actuators	Specific actuator arrangements, substrate integration, FPGA control architecture, fabrication methods	Virtual apertures require no substrate, no actuators, no fabrication—nodes in space already have wave-response properties
CN116482609A (2023): Time-modulated metasurface for radar invisibility	Temporal modulation circuits, element design, switching mechanisms	Virtual apertures don't need modulation circuits—moving nodes creates equivalent effects through geometry alone
Nature Comms (2025): Chaotic metasurface for secure communications	Chaos generation circuits, substrate-fixed element arrays, synchronization hardware	Virtual apertures can exhibit chaotic behavior simply by chaotic node motion—no circuits required

The pattern: Physical metasurfaces require invention at every level. Virtual apertures require only *recognition* that the physics already exists.

The 35 U.S.C. 101 Argument

Under 35 U.S.C. 101, laws of nature and natural phenomena are not patentable subject matter. This document argues the following distinction:

Category	Physical Metasurface	Virtual Aperture
What creates the effect	Engineered structures	Node distribution in space
Without human intervention	No metasurface properties	Aperture properties exist automatically
Patentability argument	Patentable (requires invention)	Arguably natural phenomenon

Boeing can patent their actuator arrangements because actuator arrangements don't occur naturally. This document argues that no one should be able to patent "the fact that distributed nodes have wave-response properties" because that's how physics works. Patent examiners will ultimately decide, but the argument is strong.

What This Means for Prior Art Strategy

The physical metamaterial patent landscape *strengthens* our position:

1. **Physical patents don't anticipate virtual apertures** — They're solving different problems (fabrication vs. coordination)
2. **Physical patents don't cover virtual apertures** — Distributed nodes are not "metasurfaces with actuators"
3. **The contrast reinforces our 101 argument** — If virtual apertures were inventions like physical metasurfaces, they would require similar engineering. They don't.
4. **Our prior art covers the exploitation gap** — Physical metamaterial patents don't teach exploiting emergent apertures from existing networks. This document does.

Real-World Precedent: Distributed Acoustic Sensing (DAS)

The fiber-optic Distributed Acoustic Sensing industry provides a real-world precedent for how patent law treats emergent sensing capabilities in existing infrastructure.

The physics: Any optical fiber exhibits Rayleigh backscattering from microscopic imperfections. When seismic waves strain the fiber, the backscatter pattern changes. This is physics—it happens whether anyone exploits it or not. Every telecommunications fiber in the world is simultaneously a seismic sensor, whether the telecom company knows it or not.

What DAS patents cover:

- Interrogator hardware designs (laser sources, detectors, signal processing)

- Specific algorithms for extracting strain from backscatter
- Engineered fiber treatments to enhance sensitivity
- Deployment methods for coupling fiber to ground motion

What DAS patents do NOT cover:

- "Fiber optic cables can sense vibrations" (natural phenomenon)
- "Existing telecom fiber constitutes a seismic array" (inherent property)
- The general principle of using fiber for distributed sensing

The parallel to virtual apertures:

Aspect	DAS (Fiber Sensing)	Virtual Apertures (Distributed Nodes)
Emergent property	Fiber senses strain via backscatter	Synchronized nodes have wave-response
Exists in deployed infrastructure	Every telecom fiber	Every synchronized IoT network
Patent treatment of phenomenon	Not patentable (physics)	Not patentable (physics)
Patent treatment of exploitation	Interrogator designs, algorithms	This document: prior art

Recent research demonstrates this principle dramatically: a 2025 Nature Communications paper showed that repurposing a 50-kilometer telecommunications fiber in San Jose created an "ultra-dense seismic array" that could map urban activities, land use patterns, and demographic trends—capabilities that existed inherently in the infrastructure, waiting to be exploited.

The lesson for virtual apertures: Just as DAS patents cover interrogator hardware but not the fact that fiber senses vibrations, patents in the virtual aperture space should cover specific exploitation hardware but not the fact that distributed nodes form apertures. This document establishes prior art for the exploitation architecture, ensuring that both the phenomenon and the general exploitation method remain in the public domain.

The Emergent Aperture Is Not an Invention

Consider the analogy:

- **Physical lens:** Must be ground, polished, designed with specific curvature. Patentable.
- **Gravitational lens:** Mass in space bends light automatically. Natural phenomenon. Not patentable.

Similarly:

- **Physical metasurface:** Must be fabricated with engineered elements. Patentable.
- **Virtual aperture:** Distributed synchronized nodes bend/focus waves automatically. Natural phenomenon. Not patentable.

What CAN be patented (and what this document establishes prior art for):

- Specific methods of exploiting virtual apertures
- Novel coordination architectures (UTLP/RFIP/SMSP)
- Particular sensing or actuation applications

What CANNOT be patented:

- The fact that virtual apertures exist
- The general principle of exploiting them
- The mathematical relationship between node spacing and wavelength

This document ensures both the phenomenon (natural law) and the general exploitation framework (prior art) remain in the public domain.

The Event Horizon Telescope: A Planet-Sized Virtual Aperture

The most famous virtual aperture in existence is the Event Horizon Telescope (EHT), which imaged a black hole in 2019 using radio telescopes distributed across Earth. The EHT is not a single instrument—it is a planet-sized virtual aperture created by synchronized, position-known nodes.

The physics:

- 8 radio telescopes on 4 continents
- Earth-diameter baseline (~12,742 km)
- Hydrogen maser atomic clocks for nanosecond synchronization
- Hard drives shipped physically for correlation (bandwidth exceeds internet capacity)

What the EHT demonstrates:

- Virtual apertures work at planetary scale
- The aperture is not an invention—it's a mathematical consequence of distributed coherent receivers
- Synchronization precision determines usable frequency (230 GHz for EHT)
- Position knowledge (surveyed to millimeter precision) enables aperture synthesis

What was NOT patented:

- "Using distributed radio telescopes as a single aperture" (that's physics)
- "Planet-scale virtual aperture" (that's physics)
- The general principle of VLBI (established 1967)

What WAS developed (and could be patented):

- Specific correlator hardware designs
- Hydrogen maser timing systems
- Data recording formats (Mark 6 VLBI system)
- Post-processing algorithms for image reconstruction

The EHT follows the same pattern:

Aspect	EHT (Radio Astronomy)	This Architecture (General)
Aperture phenomenon	Earth-diameter radio aperture	Any distributed node aperture
Why it exists	Physics (wave coherence)	Physics (wave coherence)
Patentable?	No	No
Exploitation hardware	H-maser clocks, correlators	UTLP/RFIP/SMSP + commodity hardware
Exploitation patentable?	Specific implementations, yes	This document: prior art

The Death Star connection:

The EHT and Star Wars' Death Star are time-reversed versions of the same physics:

- **Death Star (1977):** Distributed emitters → coherent outbound beam (transmission beamforming)
- **EHT (1967/2019):** Distributed receivers → coherent inbound aperture (reception aperture synthesis)

The architecture documented here does both—SMSP operates bidirectionally. Scores flow out to nodes (transmission coordination), observations flow back from nodes (reception aperture synthesis). Same math, same physics, opposite directions.

The precedent:

If the EHT collaboration—with funding from NSF, international space agencies, and major research institutions—did not patent "planet-scale virtual aperture," it is because such a patent would fail under 35 U.S.C. 101. The aperture is physics. Only the specific exploitation methods are patentable subject matter.

This document establishes prior art for a general exploitation architecture, ensuring that the UTLIP/RFIP/SMSP framework for creating and exploiting virtual apertures at any scale remains in the public domain—just as VLBI remains in the public domain while specific correlator designs may be proprietary.

7. Why This Wasn't Done Before

7.1 The BLE Stack Abstraction (And Why We Left Anyway)

BLE was designed for reliable data transfer with minimal power, not precision timing. The stack abstracts away the RF-level timing that would enable tight synchronization. The controller knows the connection event anchor with microsecond precision; the application learns "a packet arrived" with millisecond uncertainty.

Important clarification: Tight timing *can* be achieved over BLE. With careful stack configuration, statistical filtering, and jitter compensation, sub-millisecond synchronization is possible. Research implementations have demonstrated this.

We chose the connectionless RF path deliberately, not because BLE timing was impossible, but because:

1. **Connectionless eliminates a failure mode:** No connection to drop during therapy
2. **Lower latency jitter:** ESP-NOW's $\sim 100\mu\text{s}$ jitter vs BLE's $\sim 10\text{-}50\text{ms}$ simplifies the timing stack
3. **Independence during execution:** Devices don't need to maintain a link while operating
4. **Power efficiency:** When everyone has the same script, radio silence until something changes
5. **Phone compatibility preserved:** BLE handles the user-facing interface; ESP-NOW handles peer coordination

The architecture uses BLE for what it's good at (phone pairing, trust establishment) and ESP-NOW for what it's good at (low-jitter peer communication).

7.2 The Coordination Assumption

Distributed systems literature focuses heavily on consensus and coordination: how do nodes agree? BFT, Raft, Paxos—all assume ongoing communication is necessary for agreement.

For many applications, this assumption is unnecessary. If nodes can agree *once* on time and plan, they don't need to agree *continuously* during execution. The FLP impossibility result (consensus is impossible in asynchronous systems) doesn't apply when you've already established synchronous time.

7.3 GPS as Crutch

Professional systems (emergency vehicle lighting, broadcast synchronization) solved this problem with GPS. Every node gets atomic time from satellites; coordination is implicit.

This works but creates dependencies: satellite visibility, antenna placement, acquisition time. The connectionless architecture provides an alternative path that works indoors, underground, and in RF-challenged environments.

8. Implementation Availability

Reference implementations are available under MIT license:

- **UTLP specification and ESP32 implementation:** github.com/mlehaptics
- **RFIP addendum with 802.11mc FTM integration:** Included in UTLF repository
- **Bilateral stimulation firmware:** ESP32-C6 reference design with BLE+ESP-NOW

Hardware requirements (minimum):

- ESP32-C6 (recommended) or ESP32-S3/C3: $\sim \$5\text{-}8$
- Standard BLE and WiFi capabilities

- No specialized timing hardware required
- **Total BOM for bilateral device: under \$15/node**

Hardware options (enhanced capabilities):

- Seeed XIAO MG24 Sense (~\$15): Adds 6-axis IMU for orientation/dead reckoning
- ESP32-C6 v0.2+ silicon: Enables FTM initiator role (earlier revisions: responder only)
- External GPS module: Stratum 0 time source for outdoor applications

9. Prior Art Claims

This document establishes prior art for the following techniques, ensuring they remain available for public use:

9.1 Architectural Patterns

1. **Connectionless synchronized actuation:** Devices sharing time reference and script execute in coordination without runtime communication
2. **Bootstrap/Configuration/Execution phase separation:** BLE for trust and setup, connectionless for timing-critical operation
3. **Script-based distributed execution:** Deterministic event sequences calculated locally from shared parameters
4. **Shared-clock execution model:** Devices calculate state from synchronized time rather than exchanging coordination messages
5. **Local jitter characterization:** Treating synchronization error as a property of local software stack, not network

9.2 Protocol Techniques

6. **BLE bootstrap for ESP-NOW security:** Deriving ESP-NOW encryption keys from BLE pairing material, then releasing peer BLE connection
7. **UTLP time as public utility:** Unencrypted broadcast time with Glass Wall isolation from application data
8. **Common Mode Rejection security:** Spoofed time affects all nodes equally, preserving relative synchronization
9. **Stratum-based opportunistic upgrade:** Automatic precision improvement when better sources become available
10. **Kalman-filtered holdover:** Joint offset/drift estimation for graceful degradation during source loss

11. **HKDF for high-entropy key derivation:** Using HKDF-SHA256 (not PBKDF2/Argon2) for secrets that are already high-entropy—password stretching is inappropriate when input has 64+ bits of entropy
12. **Multi-layer replay protection:** Session nonce (session uniqueness) + sequence numbers (intra-session ordering) + TOTP (time-binding) + CCMP (transport integrity)—four independent layers
13. **Defense-in-depth security architecture:** Physical, transport, key derivation, and application layer security with each layer providing independent protection
14. **Threat-proportional security design:** Cryptographic strength appropriate to actual threat model, avoiding over-engineering that increases complexity and attack surface

9.3 Application Patterns

15. **Swarm-emergent warning systems:** Distributed nodes forming coherent visual signals without central coordination
16. **Aerial extension of ground-level warnings:** Drone swarms providing elevated visibility for traffic incidents
17. **Zone/Role architectural separation:** Identical firmware, runtime-assigned function based on position or configuration
18. **RFIP intrinsic positioning:** Spatial awareness without Earth-referenced infrastructure

9.4 Validation Methods

19. **High-speed video validation of distributed timing:** Using frame-accurate capture to verify synchronization precision
20. **SAE J845 compliance testing for swarm systems:** Applying emergency vehicle lighting standards to distributed architectures

9.5 Techniques Extending Beyond Existing Patents

21. **Wireless connectionless sync vs. wired sync lines:** US7116294B2 requires physical SYNC wire; this work achieves equivalent coordination over RF without wired connection
22. **Script-based execution vs. continuous mesh coordination:** EP3535629A1 requires ongoing timestamp exchange between "fully meshed" devices; this work distributes script once, then executes independently
23. **Peer-derived time vs. GPS dependency:** Emergency vehicle systems require GPS receivers; UTLP stratum hierarchy achieves equivalent precision from peer sources
24. **Pattern-boundary resync generalization:** Extending Feniex's per-cycle resync concept to arbitrary script boundaries and multi-modal actuation
25. **Infrastructure-free spatial awareness:** RFIP provides relative positioning without surveyed anchor points, fixed infrastructure, or Earth-referenced coordinates

26. **Ranging-based autonomous zone assignment:** Using 802.11mc FTM or equivalent ranging to derive zone assignments from spatial position rather than pre-configuration or connection order
27. **Pile-to-swarm self-organization:** Undifferentiated identical devices establishing spatially-coherent role topology through peer ranging without central assignment authority
28. **Spatial-semantic zone mapping:** Zone assignments that reflect physical relationships (leftmost, northernmost, highest) rather than arbitrary identifiers
29. **Self-mapping search patterns in GPS-denied environments:** Swarm builds and tracks searched areas using only peer-derived RFIP coordinates, enabling coordinated coverage without external positioning infrastructure
30. **Distributed IMU from ranging geometry:** 3+ nodes with peer ranging provide 6-DOF swarm orientation (translation, rotation, scale) without per-node inertial sensors—the swarm's geometry is itself an inertial reference
31. **IMU-augmented peer ranging:** Combining 802.11mc FTM with per-node inertial measurement for reflection ambiguity resolution, dead reckoning between ranging updates, and orientation awareness in mobile swarms
32. **Connection-oriented sync bootstrapping connectionless execution:** Using PTP/NTP-style timestamp exchange over connection-oriented transports (BLE, WiFi, etc.) to establish a persistent time reference that outlives the connection—the sync method is scaffolding, removed after use, while the time agreement enables indefinite connectionless coordination

9.6 Score Protocol Techniques (SMSP)

33. **Three-layer score architecture:** Separating declarative intent (human-readable parameters), compiler layer (PWA/tool transforming intent to timeline), and imperative execution (dumb engine playing time-indexed events)
34. **Time-indexed score format:** Defining actuator state at absolute/relative timestamps rather than frequencies—frequency becomes implicit in timeline spacing, eliminating runtime waveform calculation
35. **Transition-aware keyframes:** Score lines include interpolation duration and easing specification, enabling smooth crossfades as first-class operations rather than engine complexity
36. **Multimodal channel abstraction:** LED RGB, brightness, haptic intensity, audio frequency/amplitude as parallel channels in unified timeline—modality is a channel property, not a protocol distinction
37. **Pattern classification metadata:** Score-level enum (BILATERAL, EMERGENCY, SWARM_SYNC, PURSUIT, CUSTOM) enabling UI hints, validation rules, and zone logic optimization without parsing the timeline
38. **Scale-invariant score execution:** Identical score format from PCB-mounted LEDs (zones = GPIO pins) to field-deployed swarms (zones = node IDs)—playback engine unaware of physical scale

- 39. **Transport-agnostic score delivery:** Score format independent of delivery mechanism (ESP-NOW, BLE, wired bus, flash-at-build-time)—protocol complete when node has score + time + zone
- 40. **8-bit capable score engine:** Playback loop (time check → segment advance → interpolate → output) simple enough for ATtiny-class MCUs; wireless sync capability determines cost, not execution capability
- 41. **Conductor/performer topology:** Single "smart" node handles UTLP sync and score distribution; multiple "dumb" nodes execute scores with minimal hardware—enables \$0.50 per additional node in wired deployments
- 42. **Synthesized audio as score channel:** Audio frequency and amplitude as score parameters for real-time synthesis (no sample storage), enabling binaural/bilateral audio patterns with same connectionless execution model

9.7 Wave Domain Techniques (Beamforming)

- 43. **Distributed beamforming via connectionless phase coordination:** Nodes with synchronized time and known geometry execute scores containing per-node phase offsets, enabling steered wave emission without real-time coordination during transmission
- 44. **Domain-invariant phased array architecture:** Same UTLP/RFIP/SMSP stack applies from acoustic (ms periods, MCU-achievable) to RF (ns periods, FPGA-required)—architecture unchanged, timing precision determines applicable domain
- 45. **Swarm geometry feeding beam steering calculations:** RFIP-derived inter-node distances as direct input to phase offset computation ($\text{delay} = n \times d \times \sin(\theta) / v$), eliminating pre-surveyed array geometry requirements
- 46. **Phase offset as SMSP score parameter:** Beam steering angles compiled to per-node microsecond delays, distributed as standard score timing, executed via normal connectionless playback
- 47. **Dynamic beam steering via score update:** Changing target bearing requires only new score distribution, not architectural modification—same connectionless execution model for fixed, scanning, or tracking beams
- 48. **Acoustic validation of RF-applicable architecture:** Proving distributed phase coordination at human-observable timescales (kHz acoustic) to validate control logic for RF timescales (GHz)—architecture identical, only clock hardware differs
- 49. **Enclosure acoustic effects as radome simulation:** Structural acoustic non-idealities (phase distortion, directivity modification, internal reflections) modeling RF antenna detuning, body shadowing, and boresight error in aerospace deployments

9.8 Implementation Philosophy

- 50. **Score generation method independence:** Score authoring by any means—manual, algorithmic, AI/LLM-assisted, or real-time sensor-driven compilation—is implementation detail; the protocol and execution model are the contribution, not the generation method

9.9 Dynamic Aperture Techniques (Time-Varying Geometry)

51. **Dynamic aperture beamforming via synchronized geometry change:** Swarm nodes on flexible/deformable substrate where RFIP provides continuous geometry updates enabling coherent beamforming despite time-varying node positions
52. **True time delay beamforming via mechanical displacement:** Physical node motion providing frequency-independent signal delay, enabling wideband/pulsed beam focusing without beam squint—all frequencies experience identical delay from physical path length change
53. **Mechanical wave beam scanning:** Traveling wave across swarm array physically steering beam direction without electronic phase control, scan rate determined by mechanical wave velocity, UTLP ensuring deterministic wave phase
54. **Phase-coherent mechanical and electromagnetic waves:** UTLP synchronization ensuring deterministic phase relationship between array mechanical deformation and RF/acoustic emission—the mechanical wave becomes part of the modulation scheme
55. **Non-reciprocal array via time-varying geometry:** Doppler encoding of transmission angle into signal frequency, where array state change between transmission and potential countermeasure arrival breaks reciprocal path—inherent jamming resistance without cryptographic complexity
56. **Synthetic aperture from element motion:** Array elements tracing paths over time synthesize larger effective aperture without platform motion—SAR-like resolution enhancement from mechanical wave displacement
57. **Integrated dynamic aperture device:** Single physical device with distributed actuation points (piezoelectric membrane, MEMS mirror array, tensioned mesh) implementing coordinated true-time-delay beamforming via embedded geometry sensing and synchronized actuation
58. **Active radome beam compensation:** Mechanically actuated radome surface providing real-time correction for radome-induced beam distortion plus additional beam steering capability beyond the antenna element
59. **Scale-invariant aperture architecture:** Same coordination model (time sync, geometry knowledge, score-based actuation) spanning from interstellar spacecraft constellations (light-year scale, plasma wave detection) through distributed field swarms (km scale) to integrated MEMS surfaces (μm scale) to optical metamaterials (nm scale)—"swarm" generalizes to "distributed actuation points," "RFIP" generalizes to "geometry sensing," wave physics changes but coordination architecture doesn't
60. **Space-time modulated metasurface via distributed coordination:** Implementing space-time modulated metasurface effects (frequency conversion, non-reciprocal propagation, wideband operation) through connectionless swarm coordination rather than centralized control

9.10 Passive Acoustic Detection (Stealth-Independent Sensing)

- 61. **Passive acoustic detection of aerodynamic disturbances:** Distributed infrasound/acoustic sensor array using UTLP time synchronization for coherent receive beamforming, detecting aircraft, missiles, or other airborne targets via the pressure disturbances they must create by moving through atmosphere—RF-stealth-independent detection where radar cross section reduction provides no protection against acoustic wake signature
- 62. **Infrasound synthetic aperture via distributed MEMS arrays:** Multiple UTLP-synchronized passive acoustic arrays with RFIP-known positions performing coherent integration of infrasound signals (<20 Hz) for directional detection and tracking of targets at ranges where active acoustic ranging is impractical—exploiting signals the CTBTO network treats as nuisance detections

9.11 Oscillating Aperture Modes

- 63. **Switchable wave/rigid aperture modes:** Dynamic aperture array capable of transitioning between traveling wave mode (continuous scanning, time-averaged sidelobes) and frozen mode (static curvature optimized for specific bearing)—enabling power-efficient focused transmission after initial scanning acquisition
- 64. **Oscillating partial-cycle beam dithering:** Transverse wave driven in forward/reverse oscillation over limited amplitude range, creating beam that rocks across target bearing rather than sweeping past—increasing dwell time on target, enabling resonant amplification at mechanical resonance frequency, and generating FM Doppler signature from sinusoidal element velocity
- 65. **Mechanically-generated Doppler diversity:** Array element oscillation creating frequency modulation of transmitted signal where the FM pattern is determined by mechanical oscillation frequency and amplitude—providing spread-spectrum-like properties, FMCW-style ranging capability, and jamming resistance through unpredictable Doppler structure

9.12 Atmospheric Sensing and Meteorology

- 66. **Acoustic tomography of atmosphere via distributed synchronized arrays:** UTLP-synchronized nodes with RFIP-known positions measuring acoustic travel times to extract temperature, humidity, and wind fields through sound speed inversion—providing dense volumetric atmospheric sounding without expendable sensors
- 67. **Passive infrasound severe weather detection network:** Distributed infrasound sensor array using coherent beamforming to detect, locate, and track tornadoes, microbursts, and severe convection via their characteristic low-frequency acoustic signatures—providing detection lead time beyond Doppler radar capability
- 68. **Clear-air turbulence detection via passive acoustic sensing:** UTLP-synchronized acoustic arrays along flight corridors detecting CAT signatures (density and velocity discontinuities) that are invisible to radar—addressing a gap in aviation safety where current detection relies on pilot reports and model prediction
- 69. **Distributed wind field extraction from acoustic time-of-flight:** Bidirectional acoustic travel time measurements between synchronized nodes extracting vector wind fields, operational in clear air and at ground level where radar has limitations—applicable to agricultural, aviation, and urban meteorology

- 70. **Hyperlocal agricultural microclimate monitoring:** Dense mesh of synchronized acoustic/environmental sensors providing field-scale weather prediction (frost pocket detection, cold air drainage, precipitation approach) at resolution finer than national radar networks
- 71. **Remote tornado structure sensing via acoustic tomography:** Distributed UTLP-synchronized infrasound arrays extracting tornado pressure field, vortex geometry, and rotation rate from safe standoff distance—providing volumetric structure data previously requiring hazardous in-situ deployment (Dorothy/TOTO-class instruments)
- 72. **Tornado precursor detection via mesocyclone acoustic signature:** Infrasound network detecting rotating updraft and pressure deficit signatures 15-30 minutes before visible funnel formation or clear radar hook echo—extending warning lead time beyond current Doppler radar capability
- 73. **Multi-sensor tornado warning fusion:** Combining Doppler radar velocity data with distributed infrasound pressure/precursor detection for reduced false alarm rate and extended lead time—acoustic confirmation of radar-indicated rotation before warning issuance
- 74. **Humidity field extraction via differential acoustic absorption:** Measuring frequency-dependent acoustic attenuation across distributed array to extract atmospheric humidity distribution—exploiting the physical phenomenon where water vapor molecular resonances create humidity-dependent absorption spectra
- 75. **Precipitation tracking via acoustic emission localization:** Passive detection and tracking of rain cells, hail cores, and precipitation boundaries using the broadband acoustic signature of hydrometeors—rain announces itself acoustically, enabling tracking without radar

9.13 Seismoacoustic Detection (Ground-Atmosphere Coupling)

- 76. **Seismic event detection via atmospheric infrasound:** Using distributed infrasound arrays to detect earthquakes, explosions, and other seismic events through ground-atmosphere acoustic coupling—seismic waves cause surface displacement that radiates infrasound, enabling seismic monitoring without ground-coupled equipment
- 77. **Multi-phenomenology environmental monitoring:** Single distributed infrasound array simultaneously serving as severe weather detector, wind field mapper, aircraft tracker, and seismic monitor—same hardware and processing architecture applied to multiple detection domains without modification
- 78. **Complementary seismic-acoustic event characterization:** Combining detection of events via both seismic ground-truth (if available) and atmospheric infrasound signature to improve event classification, location accuracy, and false alarm rejection—exploiting the different propagation characteristics of solid-earth and atmospheric waves

9.14 Architectural Scaling (Capstone Claims)

79. **Localized-to-planetary warning system architecture:** The connectionless distributed timing architecture validated at minimum scale (bilateral therapeutic device, 2 nodes, centimeter spacing) is mathematically identical to the architecture required for planetary-scale early warning systems (continental sensor networks, thousands of nodes, megameter spacing)—scale changes node count and spacing, not the underlying coordination model of synchronized time, known geometry, and scripted actuation
80. **Interstellar-capable coordination architecture:** Extension of UTLP/RFIP/SMSP protocols to spacecraft constellation scale where light-time delays become significant—the architecture accommodates propagation delay as a parameter, not a fundamental limit, enabling coherent coordination across solar-system and potentially interstellar distances with appropriate clock stability
81. **Minimum viable instantiation as architectural proof:** A functioning 2-node bilateral stimulation device constitutes complete validation of the distributed coordination architecture, with all larger deployments (emergency lighting, swarm robotics, atmospheric sensing, planetary defense) being scale variations requiring no architectural modification—the therapy device is not a precursor to the warning system, it IS the warning system at minimum viable scale

9.15 Bidirectional SMSP (Observation and Feedback)

82. **Symmetric instruction/observation format:** SMSP extended with observation messages structurally identical to instructions—same timestamp, node ID, and payload semantics, but reversed direction—enabling sensing applications where observations flow back using the same protocol that distributes actuation commands
83. **Conductor-targeted sync correction:** Fine-grained clock correction messages sent to specific drifting nodes, complementing UTLP broadcast sync—the coordinator (conductor) observes timing errors in returned observations and sends targeted corrections ("you're 450µs late") to individual nodes
84. **Query-driven swarm sensing:** Swarm operates as instrument that responds to queries rather than continuously streaming data—query becomes SMSP sampling schedule, nodes execute coordinated measurement, observations flow back, correlation produces answer—on-demand rather than always-on
85. **Orchestra feedback model for distributed systems:** Explicit architectural pattern where score distribution (conductor → musicians) and observation return (music → conductor's ears) use the same transport medium and protocol structure—the output IS the feedback, eliminating need for separate telemetry channel
86. **Multi-level observation access:** Single observation stream supporting operator-level queries ("where's the tornado?"), analyst-level aggregations (confidence distributions), researcher-level raw data (timestamped samples), and debug-level diagnostics (per-node clock state)—abstraction serves users without hiding data

9.16 Dynamic Metasurface and Configuration Space

87. **Cryptographically large configuration space via continuous wave parameters:** Wave-shaped or wave-controlled metamaterial apertures achieving effectively infinite distinct configurations through continuous parameter variation (frequency, amplitude, phase, direction of multiple simultaneous waves)—configuration count exceeds any feasible catalog, preventing signature-based identification
88. **Round-trip latency guaranteeing configuration change:** Time-varying surfaces where the round-trip delay between adversary detection and response ensures the surface configuration has changed—adversary always interacts with a configuration they haven't previously characterized
89. **Position-dependent response from time-varying surfaces:** Dynamic metasurfaces where observers at different spatial positions perceive different apparent configurations due to combined spatial and temporal modulation—security emerges from physics, not encryption
90. **Chaotic modulation for keyless physical-layer security:** Metasurface configurations driven by chaotic sequences achieving secure communication without shared encryption keys—legitimate receiver at correct position receives clear signal, all others receive irreversibly scrambled noise
91. **Distributed time-varying metasurface coordination:** Multiple physically separate time-varying metasurfaces coordinated via connectionless UTLP/RFIP/SMSP architecture—extending single-surface capabilities to spatially distributed aperture synthesis, coherent multi-platform operation, and self-healing arrays
92. **Cross-domain metasurface architecture:** Same coordination architecture (synchronized time, known geometry, scripted actuation) applied to both electromagnetic and acoustic metasurfaces—principles validated in RF radar/communications transfer directly to acoustic sensing/beamforming and vice versa

9.17 Deformable Virtual Metasurface

93. **Position as primary control variable:** Swarm metasurface where node physical position is treated as a primary control variable rather than error to compensate—RFIP-tracked geometry changes are commanded and exploited for wavefront manipulation, not merely measured and corrected
94. **Three-axis metasurface control:** Combined control of phase timing (SMSP), physical position (RFIP), and local node density in a single virtual metasurface—providing $e^{i\phi} \times \Delta r \times \rho(x,y,z)$ control space impossible with fixed-geometry metasurfaces
95. **Adaptive resolution via density modulation:** Swarm metasurface that concentrates node density in regions of interest while maintaining sparse coverage elsewhere—real-time redistribution based on detected signal characteristics without substrate constraints
96. **Topological reconfiguration:** Swarm metasurface capable of complete topological restructuring—single aperture to multiple sub-apertures, planar to volumetric, flat to arbitrary curvature—not merely deformation within substrate limits

97. **Dual-mode longitudinal/transverse metasurface:** Single swarm infrastructure with appropriate sensor/actuator hardware handling both longitudinal (acoustic) and transverse (electromagnetic) wave physics simultaneously—horizontal deployment for acoustic, vertical extent for EM, combined for multi-physics
98. **Wave-orientation-aware deployment geometry:** Automatic or commanded swarm reconfiguration between horizontal-dominated (longitudinal wave optimal) and vertical-dominated (transverse wave optimal) geometries based on target wave physics—same nodes, different topology, different wave interaction
99. **Dynamic aperture focusing:** Swarm metasurface with real-time variable focal distance achieved through physical curvature change—unlike fixed dishes, can transition between parabolic, flat, and hyperbolic profiles, or create multiple simultaneous foci
100. **Substrate-free deformable metasurface:** Metasurface composed of fully mobile nodes with no physical substrate connecting them—enables topology changes impossible with kirigami, MEMS, or stretchable substrate approaches; geometry limited only by node mobility, not material strain
101. **Dynamic macroscopic lattice:** Distributed synchronized nodes forming a programmable lattice structure at macro scale (meters) that exhibits the same wave physics as atomic crystal lattices at micro scale (angstroms)—Bragg reflection, band gaps, refraction—but with runtime reconfigurability impossible in fixed crystal structures; the term "virtual metasurface" used in literature derives from fabrication constraints that don't apply; "dynamic macroscopic lattice" more accurately describes the physics
102. **Solid-state physics at macro scale:** Node spacing determining band gaps (frequencies blocked/passed) exactly as atomic spacing determines band gaps in photonic/phononic crystals; $\lambda/2$ node spacing creates Bragg reflection (selective mirror); true time delay across nodes creates refraction (wavefront steering); same Bragg's Law ($n\lambda = 2d \sin \theta$) applies at both scales; material properties (transparency, reflectivity, impedance) programmable rather than fixed
103. **Unified interference pattern coordination:** All dynamic macroscopic lattice wavefront manipulation—beamforming, null steering, band-pass filtering, band-stop filtering, focusing, defocusing, scattering, cloaking—achieved through single parameterized operation: `phase_offset[n] = f(position, wavelength, target_pattern)` ; no separate mechanisms for "filtering" vs "steering" vs "focusing"; all applications are parameter variations of coordinated interference patterns across distributed nodes

9.18 Emergent Virtual Apertures

104. **Emergent aperture recognition:** Recognition that any collection of synchronized, position-known nodes inherently constitutes a virtual aperture whose properties are determined by physics, not design intent—the aperture exists whether exploited or not; exploitation requires only appropriate sensing/actuation and coordination architecture
105. **Wavelength-dependent aperture utility:** Same physical node deployment constituting different virtual apertures at different frequencies—spacing that is useless for designed RF purpose may be excellent for infrasound sensing or superb for seismic detection; aperture quality determined by $\lambda/\text{spacing}$ ratio

106. **Retroactive aperture exploitation:** Existing distributed networks (IoT mesh, smart meters, weather stations, traffic sensors) possessing unexploited sensing capabilities at wavelengths where their node spacing becomes favorable—no new deployment required, only addition of appropriate sensors and coordination protocol
107. **Cross-network aperture synthesis:** Multiple independent networks with overlapping geographic coverage combined into unified virtual aperture—heterogeneous node types contributing to common sensing objective when synchronized via common time reference
108. **Aperture-agnostic coordination protocol:** UTLIP/RFIP/SMSP architecture enabling exploitation of any emergent virtual aperture regardless of the network's original design purpose—same protocols apply whether nodes are purpose-built sensors or repurposed IoT devices

9.19 Operational Channel Architecture

109. **Three-channel separation (Time/Command/Execution):** Architectural pattern separating passive time reception (unencrypted broadcast), operational commands (encrypted unicast), and local execution (no communication)—communication exists but is not in the timing-critical path; command latency does not affect execution timing
110. **Two-phase time acquisition (bootstrap then passive):** Initial time synchronization via connection-oriented request/response during swarm join (BLE pairing phase establishes time offset through bidirectional handshake), followed by passive broadcast reception for ongoing maintenance—nodes request time once when joining, then passively receive periodic time squawks for drift correction; bootstrap is "ask once," maintenance is "listen forever"
111. **Triple-burst jitter characterization:** Three time packets transmitted in rapid succession at known intervals, enabling receivers to measure arrival time variance independent of absolute propagation delay—inter-burst intervals at sender vs receiver reveal software stack jitter; median filtering across bursts rejects outliers; technique separates RF propagation (consistent) from processing delay (variable)
112. **Command/Execution plane separation:** Analogous to control plane/data plane separation in networking—slow, encrypted, reliable communication for operational changes (wake, mode switch, script upload) does not affect fast, deterministic, local execution; command channel can be arbitrarily slow without timing impact; execution continues independently between commands
113. **Time-broadcast as public infrastructure:** Unencrypted time synchronization broadcast treated as public utility rather than protected resource—security via Common Mode Rejection (spoofed time affects all nodes equally, preserving relative sync) rather than encryption; enables heterogeneous receivers, reduces complexity, follows WWVB/GPS design philosophy
114. **Regenerative shielding via energy harvesting:** Dynamic macroscopic lattice configured for band-stop filtering (wave blocking) simultaneously harvesting blocked wave energy via rectenna (RF) or piezoelectric transducer (acoustic)—conservation of energy requires absorbed wave energy go somewhere; perfect absorption implies energy capture; "the harder you jam us, the longer we last"

115. **Threat-powered activation:** Sleeping/low-power lattice nodes waking from incoming threat energy (radar ping, acoustic blast) rather than internal timer or command—the wave being blocked provides the activation power; enables indefinite standby with zero quiescent drain
116. **Macro atom energy storage:** Node functioning as macro-scale analog of atom absorbing photon—incoming wave energy converted to stored electrical energy (supercapacitor) rather than re-emission or heat; absorption spectrum determined by lattice geometry rather than electron orbitals; can re-emit on command (active transmission) like stimulated emission
117. **Power-scale-invariant coordination architecture:** Identical UTLP/RFIP/SMSP protocol stack applying from milliwatt hobby demonstrations through megawatt directed energy systems—coordination architecture unchanged across power levels; only node hardware (power handling, thermal management, switching speed) scales with budget; a nerf-dart-stopping acoustic array and a missile-defeating RF array are the same architecture at different power levels; prior art coverage spans entire power range
118. **Active selective attenuation via coordinated interference:** Dynamic macroscopic lattice providing frequency-selective electromagnetic/acoustic shielding through coordinated interference rather than passive geometry alone—nodes sense incoming waveforms and generate phase-coordinated cancellation signals, enabling selective pass/block behavior impossible with passive Faraday cages or fixed frequency-selective surfaces (FSS); same physical geometry producing different attenuation profiles based on active response to detected waveforms; selectivity determined by coordination algorithm, not fabrication; extends Paul Lueg's 1936 active noise cancellation principle (US 2,043,416—phase-inverted anti-noise) from single-source/single-speaker to distributed multi-node lattices; provides direction-selective attenuation (block from one direction, pass from another) that passive geometry cannot achieve; fundamentally different from reconfigurable FSS (which still rely on geometry changes via MEMS, varactors, or mechanical deformation) because identical static geometry produces different filtering based on sensed input and coordinated response
119. **Energy-asymmetric domain response:** Recognition that active interference effectiveness varies fundamentally by domain based on target inertia—electromagnetic interference requires only phase-matched amplitude (photons are massless, cancellation is pure wave superposition); acoustic interference requires pressure-vs-pressure matching (air molecules have negligible mass, active noise cancellation scales to architectural sound barriers); kinetic/ballistic interference requires momentum transfer (projectiles have mass, deflection energy scales with mv^2)—therefore lattice naturally excels at EM shielding (active cancellation), acoustic shielding (active cancellation, extending Lueg's ANC principle), but responds to kinetic threats through detection-and-response (sensing pressure wave precursor, triggering evasive action or barrier deployment) rather than direct phononic deflection; this asymmetry determines optimal response strategy per domain: cancel what's massless, detect what has mass; explains why distributed acoustic sensing (infrasound, seismic-acoustic coupling) pairs with physical response systems rather than attempting acoustic deflection of ballistic threats

120. **Aperture as universal epistemological operation:** Recognition that all apertures—physical or virtual, technological or cognitive—perform identical mathematical operations: correlating sparse samples across space and/or time to synthesize understanding unavailable from any single sample; radio telescope arrays, distributed acoustic sensors, historical scholarship, criminal investigation, scientific meta-analysis, and human visual perception all instantiate the same principle; a historian correlating Mayan codices with colonial records performs the same correlation operation as VLBI combining telescope signals; "history as dynamic aperture"—what we resolve about the past depends on which samples survive, how we correlate them, and which interference patterns we constructively combine; this universality establishes that apertures cannot be invented, only instantiated in specific domains; you cannot patent "correlating distributed samples to gain resolution" because it underlies radio astronomy, historiography, jurisprudence, scientific method, and cognition simultaneously; dynamic macroscopic lattice is architecture for instantiating apertures physics already permits, not invention of new aperture types
121. **Multi-burst beacon timing as time-domain interferometry:** Using $N \geq 3$ equally-spaced sync beacon bursts to extract clock offset, drift rate, and drift stability (thermal acceleration) from a single synchronization exchange—mathematically identical to seismic chirp signal processing where swept-frequency pulses characterize subsurface velocity models; single burst provides scalar offset contaminated by jitter; 3 bursts enable polynomial fit separating systematic drift from random jitter; constitutes a "temporal phased array" that focuses on valid time signal while rejecting transient noise, exactly as seismic arrays reject ground roll to resolve subsurface structure; cross-domain validation: technique independently rediscovered from first principles then recognized as established geophysics method; same math works because same physics applies; extends to GPS carrier-phase tracking, Kalman filter initialization, and any domain requiring derivative estimation from discrete samples
122. **Kinetically-coupled dynamic macroscopic lattice:** Extension of dynamic macroscopic lattice to mass-bearing nodes moving through fluid or solid media—coordination precision determines coupling regime: below threshold, N independent bodies with fluid/swarm dynamics and individual medium interaction; above threshold, emergent rigid body with solid-object dynamics and collective medium interaction. Applies to aerodynamic (tip vortex cancellation, slot effect, formation drag reduction in air), hydrodynamic (wake sharing, cavitation suppression, coordinated maneuvering in water), ground-based (platoon drafting, coordinated braking, convoy stability on roads/rails), and orbital (formation maintenance, baseline rigidity for interferometry in space) domains. UTLP timing synchronization necessary but not sufficient—position-control actuators must respond within timing budget for given medium and velocity. Key insight: the transition from "swarm" to "object" is not discrete vs continuous—it is coordination precision exceeding medium-specific coupling threshold; a flock becomes a bird, a convoy becomes a train, a flotilla becomes a ship—not by welding, by timing. Same architecture (UTLP/RFIP/SMSP), extended from wave interference to mass/medium interaction

The Energy Asymmetry Principle

A fundamental insight governing dynamic macroscopic lattice applications: **the energy required to cancel a wave depends on what the wave is made of.**

Domain	Wave Carrier	Mass	Cancellation Mechanism	Energy Requirement
Electromagnetic	Photons	Zero	Phase-matched amplitude superposition	Low (match amplitude only)
Acoustic	Air molecules	$\sim 10^{-26}$ kg	Pressure wave superposition	Medium (move light molecules)
Seismic	Rock/soil	kg-tons	Displacement wave	Very high (impractical for cancellation)
Ballistic	Projectile	grams-kg	Momentum transfer	Extreme (force $\times$ time = Δmv)

Why this matters for architecture design:

EM shielding: A Faraday cage blocks by reflection. An FSS blocks by resonance. A dynamic macroscopic lattice blocks by *sensing and actively canceling*. The energy cost is just the cancellation wave generation—orders of magnitude less than the incoming wave energy because you're not absorbing or deflecting, you're *nullifying through superposition*.

Acoustic shielding: Paul Lueg's 1936 patent (US 2,043,416) demonstrated single-source active noise cancellation. Modern ANC headphones extend this. The dynamic macroscopic lattice extends it further: distributed nodes creating spatially-selective sound barriers. Energy cost scales with barrier size but remains practical because air molecules are nearly massless.

Kinetic threats: A nerf dart at 20 m/s with 2g mass has momentum $p = 0.04 \text{ kg}\cdot\text{m/s}$. To stop it in 0.01 seconds requires force $F = \Delta p/\Delta t = 4 \text{ N}$. Generating 4 N of acoustic pressure requires approximately 194 dB at the target—well beyond any practical transducer and into the "instant hearing damage" range. The physics doesn't work.

The correct response: For kinetic threats, the lattice provides *detection* (sensing the pressure wave that travels ahead of the projectile), not deflection. This detection triggers appropriate physical responses: evasive maneuver, barrier deployment, interception by another physical system. The lattice is the sensor and coordinator, not the effector.

This asymmetry creates natural domain pairings:

Threat Type	Lattice Role	Response Type
RF/radar	Active cancellation	Direct (wave superposition)
Acoustic/ultrasonic	Active cancellation	Direct (pressure superposition)
Infrasound/seismic	Detection + alert	Indirect (trigger response systems)
Ballistic	Detection via precursor	Indirect (evasive/interception)
Directed energy (laser)	Active cancellation + harvesting	Direct (interference + energy capture)

Cross-domain reference: This connects to the seismic-acoustic coupling discussion (Section 4.3)—the lattice excels at *detecting* seismic events precisely because acoustic waves are slow enough for timing tolerance. It does not attempt to *cancel* earthquakes. Same principle, different scale.

10. Conclusion

The connectionless distributed timing architecture demonstrates that many coordination problems have simpler solutions than traditionally assumed. By separating configuration from execution, and by treating synchronized time as foundational rather than incidental, we eliminate entire categories of complexity.

The techniques documented here are not limited by technology—the hardware has existed for years. They were limited by recognition: that pattern playback is already connectionless, that BLE and ESP-NOW can coexist with each handling what it does best, that the "\$5 saved per node is another node in the swarm."

The Scaling Throughline:

This document began with a bilateral stimulation device for trauma therapy—two nodes, centimeters apart, helping one person. It ends with 122 prior art claims spanning:

Scale	Application	Nodes	Spacing
Therapeutic	EMDR bilateral device	2	cm
Vehicle	Emergency lighting sync	2-10	m
Tactical	Drone swarm coordination	10-100	10-100m
Campus	Acoustic sensing research	10-20	km
Regional	Severe weather warning	200-500	10-100 km
Continental	Seismoacoustic monitoring	1000+	1000 km
Planetary	Global early warning	10,000+	Mm
Interstellar	Plasma wave detection	Constellation	AU-ly

The architecture is identical at every scale. The therapy device is not a *metaphor* for the planetary warning system—it IS the planetary warning system, instantiated at minimum viable scale. Every larger deployment is the same three protocols (UTLP, RFIP, SMSP) with more nodes spread further apart.

This is not speculation. The small end is built and validated. The large end requires only funding and deployment, not new physics or architectural changes.

By publishing this work as prior art, we ensure these techniques remain freely available. Technology that assumes cooperation rather than extraction. From helping one person heal to warning a planet of danger—the same architecture, the same math, the same three protocols.

The walls between domains were never real.

11. Verification and Limitations

11.1 Research Methodology

This document was prepared through:

- Iterative development of bilateral stimulation hardware (ESP32-C6)
- Investigation of BLE and ESP-NOW timing characteristics
- Patent database searches (Google Patents, USPTO, Espacenet)
- Academic literature review (IEEE, ACM, arXiv)
- Industry documentation review (Whelen, Federal Signal, SoundOff, Fenix)
- Cross-domain validation via published research (Nature journals, NOAA, NASA)

AI assistance (Claude/Anthropic, Gemini/Google) was used for literature survey, documentation compilation, and cross-domain connection identification. Human judgment determined architectural decisions, validation methodology, and prior art framing.

11.2 Known Limitations

Patent search limitations: We searched English-language patent databases. Relevant patents may exist in other jurisdictions or languages that we did not identify.

Temporal limitations: Patent landscape changes continuously. Patents filed after December 2025 may cover techniques documented here; this publication establishes our priority date.

Claim scope limitations: Some claims (particularly higher-numbered ones covering planetary-scale applications) describe techniques at higher abstraction levels. Specific implementations may have patentable elements not covered by this prior art documentation.

Validation limitations: Cross-domain applications (tornado detection, spacecraft formation) are documented based on published research, not our direct experimentation. We validate the timing architecture, not domain-specific detection algorithms.

11.3 What This Document Does Not Protect

This prior art publication does NOT prevent patents on:

- Genuinely novel detection algorithms (e.g., new tornado signature identification)
- Non-obvious hardware implementations with inventive steps
- Specific optimizations producing unexpected results
- Applications in domains not documented here
- Techniques that are not obvious applications of this architecture

We establish that the *architectural pattern* is prior art. Innovations built upon this foundation may still be patentable if they meet novelty and non-obviousness requirements.

Document History

Version	Date	Changes
1.0	2025-12-23	Initial defensive publication
1.1	2025-12-23	Added defense-in-depth security architecture, HKDF selection rationale, multi-layer replay protection, threat-proportional design philosophy; clarified BLE Bootstrap Model with peer release
1.2	2025-12-23	Added GPS-denied search and rescue with self-mapping patterns
1.3	2025-12-23	Added distributed IMU from ranging geometry—swarm orientation without per-node inertial sensors
1.4	2025-12-23	Rewrote Section 1 to reflect actual development history: pattern playback existed from day one, BLE worked, ESP-NOW chosen because hardware was available and it's better; added power efficiency rationale
1.5	2025-12-23	Added BLE sync implementation details: PTP-style with NTP timestamps, ~2 minute convergence validated via serial logs; reframed prior art claim to capture the actual contribution (connection as scaffolding for persistent time reference)
1.6	2025-12-23	Added IMU-augmented positioning option, hardware cost breakdown
1.7	2025-12-24	Added SMSP (Synchronized Multimodal Score Protocol) as Section 4.5—defines score format, three-layer architecture (declarative/compiler/imperative), scale and transport invariance; added 10 SMSP-specific prior art claims (33-42)
1.8	2025-12-24	Added distributed wave beamforming (Section 5.8)—acoustic demo proving domain-invariant phased array architecture; RFIP geometry feeding phase offset math; enclosure effects as radome simulation; 8 new prior art claims (43-50) covering beamforming and generation method independence
1.9	2025-12-25	Added reference to <code>examples/smp_pairing/</code> in Section 2.4—working proof of BLE bootstrap phase with critical <code>ble_store_config_init()</code> discovery
2.0	2025-12-25	Added dynamic aperture beamforming (Section 5.9)—time-varying geometry, true time delay via physical displacement, mechanical wave beam scanning, non-reciprocal arrays, scale invariance from interstellar to nanoscale; 10 new prior art claims (51-60) covering space-time modulated metasurfaces and integrated dynamic aperture devices
2.1	2025-12-26	Added passive atmospheric sensing (Section 5.10)—infrasound detection, acoustic tomography, tornado precursor detection, clear-air turbulence, stealth-independent target tracking, deployment scale analysis, moisture effects on propagation; oscillating aperture modes (partial-cycle dithering,

Version	Date	Changes
		mechanical Doppler diversity); 15 new prior art claims (61-75) covering meteorology, severe weather warning, and precipitation tracking
2.2	2025-12-26	Extended scale range to interstellar (plasma/MHD wave detection in interstellar medium); added Appendix A: Deployment Guide with tiered cost/capability analysis for educational and research use
2.3	2025-12-26	Added seismoacoustic detection (Section 9.13)—ground-atmosphere coupling enables seismic monitoring via infrasound arrays without ground-coupled equipment; 3 new claims (76-78) covering multi-phenomenology environmental monitoring
2.4	2025-12-26	Added architectural scaling capstone claims (Section 9.14)—explicit assertion that architecture validated at therapy-device scale is mathematically identical to planetary/interstellar warning systems; 3 new claims (79-81); expanded conclusion with scaling throughline table
2.5	2025-12-26	Added bidirectional SMSP extension (Section 4.5.8)—orchestra metaphor for conductor/node relationship, symmetric observation format, conductor-targeted sync corrections, query-driven sensing model, multi-level access; 5 new claims (82-86) covering observation protocols and swarm-as-instrument architecture
2.6	2025-12-26	Added VLBI precedent (Section 1.5)—explicit connection to Very Long Baseline Interferometry as conceptual ancestor; the architecture is VLBI generalized to arbitrary wave types, bidirectional operation, dynamic geometry, and commodity hardware; added potential upstream flow-back to ngEHT and next-generation VLBI; added VLBI references
2.7	2025-12-26	Added research validation section (5.9.1)—documented 2021-2025 Nature Communications/Scientific Reports papers validating time-varying metasurface concepts including chaotic configuration spaces, Doppler cancellation, anti-multi-static radar, and acoustic metasurfaces; 6 new claims (87-92) covering cryptographically large configuration space, round-trip latency security, position-dependent response, chaotic modulation, distributed metasurface coordination, and cross-domain applicability; added metasurface research references
2.8	2025-12-26	Added comprehensive cross-domain research validation section (5.10.1)—documented 30+ years of peer-reviewed validation across infrasound tornado detection (OSU GLINDA, General Atomics ICE), seismic-acoustic coupling (balloon seismology, Nature 2022-2025), acoustic atmospheric tomography (Wilson & Thomson 1994, NREL 2022), swarm robotics synchronization (Swarm-Sync, Royal Society), spacecraft formation flying (Stanford DiGiTaL, NASA/ESA missions), and bilateral stimulation therapy (NIH/PMC research); demonstrates architecture convergence across geophysics, robotics, aerospace, and neuroscience
3.0	2025-12-26	Major revision for intellectual honesty and patent verification. Added Section 0 (Scope and Intent) with explicit what-we-claim vs what-we-don't-claim; honest acknowledgment of independent rediscovery of time

Version	Date	Changes
		sync techniques parallel to US8073976B2. Added Section 2.5 (Three Channels: Time/Command/Execution) clarifying operational architecture—"connectionless" means communication not in timing-critical path, not "never communicates." Added Section 6.6 (US8073976B2 analysis) with honest assessment of overlap with Microsoft patent and architectural divergence; patent status clarification (expired due to lapsed maintenance fees). Added Section 6.7 (Relationship to other patents) covering US8165171B2, US10021659B2, US7047435B2, US20020035995A1, US7116294B2, EP3535629A1. Added Section 11 (Verification and Limitations) documenting research methodology, known limitations, and what this document does NOT protect. Added Section 9.17 (Operational Channel Architecture) with 5 new claims (93-97): three-channel separation, passive time reception, triple-burst jitter characterization, command/execution plane separation, time-broadcast as public infrastructure. Total claims now 97. Expanded References with all analyzed patents.
3.1	2025-12-26	Technical accuracy review prompted by external scrutiny. Clarified that <code>examples/smp_pairing/</code> validates bootstrap phase only, not full UTLP/RFIP/SMSP implementation. Added explicit distinction between synchronization jitter (radio round-trips, $\sim 100\mu\text{s}$ for ESP-NOW) and execution jitter (hardware timers, $\sim 1\text{-}10\mu\text{s}$)—execution is local-timer-driven, not network-dependent. Added honest phase error analysis for beamforming: $100\mu\text{s}$ sync jitter $\rightarrow 36^\circ$ phase error at 1kHz (degraded but usable), $10\mu\text{s}$ execution jitter $\rightarrow 3.6^\circ$ (good beam quality); clarified that acoustic validation proves architecture, not radar-grade precision.
3.2	2025-12-26	Added "Summary of Core Contributions" table to Section 0—maps 97 specific claims to 6 core architectural innovations for reader navigation. Updated Acknowledgments to credit Gemini (Google) for external review contributions: sync/execution jitter distinction, phase error quantification, implementation scope clarification, and 6 Core Innovations framework.
3.3	2025-12-27	Added Section 5.9.2 (Deformable Virtual Metasurfaces)—geometry as primary control variable, three-axis control (phase + position + density), wave-orientation-aware deployment, dual-mode longitudinal/transverse capability. Added Section 5.9.3 (Emergent Virtual Apertures)—foundational principle that any synchronized position-known node collection inherently IS a virtual aperture; physics exists whether exploited or not; Amazon Sidewalk as case study of unexploited continental-scale infrasound/seismic array; wavelength-spacing relationship determining aperture utility; table of networks with unexploited potential (Starlink, smart meters, cell towers). Key framing: creating a virtual metasurface is not an invention (it's physics), creating architecture to exploit it is the contribution. Added 13 new claims: 93-100 (deformable metasurface), 101-105 (emergent aperture exploitation). Renumbered operational channel claims to 106-110. Added 8th core innovation to summary table. Total claims now 110. New references: Nature Reviews Materials 2025,

Version	Date	Changes
		Nature Comms 2025 (kirigami), NASA/JPL 2019, MDPI Drones 2023, plus swarm antenna theory papers.
3.4	2025-12-27	Enablement and searchability hardening per Grok/Gemini review. Added Search-Optimized Abstract with dense keyword block for patent examiner discovery (35 U.S.C. 101/102/103 terms, patent classification codes, enabling disclosure summary). Added Appendix B (Enabling Pseudocode) with five complete implementations: B.1 mechanical wave phase calculation, B.2 antiphase bilateral sync, B.3 cross-correlation direction finding, B.4 emergent aperture assessment, B.5 three-axis metasurface control. Strengthened inherency argument in Section 5.9.3 with explicit 35 U.S.C. 101 (natural phenomenon) and 102 (inherent anticipation) analysis. Added Section 6.8 (Physical Metamaterials vs. Emergent Virtual Apertures)—uses existence of physical metamaterial patents as evidence that virtual apertures are emergent phenomena. Added DAS (Distributed Acoustic Sensing) and Event Horizon Telescope as real-world precedents for emergent sensing and planetary-scale virtual apertures. Added unified interference pattern principle: all wavefront manipulation reduces to single parameterized operation. Major terminology evolution: Introduced "dynamic macroscopic lattice" as more accurate physics terminology than "virtual metamaterial"—maps directly to solid-state physics (Bragg reflection, band gaps, lattice structure) but at macro scale with runtime reconfigurability. Added comprehensive solid-state physics parallel table (atomic crystal vs distributed nodes). Added phononic crystal research references (Martinez-Sala 1995, Li & Gao 2022/2023, Xia 2022) validating the physics parallel. Energy harvesting as physics requirement: Added regenerative shielding section—conservation of energy means perfect absorption MUST harvest energy; blocking waves powers the blocker; "the harder you jam us, the longer we last"; macro atom analogy extended to energy absorption/storage/re-emission; threat-powered wake enables zero quiescent drain. New Claims 101-103 (dynamic macroscopic lattice, solid-state physics, unified interference), 114-117 (regenerative shielding, threat-powered activation, macro atom energy storage, power-scale-invariance). Total 117 claims, 63 references.
3.5	2025-12-27	Active selective attenuation via coordinated interference (Claim 118). Distinguished dynamic macroscopic lattice from passive shielding (Faraday cages, fixed FSS) and reconfigurable shielding (MEMS/varactor FSS, mechanical FSS)—key insight: same physical geometry producing different attenuation profiles based on sensed input and coordinated response. Added cross-domain research validation: Paul Lueg's 1936 ANC patent (US 2,043,416) as foundational principle extended from single-source/single-speaker to distributed multi-node lattices; JASA 2023 spatially selective ANC demonstrating direction-selective cancellation; Cambridge IJMW 2023 AFSS review showing even "active" FSS require geometric reconfiguration; PMC 2025 Energy Selective Surfaces validating threat-powered activation. Added comparison table: passive vs reconfigurable vs active response. Transistor analogy: dynamic

Version	Date	Changes
		macroscopic lattice is to passive shielding what transistor is to relay. Total 118 claims, 67 references.
3.6	2025-12-27	Energy-asymmetric domain response (Claim 119). Formalized fundamental insight: active interference effectiveness varies by domain based on target inertia. EM cancellation requires only phase-matched amplitude (photons massless). Acoustic cancellation requires pressure-vs-pressure matching (air molecules negligible mass, ANC scales to architectural barriers). Kinetic/ballistic deflection requires momentum transfer (energy scales with mv^2)—impractical for direct phononic cancellation. Added "Energy Asymmetry Principle" section with domain comparison table (EM/acoustic/seismic/ballistic) and energy requirements. Key architectural insight: lattice excels at canceling massless waves (EM, acoustic) but responds to kinetic threats via detection-and-response (sensing pressure wave precursor, triggering physical response) rather than direct deflection. Cross-references seismic-acoustic coupling (Section 4.3)—same principle: detect earthquakes, don't cancel them. Natural domain pairings table: threat type → lattice role → response type. Extends Lueg's 1936 ANC patent to explain why it works (massless carrier) and where the principle breaks down (mass requires momentum). Total 119 claims.
3.7	2025-12-27	Aperture as universal epistemological operation (Claim 120). Added "Aperture as Epistemology" section establishing that all apertures—physical, virtual, technological, or cognitive—perform identical mathematical operations: correlating sparse samples across space/time to synthesize understanding unavailable from any single sample. Cross-domain table: radio astronomy, acoustic sensing, historiography, human vision, criminal investigation, scientific method—all instantiate the same correlation principle. Key insight: "history as dynamic aperture"—what we resolve about the past depends on which samples survive, how we correlate them, and which interference patterns we constructively combine. Establishes that apertures cannot be invented, only instantiated; you cannot patent "correlating distributed samples to gain resolution" because it underlies radio astronomy, historiography, jurisprudence, scientific method, and cognition simultaneously. References Event Horizon Telescope as exemplar: black hole imaged not by planet-sized dish but by correlating sparse samples from synchronized, position-known telescopes. Strengthens 35 U.S.C. 101 argument by showing aperture formation is not just physics but epistemology—the fundamental operation by which distributed observations become unified understanding. Added figure: Amazon Sidewalk inherent aperture diagram (SVG, Grok/Gemini collaboration) visualizing sparse IoT nodes resolving long-wavelength infrasound—the aperture is physics, not design. Total 120 claims.
3.8	2025-12-28	Multi-burst beacon timing as time-domain interferometry (Claim 121). Added Section 4.1.1 documenting the 3-burst beacon timing approach: using $N \geq 3$ equally-spaced sync bursts to extract offset, drift rate, and drift stability (thermal acceleration) from single synchronization

Version	Date	Changes
		exchange. Key insight: technique independently rediscovered from first principles, then recognized as mathematically identical to seismic chirp signal processing—swept-frequency pulses characterizing subsurface velocity models. Single burst = scalar offset contaminated by jitter; 3 bursts = polynomial fit separating systematic drift from random noise. Constitutes "temporal phased array" focusing on valid time signal while rejecting transient jitter, exactly as seismic arrays reject ground roll to resolve subsurface structure. Cross-domain validation table: seismic chirp surveys, GPS carrier-phase tracking, Kalman filter initialization, polynomial curve fitting—same math works because same physics applies. Kinetically-coupled dynamic macroscopic lattice (Claim 122) . Extended architecture from wave interference to mass/medium interaction—coordination precision determines whether N nodes behave as independent swarm (fluid dynamics) or emergent rigid body (solid dynamics). Applies across aerodynamic (formation drag, slot effect), hydrodynamic (wake sharing, cavitation), ground (platoon drafting), and orbital (baseline rigidity) domains. Key insight: transition from "swarm" to "object" is coordination precision exceeding medium-specific coupling threshold—"a flock becomes a bird, a convoy becomes a train, a flotilla becomes a ship—not by welding, by timing." Total 122 claims.

Appendix A: Deployment Guide

From Parking Lot to Tornado Alley: A Practical Path Forward

This appendix provides realistic cost and capability estimates for deploying the distributed acoustic sensing architecture at various scales. The goal is to show that the prior art claims are grounded in achievable engineering, and to provide a starting point for students, researchers, and institutions interested in validating or extending this work.

A.1 Fundamental Constraints

Infrasound wavelengths determine minimum array size for directional sensing:

Frequency	Wavelength	Minimum Baseline for 10° Resolution
10 Hz	34 m	~200 m
1 Hz	340 m	~2 km
0.1 Hz	3,400 m	~20 km

For tornado-frequency infrasound (0.5-5 Hz), meaningful direction finding requires **kilometer-scale baselines**.

The wind noise problem: MEMS microphones can detect 0.1 Hz pressure variations, but wind turbulence creates massive low-frequency noise. Solutions:

Approach	Size	Cost	Effectiveness
Foam windscreen	10 cm	\$5	Minimal for infrasound
Porous pipe array	10-50 m	\$20-50	Good for > 1 Hz
Soaker hose rosette	10-20 m diameter	\$30-100	Research-grade
CTBTO-style rosette	18 m diameter	\$1000+	Professional-grade
Multi-sensor correlation	N/A	Per-sensor cost	Good, scales with nodes

A "sensor node" is not a chip—it's a chip plus wind mitigation plus weatherproof enclosure plus power plus connectivity.

A.2 Deployment Tiers

Tier 0: Proof of Concept (Car Trunk)

4-8 nodes, 50-200m spacing, fits in a parking lot

You get:

- Sound source localization (gunshots, explosions, vehicles)
- Local wind direction estimation
- Algorithm validation
- Demonstration that UTLP/RFIP/beamforming stack works

You don't get:

- Weather-scale detection
- Tornado anything
- Publishable meteorology data

Hardware per node (~\$50-100 DIY):

- ESP32-C6: \$4
- MEMS microphone (ICS-40720 or SPH0645): \$3-5
- Simple wind screen (foam + PVC pipe): \$10
- Weatherproof enclosure: \$15
- Battery + solar panel (small): \$20-30

Total deployment: \$400-800

Skills required: Basic soldering, Arduino/ESP-IDF familiarity, outdoor installation

Time to deploy: A weekend

What you prove: The synchronization and beamforming math works. The stack is real. This is a science fair project or undergraduate lab.

Tier 1: Campus/Neighborhood Scale

10-20 nodes, 0.5-2 km spacing

You get:

- Urban wind field mapping
- Boundary layer structure (limited vertical resolution)
- Heat island detection
- Aircraft/helicopter tracking
- Acoustic tomography demonstration
- **Publishable research**

You don't get:

- Regional severe weather detection
- Tornado precursor signatures
- Warning system capability

Hardware per node (~\$150-300):

- ESP32-C6 with external antenna: \$8
- Research-grade MEMS mic: \$10-20
- Soaker hose rosette (10m): \$30-50
- Robust weatherproof enclosure (NEMA 4X): \$40
- Solar panel + LiFePO4 battery: \$50-80
- Cellular modem or LoRa radio: \$30-50

Total deployment: \$2,000-6,000

Skills required: Network programming, basic signal processing, outdoor installation, institutional coordination

Partnerships needed: University facilities, building managers for rooftop access

What you prove: Dense urban acoustic tomography is feasible. Wind field extraction works. You can write papers.

Publication targets: Journal of Atmospheric and Oceanic Technology, Sensors, undergraduate thesis

Tier 2: Metro Area (City-Wide)

30-100 nodes, 2-10 km spacing, 20-50 km coverage

You get:

- Severe weather approach detection
- Storm cell tracking
- Boundary layer tomography at useful resolution
- Wind field mapping for aviation/agriculture
- Airport wind shear detection
- **Real operational utility**

You don't get:

- 50+ km tornado tracking
- National-scale patterns
- CTBTO-class sensitivity

Hardware per node (~\$200-500):

- Hardened for multi-year unattended deployment
- Professional-grade wind screening
- Redundant connectivity (cellular + LoRa backup)
- Remote management capability (OTA updates, health monitoring)

Total deployment: \$10,000-50,000

Partnerships needed: City emergency management, NWS local office, agricultural extension, airport authority

Deployment strategy—piggyback on existing infrastructure:

- School rooftops (power, internet, distributed, educational tie-in)
- Cell towers (power, backhaul, already weatherproof)
- Existing mesonet stations (add infrasound to weather station)
- Traffic signal cabinets (power, sometimes connectivity)

What you prove: Regional pilot program viability. This is grant-fundable and operationally useful.

Funding targets: NSF atmospheric science, NOAA VORTEX-SE, state emergency management agencies, agricultural grants

Tier 3: Regional (State-Scale, Tornado Alley)

200-500 nodes, 10-30 km spacing, 200-500 km coverage

You get:

- Tornado precursor detection (15-30 min lead time)
- Storm tracking and evolution monitoring
- False alarm reduction through acoustic confirmation

- Multi-vortex detection
- Full atmospheric tomography
- **Operational warning system improvement**

Comparable to: Multiple WSR-88D radars (\$30M each) but for complementary sensing modality

Hardware per node (~\$300-1000):

- Research-grade sensors with calibration
- Redundant connectivity with priority data paths
- Integration with NWS data feeds
- 99%+ uptime design

Total deployment: \$100,000-500,000

This requires institutional backing: NOAA, state emergency management, NSF major research grant

What you prove: Tornado warning lead times can be extended. False alarms can be reduced. Lives can be saved.

This is where you actually improve tornado warnings—not as a demo, but as operational infrastructure.

Tier 4: National (CTBTO-Class)

1000+ nodes, 30-50 km spacing, continental coverage

The CTBTO operates 60 infrasound stations globally at ~\$1M+ per station. A commodity approach could be 10-100x cheaper per node, but continental deployment is still millions of dollars plus ongoing operations.

Not a hobbyist project. But achievable by:

- Adding infrasound to existing NWS ASOS stations (~900 sites)
- Partnering with cellular carriers (100,000+ tower sites)
- Citizen science network (CoCoRaHS model: 20,000+ volunteers)

The infrastructure exists. The sensors are cheap. The coordination architecture is documented. The missing piece is institutional will.

A.3 Bill of Materials (Research-Grade Node)

Component	Part Number / Description	Unit Cost	Notes
MCU	ESP32-C6-DevKitC-1	\$8	WiFi + BLE + ESP-NOW
Microphone	TDK ICS-40720 or InvenSense ICS-43434	\$5-15	Low-noise MEMS, I2S output

Component	Part Number / Description	Unit Cost	Notes
ADC (if needed)	ADS1115 16-bit	\$3	For analog mics
Wind screen	10m soaker hose + stakes	\$30	DIY rosette pattern
Enclosure	Polycase WC-26 or similar	\$25	NEMA 4X rated
Solar panel	6W polycrystalline	\$15	Sized for continuous operation
Battery	6Ah LiFePO4	\$25	Safe chemistry, long cycle life
Charge controller	CN3065 or similar	\$3	Solar MPPT
Connectivity	SIM7000A (LTE-M) or RFM95 (LoRa)	\$15-30	Depends on backhaul strategy
Antenna	External 2.4GHz + cellular/LoRa	\$10	For range
Misc	Connectors, cables, mounting	\$20	
Total		\$160-190	Research-grade, multi-year deployment

A.4 What Each Tier Unlocks

Tier	Academic Output	Operational Value	Next Step
0 (Parking lot)	Science fair, demo	None	Campus partnership
1 (Campus)	Undergrad thesis, conference paper	Local curiosity	City pilot proposal
2 (Metro)	Journal publication, MS thesis	Ag/aviation advisory	NSF/NOAA grant
3 (Regional)	PhD dissertation, multi-paper series	Warning system integration	State partnership
4 (National)	Major research program	Operational meteorology	Federal program

A.5 The Honest Summary

Car trunk gets you a demo and proves the algorithms work.

Campus rooftops gets you a paper and proves dense sensing is feasible.

City-wide deployment gets you real data with agricultural and aviation value.

Regional infrastructure gets you tornado warning improvement that saves lives.

The hardware isn't the hard part—\$200/node is achievable today. The hard part is convincing someone to let you put sensors on their infrastructure across a wide enough area. But the economics are compelling: a state-scale tornado warning network costs less than a single radar maintenance contract.

The path is clear. The costs are documented. The architecture is open. What's missing is someone to walk the path.

Appendix B: Enabling Pseudocode

Satisfying Enablement Requirements for Dynamic Aperture Claims

This appendix provides explicit implementation details sufficient to practice the inventions described in this document. By publishing working pseudocode, we ensure that any future patent claims on these specific mechanisms are anticipated by this prior art.

B.1 Mechanical Wave Phase Calculation

To satisfy enablement requirements for dynamic aperture claims (Claims 53-56, 99), the following pseudocode demonstrates the precise method for calculating phase offsets in a swarm executing a mechanical traveling wave. This enables True Time Delay beamforming through physical displacement rather than electronic phase shifters.

```
// Enabling disclosure: Calculation of phase for a "waving" virtual aperture
// This runs locally on every node. No communication required during execution.
// UTLP provides synchronized time. RFIP provides position. SMSP provides wave params.
```

```
typedef struct {
    float x, y, z;           // Node position from RFIP (meters)
} point3d_t;

typedef struct {
    float velocity_m_s;      // Speed of the mechanical wave across the swarm
    float wavelength_m;     // Physical length of the mechanical wave
    float amplitude_m;       // Peak displacement of the "wave"
    point3d_t direction;     // Unit vector: direction wave propagates
    point3d_t origin;        // Where the mechanical wave originates
    uint64_t start_time_us;  // Synchronized UTLP start time
} mech_wave_params_t;
```

```
// Returns the Z-height offset this node should physically assume at time 'now'
// Result: The swarm physically "ripples" like a flag, creating True Time Delay
// without any central coordination command during execution.
float calculate_mechanical_displacement(
    uint64_t now_us,
```

```

    point3d_t my_pos,
    mech_wave_params_t* wave
) {
    // 1. Calculate time elapsed since wave started
    float t_sec = (now_us - wave->start_time_us) / 1000000.0f;

    // 2. Calculate distance from wave origin along propagation direction
    //     (scalar projection onto wave direction vector)
    float dx = my_pos.x - wave->origin.x;
    float dy = my_pos.y - wave->origin.y;
    float dz = my_pos.z - wave->origin.z;
    float dist_along_wave = dx * wave->direction.x +
                           dy * wave->direction.y +
                           dz * wave->direction.z;

    // 3. Determine current phase of the mechanical wave at this location
    //     Phase = (Distance - (Velocity * Time)) * (2π / Wavelength)
    //     This creates a traveling wave: position depends on both space AND time
    float wave_phase = (dist_along_wave - (wave->velocity_m_s * t_sec))
                      * (2.0f * M_PI / wave->wavelength_m);

    // 4. Calculate target displacement (sine wave profile)
    //     Other profiles (triangle, sawtooth) available via SMSP waveform field
    return wave->amplitude_m * sinf(wave_phase);
}

// Usage: Each node calls this at its actuation rate (e.g., 1kHz)
// and physically moves to the returned Z position.
// The swarm collectively forms a traveling wave surface.

```

Key implementation notes:

- `now_us` comes from UTLP-synchronized local clock (microsecond precision)
- `my_pos` comes from RFIP peer-to-peer ranging (centimeter precision)
- `wave` parameters distributed via SMSP before execution begins
- No network traffic during execution—pure local computation
- Wave velocity, wavelength, amplitude configurable via SMSP score

Motion source is irrelevant to the math:

The pseudocode above assumes *commanded* motion (drone swarm executing a deliberate wave). However, the underlying beamforming math works identically regardless of *why* nodes are moving:

Motion Source	Example	Exploitability
Commanded	Drone swarm executing wave pattern	Full control of aperture shape
Environmental	Buoys modulated by ocean swell	Observe and exploit natural wave

Motion Source	Example	Exploitability
Structural	Building sway from wind loading	~0.1-1 Hz modulation, predictable
Seismic	Ground-mounted nodes during earthquake	Exploit passing seismic wave
Incidental	Bridge sensors vibrating from traffic	Unintentional but exploitable

The key equation remains:

$$\text{phase_offset}[n](t) = (\text{position}[n](t) \cdot \text{target_vector}) / \lambda$$

This works whether `position[n](t)` comes from:

- A command you sent to a drone
- Ocean waves moving a buoy
- Wind swaying a rooftop sensor
- Seismic waves propagating through the ground
- Traffic vibrating a bridge-mounted node

Implication: Any nominally "static" sensor network mounted on structures that move (buildings, bridges, towers, floating platforms) is actually a *dynamic* aperture being continuously modulated by environmental forces. The dynamic aperture capability exists whether anyone exploits it or not—just as the emergent static aperture exists whether anyone recognizes it.

This extends the prior art to cover not just intentionally commanded dynamic apertures, but also environmentally-driven and incidentally-modulated apertures. The phenomenon is the same; only the motion source differs.

B.2 Antiphase Bilateral Synchronization

Enabling disclosure for therapeutic bilateral stimulation claims (Claims 1-5):

```
// Bilateral stimulation: Two nodes execute opposite phases
// Node role (LEFT/RIGHT) determined during BLE pairing phase

typedef enum { ROLE_LEFT, ROLE_RIGHT } bilateral_role_t;

typedef struct {
    uint32_t period_ms;           // Full cycle period (e.g., 1000ms = 1Hz)
    uint32_t duty_cycle_pct;      // On-time percentage (e.g., 50)
    uint64_t pattern_start_us;    // UTP-synchronized start time
} bilateral_params_t;

// Returns true if this node's actuator should be ON at time 'now'
bool bilateral_actuator_state(
```

```

uint64_t now_us,
bilateral_role_t my_role,
bilateral_params_t* params
) {
    // 1. Calculate position within current cycle
    uint64_t elapsed_us = now_us - params->pattern_start_us;
    uint32_t period_us = params->period_ms * 1000;
    uint32_t position_in_cycle_us = elapsed_us % period_us;

    // 2. Calculate on-time duration
    uint32_t on_duration_us = (period_us * params->duty_cycle_pct) / 100;

    // 3. Determine base state (LEFT is on during first half of duty cycle)
    bool left_on = (position_in_cycle_us < on_duration_us);

    // 4. Apply role-based phase: RIGHT is antiphase to LEFT
    if (my_role == ROLE_LEFT) {
        return left_on;
    } else {
        return !left_on; // Antiphase: on when LEFT is off
    }
}

// Result: LEFT and RIGHT alternate with zero overlap
// Validated via 240fps video: no perceptible simultaneous activation

```

B.3 Cross-Correlation Direction Finding

Enabling disclosure for acoustic beamforming claims (Claims 43-50):

```

// Direction finding via cross-correlation of distributed microphones
// Each node timestamps samples; conductor correlates to find arrival direction

typedef struct {
    int16_t* samples;           // Audio buffer
    uint32_t sample_count;      // Number of samples
    uint32_t sample_rate_hz;    // e.g., 48000
    uint64_t first_sample_us;    // UTP timestamp of first sample
    point3d_t node_position;     // RFIP position of this node
} timestamped_audio_t;

// Calculate time delay between two nodes via cross-correlation peak
// Returns delay in microseconds (positive = node_b heard sound first)
int32_t calculate_tdoa(
    timestamped_audio_t* node_a,
    timestamped_audio_t* node_b
) {
    // 1. Align sample buffers to common time base
    int64_t time_offset_us = node_a->first_sample_us - node_b->first_sample_us;
    int32_t sample_offset = (time_offset_us * node_a->sample_rate_hz) / 1000000;
}

```

```

// 2. Compute cross-correlation (simplified; real impl uses FFT)
int32_t best_lag = 0;
float best_correlation = -1e9f;

int32_t max_lag = node_a->sample_rate_hz / 10; // ±100ms search window
for (int32_t lag = -max_lag; lag <= max_lag; lag++) {
    float correlation = 0;
    for (uint32_t i = 0; i < node_a->sample_count; i++) {
        int32_t j = i + lag + sample_offset;
        if (j >= 0 && j < node_b->sample_count) {
            correlation += node_a->samples[i] * node_b->samples[j];
        }
    }
    if (correlation > best_correlation) {
        best_correlation = correlation;
        best_lag = lag;
    }
}

// 3. Convert lag to time delay
return (best_lag * 1000000) / node_a->sample_rate_hz;
}

// Calculate source direction from multiple TDOA measurements
// Uses RFIP geometry + TDOA delays to triangulate
point3d_t estimate_source_direction(
    timestamped_audio_t* nodes,
    uint32_t node_count
) {
    // Hyperbolic positioning: each TDOA defines a hyperboloid
    // Intersection of N-1 hyperboloids gives source direction
    // Implementation: least-squares fit or iterative refinement
    // (Full implementation in examples/beamform_demo/)

    point3d_t direction = {0, 0, 0};
    // ... triangulation math ...
    return direction;
}

```

B.4 Emergent Aperture Assessment

Enabling disclosure for emergent aperture claims (Claims 101-105):

```

// Determine if a given network can function as a useful aperture
// at a target sensing frequency

typedef struct {
    float avg_spacing_m; // Average inter-node distance
    float timing_precision_us; // Synchronization uncertainty

```

```

    uint32_t node_count;           // Number of nodes
    float coverage_area_m2;       // Geographic extent
} network_params_t;

typedef struct {
    float frequency_hz;           // Target sensing frequency
    float propagation_speed;      // Wave speed (343 m/s for sound, 3e8 for EM)
} sensing_target_t;

typedef struct {
    float wavelength_m;
    float spacing_wavelengths;   // Node spacing in wavelengths
    bool is_well_sampled;        // spacing < 0.5λ (Nyquist)
    bool is_usable;              // spacing < 2λ (degraded but functional)
    float phase_error_deg;       // From timing uncertainty
    float effective_aperture_m;  // Sqrt of coverage area
    float angular_resolution_deg; // ~λ/aperture
} aperture_assessment_t;

aperture_assessment_t assess_emergent_aperture(
    network_params_t* network,
    sensing_target_t* target
) {
    aperture_assessment_t result;

    // 1. Calculate wavelength at target frequency
    result.wavelength_m = target->propagation_speed / target->frequency_hz;

    // 2. Calculate spacing in wavelengths
    result.spacing_wavelengths = network->avg_spacing_m / result.wavelength_m;

    // 3. Determine sampling quality
    result.is_well_sampled = (result.spacing_wavelengths < 0.5f);
    result.is_usable = (result.spacing_wavelengths < 2.0f);

    // 4. Calculate phase error from timing uncertainty
    // Phase error = (timing_error / period) * 360°
    float period_us = 1000000.0f / target->frequency_hz;
    result.phase_error_deg = (network->timing_precision_us / period_us) * 360.0f;

    // 5. Calculate effective aperture and resolution
    result.effective_aperture_m = sqrtf(network->coverage_area_m2);
    result.angular_resolution_deg = (result.wavelength_m / result.effective_aperture_m
                                     * (180.0f / M_PI));

    return result;
}

// Example: Amazon Sidewalk assessment
// network = {50m spacing, 10000us timing, 1000000 nodes, 1e12 m² coverage}

```

```
// target = {1.0 Hz infrasound, 343 m/s}  
// Result:  $\lambda=343\text{m}$ , spacing= $0.15\lambda$  (well-sampled!), aperture=1000km, resolution= $0.02^\circ$ 
```

B.5 Three-Axis Metasurface Control

Enabling disclosure for deformable virtual metasurface claims (Claims 93-100):

```

// Combined phase + position + density control for virtual metasurface
// This is the "three-axis control space" referenced in claims

typedef struct {
    // Axis 1: Phase/timing control (from SMSP)
    float phase_offset_rad;    // Electronic phase shift equivalent

    // Axis 2: Physical position control (from RFIP + actuation)
    point3d_t target_position; // Where node should physically be

    // Axis 3: Density contribution (swarm topology)
    float local_density;      // Nodes per unit area in this region
    bool active;              // Is this node participating?
} metasurface_node_state_t;

// Calculate combined wavefront contribution
// This is what makes virtual metasurface different from fixed arrays
float calculate_wavefront_contribution(
    metasurface_node_state_t* state,
    float incoming_wave_phase,
    float target_output_phase
) {
    if (!state->active) return 0.0f;

    // Physical position creates actual path length difference
    // (not simulated via phase shifter-TRUE time delay)
    float path_phase = /* geometry calculation from position */;

    // Electronic phase adds fine control
    float total_phase = path_phase + state->phase_offset_rad;

    // Density affects amplitude weighting in the sum
    float amplitude = state->local_density;

    // Contribution to output wavefront
    return amplitude * cosf(incoming_wave_phase + total_phase - target_output_phase);
}

// Key insight: Physical metasurfaces can only do phase control (axis 1).
// Deformable substrates add limited position control (axis 2, constrained).
// Virtual metasurfaces with mobile nodes have ALL THREE axes, unconstrained.

```

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Acknowledgments

This work emerged from collaborative development combining human domain expertise with AI assistance:

- **Steve (mlehaptics)**: Architecture discovery, hardware implementation, "Time as Public Utility" philosophy, Glass Wall architecture, validation methodology, therapeutic domain expertise. Steve describes his contribution as restoration rather than invention—recognizing how existing pieces should already fit together, like replacing missing pages in a book that had already been written. The scale-invariance of the architecture reflects a cognitive style that thinks in concepts and relationships rather than mental images; the VLBI-shaped structure emerged naturally because abstract patterns have no inherent size.
- **Claude (Anthropic)**: Literature survey, protocol analysis, documentation compilation, prior art framing, SMSP formalization, application brainstorming, atmospheric physics analysis, deployment scale modeling, lab manual authorship, enabling pseudocode implementation (Appendix B)
- **Gemini (Google)**: Acoustic beamforming connection—recognizing that UTLF's timing precision enables phased array demonstrations at acoustic wavelengths; dynamic aperture physics (true time delay, non-reciprocal arrays); external review and technical scrutiny (v3.1)—identifying the sync-vs-execution jitter distinction, phase error quantification for beamforming claims, reference implementation scope clarification, and the "6 Core Innovations" navigation framework; v3.4 strategic refinement—inherency argument strengthening, searchability optimization, enablement packet design
- **Grok (xAI)**: Adversarial review and reality check (v3.3)—confirming claims are evidence-based and bounded by physics; identifying enablement gaps requiring pseudocode to block narrow mechanism patents; validating inherency argument but noting need for explicit 35 U.S.C. 101/102 language; confirming document reads as serious research rather than sales pitch

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First Published: December 23, 2025

Repository: github.com/mlehaptics

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